

Computer Science XI

Exam revision cards — definitions, tables, mnemonics and ★
Golden topics on one glance each.

Bilingual support: English + اردو

امتحان چیت شیٹ — تعریفیں، جدول، یادداشت اور گولڈن موضوعات

01 Computer Systems

02 Computational Thinking & Algorithm

03 Programming Fundamentals

04 Data and Analysis

05 Application and Impacts of Computing

06 Digital Literacy

XI

Contents

45 Golden cards are listed below. Start there if you have less than two hours.

Chapter 1: Computer Systems — 20 cards

Chapter 2: Computational Thinking & Algorithm — 13 cards

Chapter 3: Programming Fundamentals — 24 cards

Chapter 4: Data and Analysis — 15 cards

Chapter 5: Application and Impacts of Computing — 13 cards

Chapter 6: Digital Literacy — 17 cards

Night-before recaps: one-page summaries of every chapter

★ Golden index — if you only have two hours

These are the high-yield topics. Say each definition aloud, then draw the table or diagram from memory.

صرف دو گھنٹے ہوں تو یہ گولڈن کارڈز زبانی دہرائیں، پھر جدول حافظے سے بنائیں۔

1. ★ **Ch 1.** Analog and Digital Signals
2. ★ **Ch 1.** Logic Gates (Basic Gates: AND, OR, NOT)
3. ★ **Ch 1.** Logic Diagrams
4. ★ **Ch 1.** Karnaugh Maps (K-Maps)
5. ★ **Ch 1.** Software Development Life Cycle (SDLC)
6. ★ **Ch 1.** The Waterfall Model
7. ★ **Ch 1.** The Agile Model
8. ★ **Ch 1.** Communication Models and the OSI 7-Layer Model
9. ★ **Ch 2.** Algorithm
10. ★ **Ch 2.** Algorithm vs Pseudocode
11. ★ **Ch 2.** Decomposition
12. ★ **Ch 2.** Sorting Algorithms: Bubble Sort
13. ★ **Ch 2.** Sorting Algorithms: Selection Sort
14. ★ **Ch 2.** Searching Algorithms: Binary Search
15. ★ **Ch 3.** Fundamentals of Python Programming
16. ★ **Ch 3.** Variables
17. ★ **Ch 3.** Data Types in Python
18. ★ **Ch 3.** Operators and Operands: Arithmetic and Assignment
19. ★ **Ch 3.** Relational, Logical and Membership Operators
20. ★ **Ch 3.** Bitwise Operators
21. ★ **Ch 3.** Control Structures
22. ★ **Ch 3.** Selection Statements: if and if-else
23. ★ **Ch 3.** Selection Statements: if-elif-else and Nested if
24. ★ **Ch 3.** Repetition Statements: the for Loop and range()
25. ★ **Ch 3.** The while Loop, break and continue
26. ★ **Ch 3.** Nested Loops and for vs while
27. ★ **Ch 4.** Keys and Integrity Constraints
28. ★ **Ch 4.** Entity Relationship Model (ER-Model)
29. ★ **Ch 4.** Referential Integrity
30. ★ **Ch 4.** Creating Queries in MS Access
31. ★ **Ch 5.** Artificial Intelligence (AI)
32. ★ **Ch 5.** Internet of Things (IoT) & Components
33. ★ **Ch 5.** Information Sources (Primary, Secondary, Tertiary)
34. ★ **Ch 5.** Assistive Technologies
35. ★ **Ch 6.** Types of Data
36. ★ **Ch 6.** Data-Collection Strategies
37. ★ **Ch 6.** Interviews and Surveys
38. ★ **Ch 6.** Prototypes, Observation and Simulations
39. ★ **Ch 6.** Primary and Secondary Data
40. ★ **Ch 6.** Case Study: Digital Inquiry Project
41. ★ **Ch 6.** Steps 1 – 3: Advanced Search and Methodology
42. ★ **Ch 6.** Steps 4 – 6: Survey Preparation and Primary Data Collection
43. ★ **Ch 6.** Steps 7 – 9: Secondary Data and Spreadsheet Organisation
44. ★ **Ch 6.** Steps 10 – 12: Data Analysis, Conclusion and Recommendations
45. ★ **Ch 6.** The Final Digital Artefact

How to use these Cheat Sheets

This booklet is the **night-before and last-week** revision pack. Each topic is one card: the definition you must write, the table or formula, a memory trick, the usual exam wording, and one Urdu line. It is not a textbook. Open the Student's Edition when you need a worked example.

یہ چیٹ شیٹ امتحان سے پہلے رات اور آخری ہفتے کے لیے ہے۔ ہر موضوع ایک کارڈ ہے۔ مثالیں اصل کتاب میں ہیں۔

How to revise with this booklet

1. **Two hours left.** Read only the ★ Golden cards and the Golden index. Say each definition aloud.
2. **One day left.** One chapter at a sitting. Cover the card, write the table from memory, then check.
3. **The night before.** Read the night-before recaps. Do not start a new chapter.
4. **In the hall.** Remember the recipe: *Define. Explain. Example. Diagram. Working.*

What each card holds

- **Definition** — the 1-mark sentence. Write it exactly.
- **Points / table / snippet** — the 3-mark and 5-mark body.
- **Remember** — a mnemonic or trap to avoid.
- **Exam** — how the board asks it.
- **اردو** — the same idea in one spoken line.

★ **Golden** cards are high-yield. If time is short, skip non-Golden cards first, not these.

CHAPTER 1

Computer Systems

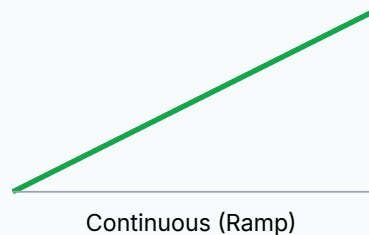
★ 8 Golden · 20 cards · Analog and Digital Signals · Logic Gates (Basic Gates: AND, OR, NOT) · Logic Diagrams · Karnaugh Maps (K-Maps) · Software Development Life Cycle (SDLC) · The Waterfall Model · The Agile Model · Communication Models and the OSI 7-Layer Model

1.1.1 Discrete and Continuous Quantities

Everything around you that can be measured or counted is a quantity.

- Quantities come in two kinds.
- Discrete quantities are separate, countable items with specific values.
- You can count them one by one: 1, 2, 3 ... There is nothing "in between".

Discrete (counted)	Continuous (measured)
Number of students in a classroom	Temperature of water
Number of cars in a parking lot	Speed of a moving car
Has fixed, separate values	Has any value in a range



Stairs have fixed steps (discrete). A ramp lets you stand at any point (continuous).

Remember: Stairs vs Ramp. On stairs you can only stand on step 1, 2 or 3 — that is discrete data. On a ramp you can stand at any height — that is continuous data. Ask...

منفصل (Discrete) مقداریں وہ ہیں جنہیں ایک ایک کر کے گنا جا سکے، جیسے کلاس میں طلبہ کی تعداد۔

1.1.2 Introduction to Digital Systems

A digital system is an electronic system that uses only two values: 0 and 1.

- 0 represents OFF, LOW, or FALSE.
- 1 represents ON, HIGH, or TRUE.

Advantage	Detail
Reliable	Less affected by noise and interference.
Accurate	No loss of quality when copying data.
Easy to design	Only two states (0 and 1) to handle.
Programmable	Can be easily controlled by software.

Remember: Think of a bit as a light switch: it is either ON (1) or OFF (0). A digital system is a huge panel of such switches.

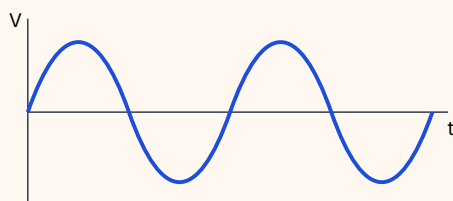
ڈیجیٹل نظام صرف دو قیمتیں استعمال کرتا ہے: 0 اور 1، جنہیں بیت (Bit) کہتے ہیں۔

1.1.3 Analog and Digital Signals ★ Golden

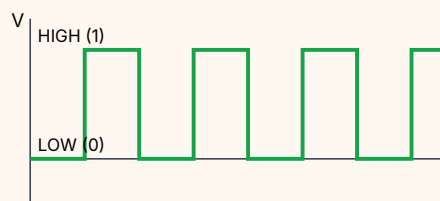
A signal is a changing voltage (or other quantity) that carries information.

- Analog signals are continuous signals that change smoothly over time, like a sine wave.
- Examples: the human voice, a traditional mercury thermometer.
- Digital signals have only two levels: HIGH (1) and LOW (0).

Feature	Analog data	Digital data
Nature	Continuous flow.	Discrete, separate values (bits).
Wave shape	Sine wave (smooth curve).	Square wave (sudden jumps).
Reliability	Low (affected by noise).	High (resistant to noise).
Examples	Human voice, traditional thermometer.	Computer data.



Analog Signal (Sine Wave)



Digital Signal (Square Wave)

An analog signal changes smoothly; a digital signal jumps between only two levels.

Remember: Analog = A smooth curve. Digital = Doorsteps. Analog looks like ocean waves; digital looks like a castle wall (square teeth).

Exam: Common questions: "Differentiate between analog and digital signals" (3–4 marks) and "Draw an analog and a digital signal." Write the comparison as a table with at least...

اینالاگ سگنل وقت کے ساتھ ہموار انداز میں بدلتا ہے، جیسے سائن ویو (Sine Wave)؛ انسانی آواز اس کی مثال ہے۔

1.1.4 Boolean Algebra

Boolean algebra was developed by the English mathematician George Boole.

- It is a kind of algebra that deals with only two values: TRUE (1) and FALSE (0).
- It is the foundation of digital electronics and computer programming — every if condition in a program is a small piece of Boolean algebra.

Concept	Description	Example
Variables	Letters that hold the value 0 or 1.	A, B, C, X, Y
Operations	Actions performed on variables.	AND, OR, NOT
Expressions	Combinations of variables and operations.	A AND B, Y = A + B

Trap: In Boolean algebra "+" means OR, not addition.

بولین الجبرا جارج بول (George Boole) نے بنایا۔

1.1.5 Boolean Operations

There are three basic Boolean operations. Every other operation can be made from them.

- Every other operation can be made from them.
- AND operation ($\cdot$) — gives output 1 only when ALL inputs are 1.
- Expression: $Y = A \cdot B$ (often written just AB).

Operation	Real-life analogy
AND	A door with two locks. The door opens only when both locks are unlocked.
OR	A room with two doors. You can enter if either door is open.
NOT	A room light switch. When the switch is ON the room is lit; when it is OFF the room is dark — the state is the opposite of "dark".

Remember: AND is "strict" — everyone must agree (all 1s). OR is "easy-going" — one 1 is enough. NOT is "the rebel" — it always says the opposite.

AND آپریشن تب ہی 1 دیتا ہے جب تمام ان پٹ 1 ہوں (دو تالوں والا دروازہ).

1.1.6 Truth Tables

A truth table shows all possible input combinations and the output for each one, for a logic gate or a circuit.

- Count the number of inputs (n).
- Calculate total rows = 2^n .
- List all input combinations in binary counting order (0, 1, 2, 3 ... in binary).
- Calculate the output for each combination.

A	B	Combination
0	0	Row 1
0	1	Row 2
1	0	Row 3
1	1	Row 4

Remember: Quick fill trick for 3 inputs: In column C write 0,1,0,1... (change every row). In column B write 0,0,1,1... (change every 2 rows). In column A write...

ٹروٹھ ٹیبل (Truth Table) کسی گیٹ یا سرکٹ کے تمام ممکنہ ان پٹ اور ان کے آؤٹ پٹ دکھاتا ہے۔

1.1.7 Logic Gates (Basic Gates: AND, OR, NOT) ★ Golden

Logic gates are electronic circuits that perform Boolean operations.

- They take one or more input bits and produce one output bit.
- They are the building blocks of all digital systems — a processor contains billions of them.
- The three basic gates match the three basic operations: the AND gate, the OR gate and the NOT gate (also called an inverter).

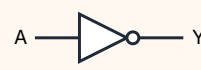
A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1



$$\text{AND: } Y = A \cdot B$$



$$\text{OR: } Y = A + B$$



$$\text{NOT: } Y = A'$$

Symbols of the basic gates. AND has a flat back and round front ("D" shape); OR has a curved back and pointed front; NOT is a triangle with a small circle (bubble).

Remember: Shortcut: AND output is 0 if any input is 0. OR output is 1 if any input is 1. Learn just the "special row": AND is 1 only at 11; OR is 0 only at 00.

Exam: You are often asked to "draw the symbol, write the expression and truth table" of a gate (2–3 marks each). Draw neat symbols with labelled inputs A, B and output Y....

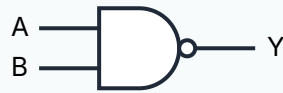
لاچک گیٹس الیکٹرانک سرکٹ ہیں جو بولین آپریشن انجام دیتے ہیں اور تمام ڈیجیٹل نظاموں کی بنیادی اینٹیں ہیں۔

1.1.7 Universal Logic Gates (NAND and NOR)

NAND means "NOT AND": it is an AND gate followed by a NOT.

- NOR means "NOT OR": an OR gate followed by a NOT.
- On the symbol, the NOT is shown as a small bubble at the output.
- NAND and NOR are called universal gates because any logic circuit in the world can be built using only NAND gates (or only NOR gates).

A	B	A · B	Y
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0



$$\text{NAND: } Y = (A \cdot B)'$$



$$\text{NOR: } Y = (A + B)'$$

NAND = AND symbol + bubble. NOR = OR symbol + bubble.

Remember: NAND is the exact opposite of AND (0 only at 11). NOR is the exact opposite of OR (1 only at 00). Just flip the AND/OR output column.

NAND کا مطلب ہے NOT AND اور NOR کا مطلب ہے NOT OR.

1.1.7 Advanced Logic Gates (XOR and XNOR)

XOR (Exclusive OR) gives output 1 when the inputs are different.

- In terms of basic operations, $A \oplus B = A'B + AB'$.
- XNOR (Exclusive NOR) is XOR followed by NOT.
- It gives output 1 when the inputs are the same.

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0



$$\text{XOR: } Y = A \oplus B$$



$$\text{XNOR: } Y = (A \oplus B)'$$

XOR = OR symbol with an extra curve at the back. XNOR = XOR symbol + bubble.

Remember: Memory trick: XOR outputs 1 when inputs are "eXclusively" different. XNOR outputs 1 when inputs are the same. XNOR is sometimes called the "equality gate".

XOR گیٹ تب 1 دیتا ہے جب دونوں ان پٹ مختلف ہوں، اور XNOR گیٹ تب 1 دیتا ہے جب دونوں ان پٹ ایک جیسے ہوں۔

1.1.8 – 1.1.10 Logic Expressions, Minterms and Maxterms

A logic expression (Boolean expression) combines variables with operations, for example $Y = A + B'C$.

1. Parentheses () → 2. NOT (') → 3. AND (·) → 4. OR (+)

- To evaluate it correctly you must follow the order of precedence, just like BODMAS in maths: 1.

A	B	C	B'	B'·C	Y = A + B'·C
0	0	0	1	0	0
0	0	1	1	1	1
0	1	0	0	0	0
0	1	1	0	0	0
1	0	0	1	0	1
1	0	1	1	1	1
1	1	0	0	0	1
1	1	1	0	0	1

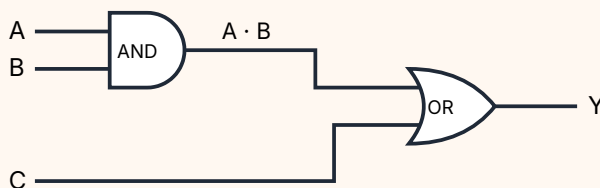
Remember: SOP = 1s, POS = 0s. Minterm: 0 → bar, 1 → plain. Maxterm: the reverse (1 → bar, 0 → plain). Think "Min likes 1s, Max likes 0s".

ایکسپریشن حل کرتے وقت ترتیب یہ ہے: پہلے قوسین، پھر NOT، پھر AND اور آخر میں OR.

1.1.11 Logic Diagrams ★ Golden

A logic diagram shows how logic gates are connected to create a circuit.

- It is a "picture" of a Boolean expression.
- To draw one, use the order of precedence: the operation with the highest priority is drawn first (nearest the inputs), and the last...



$Y = (A \cdot B) + C$: AND first, then OR.

Remember: Read the expression and ask "What is done last?" That gate goes on the far right, next to Y. Work backwards to the inputs. Count gates: $Y = A \cdot B + C$ needs 2...

Exam: Typical question: "Draw the logic diagram of $Y = \dots$ " and sometimes "also make its truth table" (4–5 marks). Write the precedence steps in one line, draw neat gate symbols...

لاجک ڈایاگرام دکھاتا ہے کہ گیٹس آپس میں کیسے جڑ کر سرکٹ بناتے ہیں۔

1.1.13 Karnaugh Maps (K-Maps) ★ Golden

A Karnaugh map (K-map) is a graphical tool used to simplify Boolean expressions.

- It is a grid of cells — one cell for each row of the truth table.
- The cells are arranged so that neighbouring (adjacent) cells differ by only ONE bit.
- This ordering is called Gray code.

	B = 0	B = 1
A = 0	Cell 0 (A'B')	Cell 1 (A'B)
A = 1	Cell 2 (AB')	Cell 3 (AB)

Remember: Gray code song: "zero-zero, zero-one, one-one, one-zero" — only one bit flips at each step. Imagine the K-map is drawn on a tube, so the left and right edges...

Exam: K-maps are a favourite long question (5–6 marks): "Simplify $F(A,B,C) = \sum m(\dots)$ using a K-map." Draw the map with labelled Gray-code headings, fill every cell (write the 0s...

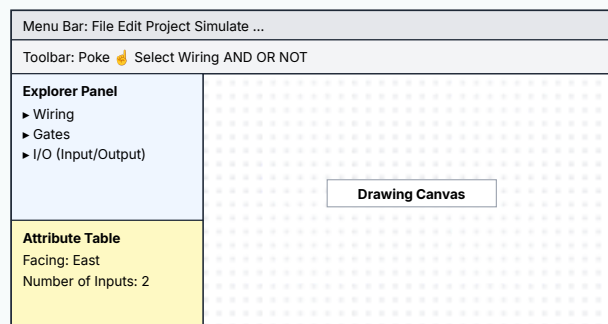
کارنو میپ (K-Map) بولین ایکسپریشن کو آسان بنانے کا تصویری طریقہ ہے۔

LogiSim Evolution: Simulation Tool

LogiSim Evolution is a free graphical tool for designing and simulating logic circuits.

- "Simulating" means the computer pretends to be the real circuit: you can switch inputs on and off and watch the output change.
- It lets you see the truth tables and Boolean expressions you learn in theory.

Area	What it does
Menu Bar (top)	File, Edit, Project, Simulate menus — open/save files and control the simulation.
Toolbar	Quick access to standard tools: Wiring, Poke, and AND, OR, NOT gates.
Explorer Panel (left)	Contains the component libraries: Gates, I/O, Wiring.
Attribute Table (bottom left)	Adjust the properties of the selected gate, e.g. Number of Inputs or Facing direction.
Drawing Canvas (right)	The dotted grid where you place and wire the gates.



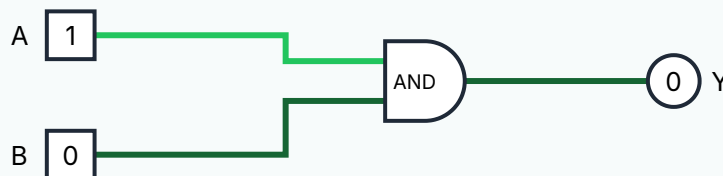
The main areas of the LogiSim Evolution window.

Remember: Colours like a traffic light: Green means "working" (bright = 1, dark = 0). Blue means "the wire is lonely — not connected". Red means "stop, something is..."

لاجی سیم ایوولوشن (LogiSim Evolution) لاجک سرکٹ بنانے اور چلا کر دیکھنے کا سافٹ ویئر ہے۔

Building a Circuit in LogiSim

Poke A	Poke B	Y shows	Output wire colour
0	0	0	Dark green
0	1	0	Dark green
1	0	0	Dark green
1	1	1	Bright green



Bright green = 1, dark green = 0

Testing with the Poke Tool: A = 1 (bright green wire), B = 0 (dark green), so $Y = 1 \cdot 0 = 0$.

Remember: Input pin = square (you push it). Output pin = circle (it shows a light). Poke = finger — you "poke" inputs to change them.

لاجی سیم میں $Y = A \cdot B$ بنانے کے لیے AND گیٹ رکھیں، بائیں جانب دو ان پٹ پن (چوکور) اور دائیں جانب ایک آؤٹ پٹ پن (گول) رکھیں، پھر وائر...

1.2.1 – 1.2.3 Software Development Life Cycle (SDLC) ★ Golden

The Software Development Life Cycle (SDLC) is a structured process that guides software development from beginning to end.

1. Requirement Analysis → 2. System Design → 3. Implementation → 4. Testing → 5. Deployment → 6. Maintenance

- It ensures clear goals, better planning, risk reduction and high quality.
- Without it, teams forget features, go over budget and deliver buggy software.
- SDLC has 6 phases: 1.

Phase	Activities and details	Deliverable (what it produces)
1. Requirement Analysis	Meet clients, understand their needs, list the features. This is the most important phase — mistakes here spread to every later phase.	SRS Document (Software Requirements Specification)
2. System Design	Create the High-Level Design (overall architecture), the Low-Level Design (database, UI screens) and flowcharts.	Design Document
3. Implementation	Write the actual code in programming languages (Python, Java, etc.) based on the design documents.	Working Source Code

Remember: Remember the order with "Real Developers Implement, Test, Deploy, Maintain" (R-D-I-T-D-M). Black box = a closed box (you can't see inside). White box = a glass...

Exam: "Explain the phases of SDLC" is a very common long question (5–8 marks). Name all 6 phases in order, write 1–2 lines of activities for each, and state the deliverable of...

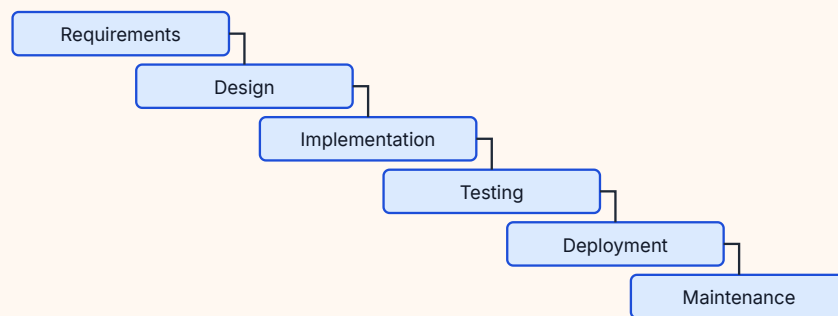
ایس ڈی ایل سی (SDLC) سافٹ ویئر بنانے کا منظم طریقہ ہے جس کے چھ مراحل ہیں: ضروریات کا تجزیہ، سسٹم ڈیزائن، کوڈنگ (Implementation)، ...

1.2.4 The Waterfall Model ★ Golden

The Waterfall model is a linear, sequential approach to the SDLC.

- Each phase must be completed before the next one begins, like water falling down steps.
- There is no easy way to go back to a previous phase.

Advantages (Pros)	Disadvantages (Cons)
Simple to understand and manage.	Not flexible to changes.
Well documented at every step.	Testing happens very late in the process.



The Waterfall model: each phase "falls" into the next, in one direction only.

Remember: Waterfall = water only flows downwards. Once water has fallen it cannot climb back up — once a phase is finished you do not go back.

Exam: Expect "Explain the Waterfall model with its advantages and disadvantages" or "When should the Waterfall model be used?" Draw the step diagram, list the three "when to..."

وائرفال (آبشار) ماڈل ایک سیدھا اور ترتیب وار طریقہ ہے؛ ہر مرحلہ مکمل ہونے کے بعد ہی اگلا شروع ہوتا ہے اور پیچھے جانا مشکل ہوتا ہے۔

1.2.4 The Agile Model ★ Golden

The Agile model is an iterative approach: the software is built in small, repeated cycles called sprints (usually 2–4 weeks long).

Plan → Design → Build → Test → Review & feedback → Next sprint ↻

- Each sprint plans, builds and tests a small part of the software, and then shows it to the customer.
- Customer feedback is gathered continuously and used in the next sprint.
- Plan→Design→Build→Test→Review & feedback→Next sprint ↻

Advantages (Pros)	Disadvantages (Cons)
Highly flexible to changes.	Less documentation.
Early delivery of working software.	Unpredictable end-time and cost.

Remember: Agile = agile athlete: quick and able to change direction. Think of making a pizza where the customer tastes each slice and you adjust the next one.

Exam: Questions ask you to define Agile, explain sprints, list pros/cons, or say when to use it. Always mention the words "iterative", "sprints (2–4 weeks)" and "continuous..."

ایجائل (Agile) ماڈل میں سافٹ ویئر چھوٹے چھوٹے ادوار میں بنتا ہے جنہیں اسپرینٹ (Sprint) کہتے ہیں، جو عموماً دو سے چار ہفتے کے ہوتے ہیں۔

1.2.5 – 1.2.7 Waterfall vs Agile: Comparison and Case Studies

Both models follow the same SDLC phases. The difference is how they move through them: Waterfall goes through each phase once, in a straight line; Agile goes through all phases many times, in small loops.

- The difference is how they move through them: Waterfall goes through each phase once, in a straight line; Agile goes through all phases...

Dimension	Waterfall model	Agile model
Flexibility	Rigid. Hard to make changes later.	Very flexible. Adapts to changes easily in the next sprint.
Customer involvement	Only at the start (requirements) and the end (delivery).	Continuous throughout the project (after every sprint).
Delivery of software	Single delivery at the very end.	Frequent, partial deliveries after every sprint.
Testing	Done late, in a dedicated testing phase.	Continuous testing within every sprint.
Documentation	Heavy documentation required.	Light documentation. Focus is on working software.

Remember: One question decides it: "Will the requirements change?" No → Waterfall. Yes / not sure → Agile.

وائرفال ماڈل سخت (Rigid) ہے، گاہک صرف شروع اور آخر میں شامل ہوتا ہے، سافٹ ویئر ایک ہی بار آخر میں ملتا ہے اور دستاویزات زیادہ ہوتی ہیں۔

1.3.1 – 1.3.4 Communication Models and the OSI 7-Layer Model ★ Golden

A communication model explains how data travels from one device to another through a network.

- Because sending data is complicated, the job is split into layers; each layer does one part of the job and passes the data to the next...
- The OSI (Open Systems Interconnection) model was developed by ISO (International Organization for Standardization) in 1984.
- It divides communication into 7 layers.

Layer	Function	Protocols / examples
7. Application	Provides network services directly to user applications (browser, email program).	HTTP, HTTPS, FTP, SMTP, DNS
6. Presentation	Data translation, encryption (security) and compression.	SSL/TLS, JPEG, MP3, ASCII
5. Session	Opens, maintains and closes communication sessions.	NetBIOS, RPC
4. Transport	Reliable end-to-end delivery, error checking, port numbering.	TCP (reliable), UDP (fast)



The 7 layers of the OSI model with the mnemonic "Please Do Not Throw Sausage Pizza Away".

Remember: Mnemonic (bottom to top, 1 → 7): "Please Do Not Throw Sausage Pizza Away" = Physical, Data Link, Network, Transport, Session, Presentation, Application. (Top...

Exam: A guaranteed long question: "Explain the seven layers of the OSI model." Draw the 7-layer stack (numbered, layer 7 on top), then write one function and one or two...

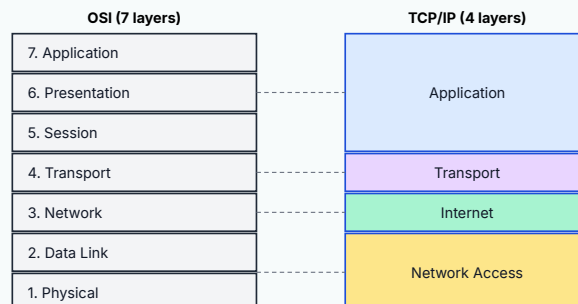
کمیونیکیشن ماڈل بتاتا ہے کہ ڈیٹا نیٹ ورک کے ذریعے ایک ڈیوائس سے دوسری تک کیسے پہنچتا ہے۔

1.3.5 – 1.3.6 TCP/IP Model vs OSI Model

The OSI model is mainly a reference model used for teaching and design.

- The TCP/IP model is the practical model used on the real Internet.
- It condenses the 7 OSI layers into just 4 layers.

TCP/IP model (4 layers)	Corresponds to OSI layers
1. Application	Application, Presentation, Session (7, 6, 5)
2. Transport	Transport (4)
3. Internet	Network (3)
4. Network Access	Data Link, Physical (2, 1)



How the 7 OSI layers map onto the 4 TCP/IP layers.

Remember: Count it as 3-1-1-2: the TCP/IP Application layer takes 3 OSI layers, Transport takes 1, Internet takes 1, Network Access takes 2. Total 3+1+1+2 = 7.

ٹی سی پی / آئی پی (TCP/IP) ماڈل حقیقی انٹرنیٹ پر استعمال ہونے والا عملی ماڈل ہے اور اس کی صرف چار تہیں ہیں۔

Key terms

Term	Meaning	اردو
Discrete quantity	A countable quantity with separate, fixed values.	منفصل مقدار
Continuous quantity	A quantity that can take any value within a range.	مسلسل مقدار
Bit	A binary digit, 0 or 1.	بائری ہندسہ
Digital system	An electronic system that uses only the values 0 and 1.	ڈیجیٹل نظام
Analog signal	A continuous signal that changes smoothly, like a sine wave.	اینالگ سگنل (مسلسل اشارہ)
Digital signal	A signal with only two levels, HIGH and LOW (square wave).	ڈیجیٹل سگنل
Noise	Unwanted interference that disturbs a signal.	شور (مداخلت)
Boolean algebra	Algebra of two values, TRUE (1) and FALSE (0), developed by George Boole.	بولین الجبرا
Truth table	A table of all input combinations and their outputs.	ٹروٹھ ٹیبل (صداقت جدول)
Logic gate	An electronic circuit that performs a Boolean operation.	منطقی گیٹ
Universal gate	A gate (NAND or NOR) from which any circuit can be built.	عالمگیر گیٹ
Unary operation	An operation with only one operand, e.g. NOT.	یک عددی عمل
Minterm	A product (AND) of all variables for a row with output 1.	منٹرم
Maxterm	A sum (OR) of all variables for a row with output 0.	میکس ٹرم
SOP	Sum of Products: minterms joined with OR.	حاصل ضرب کا مجموعہ
POS	Product of Sums: maxterms joined with AND.	مجموعوں کا حاصل ضرب
Karnaugh map (K-map)	A grid used to simplify Boolean expressions by grouping 1s.	کارنو نقشہ
Gray code	An ordering where neighbouring values differ by only one bit.	گرے کوڈ
Simulation	Running a computer model of a circuit to test it.	نقلی عمل (سمیولیشن)

Term	Meaning	اردو
SDLC	Software Development Life Cycle: the 6-phase process for building software.	سافٹ ویئر کی تیاری کا دورِ حیات
SRS	Software Requirements Specification: document listing what the software must do.	سافٹ ویئر ضروریات کی دستاویز
Black box testing	Testing inputs and outputs without seeing the code.	بلیک باکس جانچ
White box testing	Testing the internal logic while seeing the code.	وائٹ باکس جانچ
Waterfall model	A linear SDLC model; each phase is finished before the next.	آبشار ماڈل
Agile model	An iterative SDLC model using short sprints and constant feedback.	ایجنائل (لچکدار) ماڈل
Sprint	A short development cycle in Agile, usually 2–4 weeks.	اسپرینٹ (مختصر ترقیاتی دور)
Protocol	A set of rules for communication, e.g. HTTP, TCP.	پروٹوکول (ضوابطِ ابلاغ)
OSI model	ISO's 7-layer reference model of network communication (1984).	او ایس آئی ماڈل
TCP/IP model	The practical 4-layer model used on the Internet.	ٹی سی پی/آئی پی ماڈل

Chapter in one look

- Discrete quantities are counted (students, cars); continuous quantities are measured and can take any value (temperature, speed). Stairs vs ramp.
- A digital system uses only two values, 0 (OFF/LOW/FALSE) and 1 (ON/HIGH/TRUE), called bits. It is reliable, accurate, easy to design and programmable.
- Analog signals change smoothly (sine wave) and are affected by noise; digital signals jump between HIGH and LOW (square wave) and resist noise.
- Boolean algebra, by George Boole, works with only TRUE (1) and FALSE (0) using variables, operations and expressions.
- AND (·) gives 1 only when all inputs are 1; OR (+) gives 1 when any input is 1; NOT (!) inverts its single input (unary).
- A truth table lists all 2^n input combinations in binary order with their outputs.
- Logic gates are the building blocks of digital systems: AND, OR, NOT (basic); NAND, NOR (universal); XOR, XNOR (advanced).
- NAND and NOR are universal: any circuit can be built using only one of them.
- XOR gives 1 when inputs are different; XNOR gives 1 when inputs are the same.
- Order of precedence: Parentheses → NOT → AND → OR. A logic diagram is drawn in the same order.
- Minterms (SOP) come from rows with output 1; maxterms (POS) come from rows with output 0.
- K-maps use Gray code (00, 01, 11, 10); group 1s in sizes 1, 2, 4, 8, as large as possible; overlap and wrap-around are allowed; changing variables are eliminated.
- LogiSim Evolution simulates circuits: input pins are squares, output pins circles; the Poke Tool toggles inputs; bright green = 1, dark green = 0, blue = unconnected, red = error.
- SDLC has 6 phases: Requirement Analysis (SRS), Design, Implementation, Testing (black box/white box), Deployment (direct/phased/pilot), Maintenance.
- Waterfall is linear and suits fixed requirements (AI-Noor Library); Agile is iterative with 2–4 week sprints and suits changing requirements (AI-Noor Student App).
- The OSI model (ISO, 1984) has 7 layers — "Please Do Not Throw Sausage Pizza Away"; the practical TCP/IP model has 4 layers: Application, Transport, Internet, Network Access.

Computational Thinking & Algorithm

★ 6 Golden · 13 cards · Algorithm · Algorithm vs Pseudocode · Decomposition · Sorting Algorithms: Bubble Sort · Sorting Algorithms: Selection Sort · Searching Algorithms: Binary Search

2.1 Computational Thinking

Computational Thinking (CT) is a systematic approach to solving problems.

Problem → Decomposition → Pattern recognition → Abstraction → Algorithm → Program

- Decomposition — breaking a big problem down into smaller parts.
- Pattern recognition — identifying things that repeat or follow a rule.
- Abstraction — hiding (ignoring) details that do not matter.
- Algorithms — developing clear step-by-step instructions to solve the problem.

Remember: Remember the four parts of CT with the phrase "Don't Pick Any Apple": Decomposition, Pattern recognition, Abstraction, Algorithm.

کمپیوٹیشنل تھنکنگ (Computational Thinking) مسائل حل کرنے کا ایک منظم طریقہ ہے۔

2.2 Algorithm ★ Golden

An algorithm is a step-by-step logical procedure used to solve a problem.

- Input — the data it starts with.
- Clear steps — each step has only one meaning.
- A definite end — it must finish after a limited number of steps.
- Output — the result it produces.

Reason	What it means for you
Structured approach	Algorithms provide a structured approach to problem-solving, making complex tasks manageable.
Optimisation	They optimise time and memory usage, which is crucial for large-scale applications (for example, searching millions of records).
Reusability	They can be reused for similar problems, saving time and effort. A sorting algorithm for marks can also sort prices.
Accuracy and consistency	They ensure accuracy and consistency, reducing human error. The same steps always give the same result.
Blueprint	They act as a blueprint (a plan) before actual code implementation, just like an architect's map before building a house.

Remember: Remember the five reasons as S-O-R-A-B: Structured, Optimised, Reusable, Accurate, Blueprint.

Exam: Very common short question: "Define algorithm and write its importance." Write the definition word for word — a step-by-step logical procedure used to solve a problem;...

الگورتھم (Algorithm) کسی مسئلے کو حل کرنے کا مرحلہ وار منطقی طریقہ ہے۔

2.2.2 Algorithm vs Pseudocode ★ Golden

Both algorithms and pseudocode are used in problem-solving, but they serve different purposes in the design phase.

Problem → Algorithm (what) → Pseudocode (how) → Program (code) → Output

- The algorithm says what must be done.
- The pseudocode shows how it could be implemented, using programming-like keywords such as IF, FOR and WHILE, but it still cannot be run on...

Point	Algorithm	Pseudocode
Definition	A step-by-step logical procedure to solve a problem.	A human-readable representation of an algorithm using plain English & keywords.
Nature	Conceptual and language-independent.	Semi-formal and structured, but not executable code.
Purpose	To define what needs to be done logically.	To present how it can be implemented in a readable format.
Form	Usually described in plain steps, bullets or numbered points.	Written using structured constructs (IF, FOR, WHILE) like programming.
Focus	Focuses on the logic and steps of the problem.	Focuses on representing steps clearly before coding.
Audience	For anyone trying to understand the logic (including non-programmers).	Mainly for programmers or students preparing for coding.
Format rules	No fixed format; can be informal.	Follows standard pseudocode conventions for clarity.

Remember: Teacher's analogy: an algorithm is like a recipe ("Bake a cake at 350 degrees"). Pseudocode is like a structured kitchen manual using standard verbs (SET...).

Exam: "Differentiate between algorithm and pseudocode" is a favourite 4–5 mark question. Draw a two-column table and give at least five points (definition, nature, purpose,...).

الگورتھم بتاتا ہے کہ کیا کرنا ہے، جبکہ سوڈو کوڈ (Pseudocode) بتاتا ہے کہ اسے کیسے کیا جا سکتا ہے۔

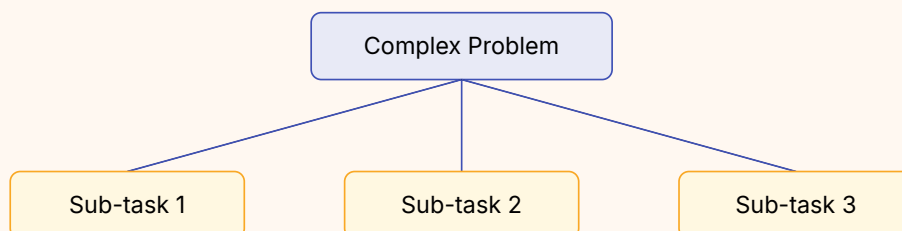
2.3.1 Decomposition ★ Golden

Decomposition is a problem-solving technique in which a complex task or system is broken down into smaller, manageable parts.

Vocabulary → Grammar rules → Tenses → Sentence building

- By dividing a big problem into simpler steps, it becomes easier to understand, develop and solve.
- Each small part is called a sub-task (or sub-problem).
- You solve each sub-task separately, then join the solutions to solve the whole problem.

Sub-task	Description
Learn vocabulary	Learn the words for people, actions and things (e.g., "I", "eat", "apple").
Grammar rules	Learn the sentence structure: Subject → Verb → Object.
Learn tenses	Present, past, future (e.g., "I eat" vs. "I ate").
Sentence building	Combine words using the rules to create valid sentences: "I ate an apple."



Decomposition: one complex problem is split into smaller sub-tasks.

Remember: Think of eating a big roti: you cannot eat it in one bite, so you break it into small pieces. Decomposition = "break the roti".

Exam: Expect "Define decomposition with a real-life example" or "Explain the benefits of decomposition". Give the definition, draw the small tree (Complex Problem → Sub-task...).

ڈیکمپوزیشن (Decomposition) کا مطلب ہے کسی بڑے اور پیچیدہ مسئلے کو چھوٹے اور آسان حصوں میں تقسیم کرنا۔

2.3.2 Pattern Recognition

Pattern recognition is the process of identifying regularities, repetitions or rules in data.

- What is repeating?
- What changes in a predictable way?
- Can I generalise this (write one rule that works for every case)?

Row	Stars
1	1
2	2
3	3
...	...
N	N

```
print("*")
print("**")
print("***")
print("****")
print("*****")
```

Remember: Whenever you see "the same thing again and again, with a small change each time", think loop. Pattern recognition is the habit of spotting that.

پیٹرن ریکگنیشن (Pattern Recognition) کا مطلب ہے ڈیٹا میں دہرائی جانے والی چیزوں یا اصولوں کو پہچاننا۔

2.3.3 Abstraction

Abstraction means hiding unnecessary features and focusing only on the essential ones needed to solve a problem.

Boil water → Add tea → Steep → Pour → Serve

- Necessary steps — the core logic that never changes.
- Changes based on the situation — these become variables (inputs) in the algorithm.
- Abstraction items (noise) — details you can ignore completely.

1. Necessary steps (core logic)	2. Changes based on situation (variables)	3. Abstraction items (ignore / noise)
1. Boil water 2. Add tea leaves/bag 3. Steep for a few minutes 4. Pour into cup 5. Serve	Ingredients (black, green, milk) Additions (sugar, lemon) Tools (stove vs kettle)	Brand of kettle Cup colour or design Where tea is made (kitchen, office)

Remember: Abstraction = "Keep the skeleton, drop the decoration." Ask: "If I remove this detail, does the solution still work?" If yes, it is noise.

ایبسٹریکشن (Abstraction) کا مطلب ہے غیر ضروری تفصیلات کو چھپا کر صرف ضروری باتوں پر توجہ دینا۔

2.4 Sorting and Searching Algorithms

Sorting and searching are both fundamental techniques of computer science.

temp = A (temp = 8) → A = B (A = 4) → B = temp (B = 8)

- They work on lists of the same data type (for example, a list of all integers or a list of all names).
- Sorting/Searching Arranging a collection of items in an order.
- The order may be ascending (0–9, A–Z) or descending (9–0, Z–A).

Sorting	Searching
Arranging a collection of items in an order. The order may be ascending (0–9, A–Z) or descending (9–0, Z–A). Used to organise data.	Locating a specific element within a list to find its position or check its existence (is it there or not?).
Examples: Bubble Sort, Selection Sort	Examples: Linear Search, Binary Search
Real life: merit list arranged from highest to lowest marks.	Real life: finding your roll number in the merit list.

Trap: Position 1 in the textbook algorithm is index 0 in Python.

سارٹنگ (Sorting) کا مطلب ہے ڈیٹا کو صعودی (چھوٹے سے بڑے) یا نزولی (بڑے سے چھوٹے) ترتیب میں رکھنا۔

2.4.1 Sorting Algorithms: Bubble Sort ★ Golden

Bubble Sort is an elementary sorting algorithm that works by repeatedly comparing adjacent (side-by-side) elements and swapping them if they are in the wrong order.

Compare DATA[j] and DATA[j+1] → Left > Right? → Yes: swap → Move one step right → End of pass: largest is at the end → Repeat N-1 passes

- In each pass, the largest remaining value "bubbles up" to the end of the list, like an air bubble rising to the top of water.
- So after pass 1 the largest value is in the last place, after pass 2 the second largest is in second-last place, and so on.
- A list of N items needs N - 1 passes, and pass K makes N - K comparisons.

Comparison	Pair	Left > Right?	Action	List after
1st vs 2nd	[8, 4], 1, 9, 3	8 > 4 Yes	Swap	4, 8, 1, 9, 3
2nd vs 3rd	4, [8, 1], 9, 3	8 > 1 Yes	Swap	4, 1, 8, 9, 3
3rd vs 4th	4, 1, [8, 9], 3	8 > 9 No	No change	4, 1, 8, 9, 3
4th vs 5th	4, 1, 8, [9, 3]	9 > 3 Yes	Swap	4, 1, 8, 3, 9

Remember: Bubble = neighbours. Bubble Sort only ever compares neighbours (j and j+1). Each pass pushes the biggest bubble to the end. For descending order, just change >...

Exam: The most common question: "Sort the list ... using Bubble Sort. Show each pass." Draw one table per pass with columns Pair, Compare, Action, List after. Mark the element...

ببل سارٹ (Bubble Sort) میں ساتھ ساتھ موجود دو عناصر کا موازنہ کیا جاتا ہے اور اگر وہ غلط ترتیب میں ہوں تو ان کی جگہ بدل دی جاتی ہے۔

2.4.1 Sorting Algorithms: Selection Sort ★ Golden

Selection Sort is a simple sorting algorithm that works by repeatedly selecting the smallest element from the unsorted portion of the list and moving it to its correct position at the front.

Fix position K → Compare DATA[K] with DATA[PTR], PTR = K+1 ... N → DATA[K] > DATA[PTR]? swap → End of pass: smallest is at K → K = K + 1

- Think of it as filling seats from the front: first find who belongs in position 1, then position 2, and so on.
- In the textbook version, the element at position K is compared with every element after it; whenever a smaller one is found, the two are...
- So at the end of pass K, the K-th smallest value sits in position K.

Compare	List	Pos1 > PosX?	Action	List after
Pos1 vs Pos2	[8], [4], 1, 9, 3	8 > 4 Yes	Swap	4, 8, 1, 9, 3
Pos1 vs Pos3	[4], 8, [1], 9, 3	4 > 1 Yes	Swap	1, 8, 4, 9, 3
Pos1 vs Pos4	[1], 8, 4, [9], 3	1 > 9 No	No change	1, 8, 4, 9, 3
Pos1 vs Pos5	[1], 8, 4, 9, [3]	1 > 3 No	No change	1, 8, 4, 9, 3

Remember: Teacher's tip — spot the difference: in Selection Sort the inner loop starts at outer + 1 and compares with the fixed position DATA[outer]. In Bubble Sort the...

Exam: Typical questions: "Sort ... using Selection Sort, showing all steps", "Differentiate between Bubble Sort and Selection Sort" and "Why is Selection Sort not stable?" In...

سلیکشن سارٹ (Selection Sort) میں فہرست کے غیر ترتیب شدہ حصے سے سب سے چھوٹا عدد چن کر اسے سامنے اس کی درست جگہ پر رکھا جاتا ہے۔

2.4.2 Searching Algorithms: Linear Search

Linear Search, also known as sequential search, works by examining each element in the list one by one, from the start, until the target is found (or the list ends).

PTR = first → DATA[PTR] = FindValue? → Yes: ValueFound = 1, exit → No: PTR = PTR + 1 → List ended? print "Not Found"

- It works on both sorted and unsorted data.
- A flag variable (here ValueFound) starts at 0 and becomes 1 when the item is found.
- After the loop, the flag tells us whether to print the item or "Not Found".

Step	Element checked	Condition (== 7)	Result
1	8	8 == 7 ?	Not found
2	3	3 == 7 ?	Not found
3	6	6 == 7 ?	Not found
4	1	1 == 7 ?	Not found
5	7	7 == 7 ?	Found! (Exit loop)

Remember: Linear Search is like looking for your friend in a queue by checking every face from the front. Best case: 1 check (first person). Worst case: N checks (last...)

لینیئر سرچ (Linear Search) کو ترتیب وار تلاش (Sequential Search) بھی کہتے ہیں۔

2.4.2 Searching Algorithms: Binary Search ★ Golden

Binary Search is a highly efficient search algorithm that works by repeatedly dividing the search interval in half.

BEG = first, END = last → MID = INT((BEG+END)/2) → Item = DATA[MID]? Found → Item < DATA[MID]: END = MID-1 → Else: BEG = MID+1 → Repeat while BEG ≤ END

- BEG (low) — first position of the current part.
- END (high) — last position of the current part.
- MID = INT((BEG + END) / 2) — the middle position. INT means "drop the decimal part".
- Equal → found, stop.

Step	BEG (low)	END (high)	MID	DATA[MID]	Compare	Action
1	1	9	INT(10/2) = 5	5	7 > 5	BEG = 6 (go right)
2	6	9	INT(15/2) = 7	7	7 = 7	Found, LOC = 7

Remember: Binary Search is how you find a word in a dictionary: open the middle, decide "before or after?", and throw away half the book each time. "Sorted first, then..."

Exam: You will usually be given a sorted list and an item and asked to "search using Binary Search, showing all steps". Always draw a table with BEG, END, MID, DATA[MID],...

بائنری سرچ (Binary Search) ایک تیز رفتار طریقہ ہے جو ہر قدم پر تلاش کے حصے کو آدھا کر دیتا ہے۔

2.5 Evaluation Criteria of Algorithm

Many algorithms can solve the same problem.

- To judge which is better, we evaluate them on three criteria: correctness, efficiency and clarity.

Input	Operation (divide by 2, find remainder)	Remainder == 0?	Result
3	3 / 2 (Quotient = 1, Remainder = 1)	No	Odd
6	6 / 2 (Quotient = 3, Remainder = 0)	Yes	Even
15	15 / 2 (Quotient = 7, Remainder = 1)	No	Odd

```
for number in [3, 6, 15]:
    remainder = number % 2
    if remainder == 0:
        print(number, "remainder", remainder, "→ Even")
    else:
        print(number, "remainder", remainder, "→ Odd")
```

Remember: Remember C-E-C: Correct (right answer, always finishes), Efficient (fast and light), Clear (easy to read).

کسی الگورتھم کو تین معیارات پر پرکھا جاتا ہے: درستگی (Correctness) یعنی درست نتیجہ دے، غیر معمولی صورتوں کو سنبھالے اور ہمیشہ ختم...

2.6 How to Decide Which Algorithm to Use

Before choosing an algorithm, ask three questions: What is the input size?

What is the task? → Need order? → Sorting algorithm → Need to find? Is data sorted? → No → Linear Search → Yes → Binary Search

- What is the input size? (10 items or 10 million?)
- Is the data sorted?
- Do we need speed or memory efficiency?
- If searching on unsorted data → use Linear Search.

Algorithm	Type	Best case	Worst case	Extra memory	Needs sorted data?	Stable?
Bubble Sort	Sorting	$O(N^2)$ as written; $O(N)$ with an early-stop check	$O(N^2)$	None — $O(1)$	—	Yes
Selection Sort	Sorting	$O(N^2)$	$O(N^2)$	None — $O(1)$	—	No
Linear Search	Searching	$O(1)$ (first item)	$O(N)$ (last item / absent)	None — $O(1)$	No	—
Binary Search	Searching	$O(1)$ (item at first MID)	$O(\log N)$	None — $O(1)$	Yes	—

Remember: Two keywords decide most exam answers: "sorted?" and "how big?". Unsorted or small → Linear. Sorted and big → Binary.

الگورتھم چننے سے پہلے دیکھیں کہ ڈیٹا کتنا بڑا ہے، ترتیب شدہ ہے یا نہیں، اور ہمیں رفتار چاہیے یا میموری کی بچت۔

Key terms

Term	Meaning	اردو
Computational thinking	A systematic approach to solving problems using decomposition, patterns, abstraction and algorithms.	کمپیوٹیشنل سوچ
Algorithm	A step-by-step logical procedure to solve a problem, independent of any language.	الگورتھم، مرحلہ وار طریقہ حل
Pseudocode	A human-readable, structured but non-executable representation of an algorithm.	سوڈو کوڈ، فرضی کوڈ
Decomposition	Breaking a complex problem into smaller, manageable parts.	تقسیم کاری
Sub-task	One small part of a decomposed problem.	ذیلی کام
Pattern recognition	Identifying regularities, repetitions or rules in data.	نمونے کی پہچان
Abstraction	Hiding unnecessary details and focusing on essentials.	تجربید، غیر ضروری تفصیل چھپانا
Noise	Details that do not affect the solution and can be ignored.	فالتو تفصیلات
Sorting	Arranging items in ascending or descending order.	ترتیب دینا
Searching	Locating an element in a list to find its position or existence.	تلاش کرنا
Ascending order	Smallest to largest (0–9, A–Z).	صعودی ترتیب
Descending order	Largest to smallest (9–0, Z–A).	نزولی ترتیب
Bubble Sort	Sort that repeatedly compares and swaps adjacent elements.	بل سارٹ
Selection Sort	Sort that repeatedly selects the smallest element and puts it at the front.	سلیکشن سارٹ
Pass	One full run of the inner loop through the list.	ایک چکر
Swap	Exchanging the values of two positions, usually with a temp variable.	جگہ کا تبادلہ
In-place sorting	Sorting without needing extra memory for another list.	اضافی میموری کے بغیر ترتیب

Term	Meaning	اردو
Stable sort	A sort that keeps equal items in their original order.	مستحکم ترتیب
Linear (sequential) search	Checking each element one by one until the target is found.	ترتیب وار تلاش
Binary search	Search that halves a sorted list at each step.	دو حصی تلاش
BEG / MID / END	First, middle and last positions of the part being searched in Binary Search.	آغاز / درمیان / اختتام
Flag	A variable (like ValueFound) that records whether something has happened.	نشان دہی کرنے والا متغیر
Correctness	Giving expected output, handling edge cases and always finishing.	درستگی
Efficiency	How well an algorithm uses time and memory as input grows.	کارکردگی
Clarity	How easy an algorithm is to read, understand and modify.	وضاحت
Trace table	A table recording values step by step during a dry run.	ٹریس ٹیبل، جانچ جدول
Edge case	An unusual or extreme input used to test an algorithm.	انتہائی صورت
Time complexity (Big O)	A measure of how the number of steps grows with input size, e.g. $O(N)$, $O(\log N)$.	وقتی پیچیدگی

Chapter in one look

- **Computational thinking** is systematic problem solving using decomposition, pattern recognition, abstraction and algorithms.
- An **algorithm** is a step-by-step logical procedure to solve a problem; it is conceptual and language-independent.
- Algorithms give structure, optimise time and memory, are reusable, ensure accuracy and act as a blueprint for code.
- **Pseudocode** is a human-readable, structured representation of an algorithm using keywords like IF, FOR and WHILE; it is not executable.
- An algorithm defines *what* to do; pseudocode shows *how* to implement it.
- **Decomposition** breaks a big problem into smaller sub-tasks; it reduces complexity, helps debugging and promotes reusability.
- **Pattern recognition** finds repetitions and rules (Row $N = N$ stars) and is the foundation of loops.
- **Abstraction** hides noise and keeps only essential steps; situational details become variables.
- Sorting arranges data (ascending or descending); searching locates an element's position or existence.
- **Bubble Sort** compares and swaps adjacent elements; the largest value moves to the end each pass; $N - 1$ passes; stable; $O(N^2)$.
- **Selection Sort** brings the smallest remaining value to the front each pass; inner loop starts at outer + 1; fewer swaps; not stable; $O(N^2)$.
- **Linear Search** checks items one by one; works on unsorted data; $O(N)$ worst case.
- **Binary Search** halves the search range using BEG, MID and END; needs sorted data; $O(\log N)$ — about 20 steps for 1 million items.
- Algorithms are evaluated on **correctness** (checked with trace tables and edge cases), **efficiency** (time and space) and **clarity**.
- Choose by asking: input size, sorted or not, speed or memory? Unsorted → Linear; sorted → Binary; needs order → sort.

Programming Fundamentals

★ 12 Golden · 24 cards · Fundamentals of Python Programming · Variables · Data Types in Python · Operators and Operands: Arithmetic and Assignment · Relational, Logical and Membership Operators · Bitwise Operators · Control Structures · Selection Statements: if and if-else · Selection Statements: if-elif-else and Nested if · Repetition Statements: the for Loop and range() · The while Loop, break and continue · Nested Loops and for vs while

3.1 Programming

Computers are "dumb" machines. They are very fast, but they cannot think or do anything on their own.

- A set of instructions given to a computer is called a program.
- The language used to write these instructions is called a programming language.
- Writing programs is called programming, and the person who writes them is a programmer.

```
print("Hello")
```

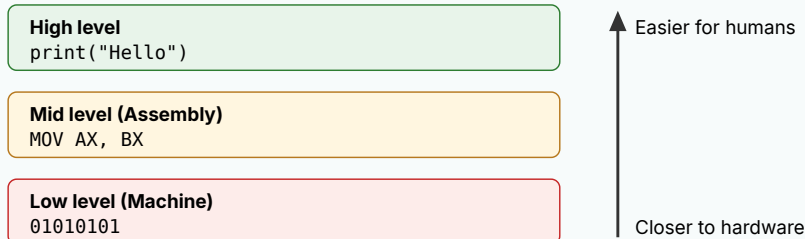
Remember: Program = recipe, programmer = recipe writer, computer = cook who follows every line exactly. If the recipe is wrong, the food is wrong. The cook does not fix...

کمپیوٹر خود سے کچھ نہیں سوچ سکتا۔

3.1.1 Programming Language Types

Programming languages are classified into three main types, depending on how close they are to the computer's own language (binary 0s and 1s) or to human language.

Type	Description	Examples
Low level	Closest to the native language of the computer (binary 0s and 1s). Very difficult for humans to read and write.	Machine language (binary)
Mid level	Bridges the gap. It can access hardware but is easier to read than binary because it uses short words called mnemonics (e.g. MOV, ADD).	Assembly language
High level	Designed to be easy for humans to read and write. Uses English-like words.	C++, Java, Python



The three levels of programming languages, with a sample line from each.

Remember: "Low = Like the machine, High = Human." Mid level sits in the Middle and uses Mnemonics (both start with M).

پروگرامنگ زبانوں کی تین اقسام ہیں: لو لیول (Low Level) جو بائری 0 اور 1 میں ہوتی ہے، مڈ لیول (Mid Level) یعنی اسمبلی جو مختصر الف...

3.2 Career Growth with Python

Python is one of the most in-demand languages in the world.

- Software Development: build desktop and web applications.
- Data Science and Data Analysis: analyse data, create reports and make predictions.
- Artificial Intelligence and Machine Learning: develop intelligent systems, chatbots and recommendation algorithms (like "videos you may...")
- Web Development: use frameworks such as Django and Flask to build websites.

```
# Python
print("Hello")
```

Remember: Six careers — remember "S-D-A-W-A-C": Software, Data, AI, Web, Automation, Cybersecurity.

پائٹھن سیکھ کر آپ سافٹ ویئر ڈیولپمنٹ، ڈیٹا سائنس، مصنوعی ذہانت (AI)، ویب ڈیولپمنٹ، آٹومیشن اور سائبر سیکیورٹی جیسے شعبوں میں کیریئر بنا سکتے ہیں۔

3.3 Integrated Development Environment (IDE)

An IDE (Integrated Development Environment) is a software tool that helps programmers write, run and manage code in one place.

- Code Editor: where you write and edit code.
- Syntax Highlighting: colours different parts of the code (keywords, strings, numbers) to make it easier to read.
- Auto-completion: suggests code while you type (type `print` and it offers `print`).
- Run/Execute Tool: runs the program with one click.

Remember: IDE features — "E-S-A-R-D": Editor, Syntax highlighting, Auto-completion, Run, Debugger.

آئی ڈی ای (IDE) ایک ایسا سافٹ ویئر ہے جس میں کوڈ لکھنا، چلانا اور غلطیاں درست کرنا ایک ہی جگہ ممکن ہوتا ہے۔

3.3.1 Visual Studio Code (VS Code)

VS Code is a free, lightweight and powerful code editor made by Microsoft and widely used for Python.

Install Python → Install VS Code → Extensions: install "Python" (Microsoft) → Explorer: new file `hello.py` → Type code → Click ► Run

- By default it is not a full IDE, but it becomes one when you install extensions (add-ons), such as the Python extension by Microsoft.

Element	Description
1. Top Menu Bar	Control over files, editing, navigation and workspace (File, Edit, Selection, View, Go, Run).
2. Side Bar	Contains the core tools: Explorer, Search, Source Control, Run & Debug and Extensions.
3. Explorer	Shows project files and folders. Helps you create, open and manage your Python files.
4. Search	Finds and replaces text in the open file or across the whole project.
5. Source Control	Connects with Git (a version control system) to track changes, undo mistakes and collaborate.
6. Run & Debug	Runs code, finds errors, pauses at specific lines and lets you inspect variables.
7. Extensions	Like an App Store for VS Code. Add features such as the Python extension, Prettier or Python Debugger.



A simplified VS Code window: menu bar, side bar, Explorer and the code editor.

```
print("Hello, World!")
```

Remember: Extensions are the "App Store" of VS Code: just as apps give your phone new abilities, extensions give VS Code new abilities.

وی ایس کوڈ (VS Code) ایک مفت اور ہلکا پھلکا کوڈ ایڈیٹر ہے۔

3.4 Fundamentals of Python Programming ★ Golden

The basic structure of a Python program is simple and clear.

- A program takes data, processes it using statements, and finally shows output.
- Python runs the statements top to bottom, one after another (sequentially).
- A statement is one complete instruction, usually written on one line, such as `print("Hi")` or `x = 5`.

```
# This is a comment. It explains the code below.
print("Hello, World!")
if 5 > 2:
    # The space before 'print' is the indentation
    print("Five is greater than two!")
```

Remember: Think of indentation like the points and sub-points in your notebook. Sub-points are written a little to the right to show they "belong" to the main point. A...

Exam: Common questions: "What is the purpose of comments?", "Why is indentation important in Python?", "How is Python different from C++/Java in defining blocks?" Write:...

پائنتھن پروگرام اوپر سے نیچے ترتیب سے چلتا ہے۔

3.4.1 Variables ★ Golden

A variable is a named space in computer memory used to store data.

- Can contain letters, numbers and underscores (_) only — no spaces or symbols like -, @, \$.
- Must start with a letter or underscore, NOT a number (1name is invalid).
- Are case-sensitive: age and Age are two completely different variables.
- Cannot be a Python keyword (a reserved word such as if, for, while, else, True).

```
name = "Ali"
age = 16
my_height = 5.7
score1 = 100
```

Remember: Choose meaningful names: total_marks is far clearer than tm or x. Use lowercase words joined by underscores (this style is called snake_case).

Exam: A favourite question: "Which of these variable names are valid? Give reasons." Check each name against all four rules and state the rule broken, e.g. "2nd_place —...

ویری ایبل (Variable) میموری میں ایک نام والی جگہ ہے جس میں ڈیٹا رکھا جاتا ہے، بالکل ایک لیبل لگے ڈیے کی طرح۔

3.4.1 Data Types in Python ★ Golden

A data type specifies the kind of data a variable holds — a whole number, a decimal, text, True/False, or a collection of items.

- The type decides what you can do with the data (you can divide numbers, but not names).
- Python has several built-in data types:

Name	Type	Description	Example
Integer	int	Whole numbers without decimals (positive, negative or zero).	age = 16
Floating point	float	Numbers with a decimal point.	height = 5.7
String	str	Text data, enclosed in quotes (" " or ' ').	name = "Sammy" or year = "2026"
Boolean	bool	Logical value: True or False (capital T / F).	is_student = True
List	list	Ordered, changeable collection in square brackets [].	marks = [85, 90, 75]
Tuple	tuple	Ordered, unchangeable sequence in parentheses ().	coords = (10, 20)
Dictionary	dict	Key : value pairs in curly braces { }.	student = {"name": "Ahmed", "age": 16}
Set	set	Unordered collection of unique items (no duplicates).	vowels = {'a', 'e', 'i'}

Remember: Brackets decide the collection: [] List (changeable), () Tuple (locked), { } with key:value = Dictionary, { } with single items = Set (unique). Think: a list...

Exam: Expect: "Name any four data types with examples" or "Differentiate list and tuple". For a difference question, use a two-column table: list — [], changeable, e.g. marks...

ڈیٹا ٹائپ (Data Type) بتاتی ہے کہ ویری ایبل میں کس قسم کا ڈیٹا ہے۔

3.4.2 Input and Output Handling

Output — the print() function. A function is a named, ready-made piece of code that does a job when you call it with brackets.

- Items separated by commas are printed with one space between them.
- An f-string (a string with f before the quote) lets you put variables directly inside { }. It is the fastest and most readable way to...

Remember: "input() always gives a string." If you plan to do maths, wrap it: int(input(...)) for whole numbers, float(input(...)) for decimals. Read it from the inside...

اسکرین پر معلومات دکھاتا ہے اور input() صارف سے معلومات لیتا ہے۔

3.4.3 Operators and Operands: Arithmetic and Assignment ★ Golden

An operand is the value or variable being worked on.

- An operator is the symbol that performs the action.
- Together they form an expression, which Python evaluates to a single value.
- In $10 + 20$: 10 is an operand, + is the operator, 20 is an operand, and the result is 30.

Operator	Purpose	Example	Result
+	Addition	$5 + 3$	8
-	Subtraction	$10 - 4$	6
*	Multiplication	$4 * 2$	8
/	Division (result is always a float)	$9 / 2$	4.5
//	Floor division (drops the decimal part)	$9 // 2$	4
%	Modulus (remainder)	$9 \% 2$	1
**	Exponentiation (power)	$2 ** 3$	8

```
print(5 + 3)
print(10 - 4)
print(4 * 2)
print(9 / 2)
print(9 // 2)
print(9 % 2)
print(2 ** 3)
print(8 / 2)
```

Remember: Three kinds of division: / = exact answer (float), // = quotient only ("floor"), % = remainder only. A number is even when $n \% 2 == 0$.

Exam: You will be asked to evaluate expressions such as $17 // 5$, $17 \% 5$, $2 ** 4$ or to trace compound assignment. Show each step and write the final value with its type (e.g....

آپریٹنڈ (Operand) وہ ویلیو ہے جس پر عمل ہوتا ہے اور آپریٹر (Operator) وہ علامت ہے جو عمل کرتی ہے۔

3.4.3 Relational, Logical and Membership Operators ★ Golden

3. Relational (comparison) operators compare two values and return a Boolean (True or False).

- Relational (comparison) operators compare two values and return a Boolean (True or False).
- They are used in conditions.

Op	Description	Example (a = 10, b = 5)	Result
=	Equal to	$a = b$	False
≠	Not equal to	$a \neq b$	True
>	Greater than	$a > b$	True
<	Less than	$a < b$	False
≥	Greater than or equal to	$a \geq 10$	True
≤	Less than or equal to	$a \leq b$	False

```
a = 10
b = 5
print(a == b)
print(a != b)
print(a > b)
print(a < b)
print(a ≥ 10)
print(a ≤ b)
```

Remember: AND is strict, OR is generous. AND is like a teacher who needs both homework and attendance; OR is like a shop that accepts cash or card. Order of evaluation:...

Exam: Typical: "Evaluate $a > b$ and $b > 3$ for $a = 10$, $b = 5$ " or "Write the truth table of and/or". Write each sub-condition's value first, then combine. Remember: == compares,...

ریشنل آپریٹرز (Relational Operators) دو ویلیوز کا موازنہ کر کے True یا False دیتے ہیں۔

3.4.3 Bitwise Operators ★ Golden

6. Bitwise operators treat numbers as binary digits (bits: 0s and 1s) and work on them bit by bit, column by column.

- AND (&) — gives 1 only if both bits are 1, otherwise 0.
- OR (|) — gives 1 if at least one bit is 1.
- XOR (^) — gives 1 if the bits are different, 0 if they are the same.
- NOT (~) — inverts all bits (0 → 1, 1 → 0). In Python the result is the number made negative, minus 1: $\sim x = -(x + 1)$. So ~ 10 gives -11.

Bit A	Bit B	A & B	A B	A ^ B	~A
0	0	0	0	0	1
0	1	0	1	1	1
1	0	0	1	1	0
1	1	1	1	0	0

```
a = 10
b = 6
print(bin(a), bin(b))
print(a & b)
print(a | b)
print(a ^ b)
print(~a)
```

Remember: Shortcuts: $x > n = x // 2^n$, $\sim x = -x - 1$. For AND/OR/XOR always write both numbers in binary, one under the other, with the same number of bits.

Exam: Exams ask you to "evaluate $12 \& 10$, $12 | 10$, $12 \wedge 10$, ~ 5 , $7 >> 1$ showing binary working". Marks are given for the binary conversion and column working, not just the...

بٹ وائز آپریٹرز (Bitwise Operators) اعداد کو بائنری میں بدل کر ہر بٹ پر الگ الگ کام کرتے ہیں۔

3.5 Control Structures ★ Golden

Control structures decide the order in which statements are executed in a program (the "flow of control").

- Sequence — do the steps one after another, top to bottom.
- Selection — choose a path depending on a condition (if, else).
- Repetition — repeat steps (loops: for, while).

Remember: Remember "S-S-R": Sequence, Selection, Repetition — "Step, Select, Repeat".

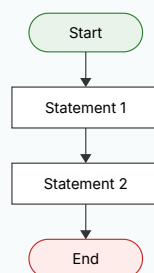
Exam: "Name and explain the three basic control structures" is a very common short question. Give a one-line definition and a two-line Python example for each.

کنٹرول اسٹرکچرز (Control Structures) طے کرتے ہیں کہ پروگرام کی ہدایات کس ترتیب سے چلیں گی۔

3.5.1 Sequence

Sequence means statements are executed one after another, strictly from top to bottom, in the exact order they are written.

- This is the default behaviour of Python.
- No statement is skipped or repeated.



Sequence: control flows straight down from one statement to the next.

```
print("Enter two numbers")
a = int(input("Enter first number: ")) # Step 1: Input
b = int(input("Enter second number: ")) # Step 2: Input
total = a + b # Step 3: Process
print("Sum =", total) # Step 4: Output
```

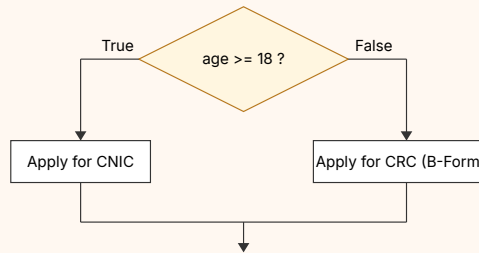
Remember: Sequence is like reading a page: line 1, then line 2, then line 3. Most programs follow the pattern Input → Process → Output.

سیکوننس (Sequence) کا مطلب ہے کہ ہدایات بالکل اسی ترتیب سے ایک کے بعد ایک اوپر سے نیچے چلتی ہیں جس ترتیب سے لکھی گئی ہیں۔

3.5.2 Selection Statements: if and if-else ★ Golden

Selection changes the flow of a program based on a condition — an expression that is either True or False.

- It helps a program respond intelligently to different situations.
- The if statement executes a block of code only if the condition is True.
- If the condition is False, the block is completely skipped.



An if-else statement: one diamond (condition), two paths, then the paths join again.

```
if condition:
    statement(s)    # runs only when condition is True
```

Remember: Formatting rule: put a colon (:) after if and else, and indent the code inside the blocks. else never has a condition of its own.

Exam: Asked as: "Write a program that inputs a number and prints whether it is even or odd / positive or negative / pass or fail." Write the full program with `int(input()),...`

سلیکشن (Selection) میں پروگرام شرط کی بنیاد پر فیصلہ کرتا ہے۔

3.5.2 Selection Statements: if-elif-else and Nested if ★ Golden

3. The if-elif-else statement handles many conditions.

- The if-elif-else statement handles many conditions. (elif is short for "else if".) Python checks the conditions from top to bottom.
- As soon as it finds a True condition, it runs that block and ignores all the rest.
- If none is True, the else block runs.

marks	>= 80?	>= 70?	>= 60?	Output
92	True	(not checked)	(not checked)	A-1 Grade
75	False	True	(not checked)	A Grade
60	False	False	True	B Grade
41	False	False	False	Fail or below B

Remember: In an if-elif-else ladder, put the strictest condition first (highest marks first). Think of a ladder: you step off at the first rung that fits.

Exam: Grade-calculator programs (if-elif-else) and "largest of three numbers" are very common long questions. Check conditions in the right order, indent nested blocks...

if-elif-else میں کئی شرطیں اوپر سے نیچے ترتیب سے دیکھی جاتی ہیں، اور جو شرط سب سے پہلے درست ہو اسی کا بلاک چلتا ہے، باقی سب چھوڑ دی جاتی ہیں۔

3.5.3 Repetition Statements: the for Loop and range() ★ Golden

A repetition statement (loop) lets a block of code run again and again.

- This greatly reduces the need to write the same code many times.
- Each single run of the loop body is called an iteration.
- The for loop is used when the exact number of iterations is known.

Syntax	How it works	Example	Values produced
<code>range(stop)</code>	From 0 up to, but not including , stop.	<code>range(3)</code>	0 1 2
<code>range(start, stop)</code>	From start up to, but not including, stop.	<code>range(2, 5)</code>	2 3 4
<code>range(start, stop, step)</code>	From start towards stop, jumping by step.	<code>range(1, 6, 2)</code>	1 3 5

```
for i in range(3):
    print(i)
```

Remember: range stops one short: `range(1, 11)` gives 1 to 10. To include a number `n`, write `range(..., n + 1)`. Number of values in `range(a, b)` = `b - a`.

Exam: Expect "predict the output" questions on `range()` with a step, and programs such as "print a table", "sum of first `n` numbers", "even numbers from 1 to 20". Always state...

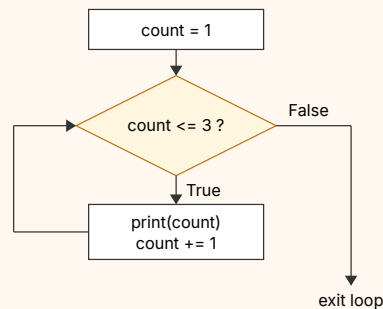
لوپ (Loop) کوڈ کے ایک بلاک کو بار بار چلاتا ہے۔

3.5.3 The while Loop, break and continue ★ Golden

2. The while loop repeats a block of code as long as a condition remains True.

- break — exits (stops) the loop entirely and immediately.
- continue — skips the rest of the current iteration and jumps to the next one.

count	count <= 3?	Printed	count after update
1	True	1	2
2	True	2	3
3	True	3	4
4	False	—	loop ends



A while loop: test the condition, run the body, go back and test again.

```

count = 1
while count <= 3:
    print(count)
    count += 1 # VERY IMPORTANT: update the condition!
  
```

Remember: break = "exit the room"; continue = "skip this turn". For every while loop, check the I-T-U rule: Initialise, Test, Update.

Exam: "Differentiate break and continue with examples" and "What is an infinite loop?" are frequent. Use the two side-by-side programs above with their outputs — examiners...

while لوپ اس وقت تک چلتا رہتا ہے جب تک شرط درست (True) رہے، اور یہ تب استعمال ہوتا ہے جب پہلے سے معلوم نہ ہو کہ کتنی بار دہرانا ہے۔

3.5.3 Nested Loops and for vs while ★ Golden

A nested loop is a loop inside another loop.

- The inner loop runs completely, from start to finish, for every single iteration of the outer loop.
- If the outer loop runs 2 times and the inner loop runs 3 times, the inner body runs $2 \times 3 = 6$ times.

for loop	while loop
Used when the number of iterations is known .	Used when the number of iterations is unknown (condition-based).
Automatically updates the loop variable.	Requires a manual update of the counter (risk of an infinite loop).
Cleaner and shorter for counting tasks.	Better for dynamic situations (like "keep playing until the user says Stop").
<code>for i in range(1, 4): print(i)</code>	<code>i = 1 / while i <= 3: / print(i) / i += 1</code>

```

for outer in range(1, 3):          # runs 2 times (1, 2)
    for inner in range(1, 4):      # runs 3 times per outer (1, 2, 3)
        print(f"Outer: {outer}, Inner: {inner}")
  
```

Remember: Ask yourself: "Do I know how many times?" Yes → for. "Do I only know when to stop?" → while. Nested loops are like a clock: the minute hand (inner loop) goes...

Exam: "Differentiate between for and while loops" (give three points and an example of each) and "How many times will the inner statement run?" (multiply the counts) are...

نیسٹڈ لوپ (Nested Loop) میں ایک لوپ دوسرے لوپ کے اندر ہوتا ہے، اور باہر والے لوپ کے ہر چکر میں اندر والا لوپ شروع سے آخر تک مکمل چلتا ہے۔

Practical Activity: Shopping Bill Calculator

This program combines all three control structures: sequence (steps in order), selection (checking "done", checking the price, checking for a discount) and repetition (a while loop that keeps adding items).

- It uses a flag — a Boolean variable (shopping) that controls when the loop stops.
- `item.lower()` converts the text to lowercase so that "done", "Done" and "DONE" all work.

Step	item	price	total	count
1	Rice	3000.0	3000.0	1
2	Soap	-50.0 (invalid, not added)	3000.0	1
3	Oil	2500.0	5500.0	2
4	DONE → shopping = False	—	5500.0	2

Remember: Label the parts of any program with comments, as here: # REPETITION, # SELECTION. It helps you, your teacher and the examiner see the structure.

یہ شاپنگ بل پروگرام تینوں کنٹرول اسٹرکچرز کو ملا کر بنایا گیا ہے: while لوپ سے اشیاء بار بار شامل ہوتی ہیں، if-else سے غلط قیمت رو...

3.6 Libraries in Python

A library (also called a module) is a collection of pre-written code that offers ready-made functions.

- Instead of writing a square-root algorithm from scratch, you import math and call `math.sqrt()`.
- Libraries save time, reduce bugs and let you stand on other programmers' work.
- There are five common ways to import a library.

Method	Syntax	How you call the function
1. Entire module	<code>import math</code>	<code>math.sqrt(25)</code>
2. Alias (nickname)	<code>import math as m</code>	<code>m.sqrt(16)</code>
3. Specific function	<code>from math import sqrt</code>	<code>sqrt(36)</code>
4. Several functions	<code>from math import sqrt, pi</code>	<code>sqrt(9)</code> , <code>pi</code>
5. Everything (*)	<code>from math import *</code>	<code>sqrt(64)</code> — avoid this

Trap: `from math import *` dumps every name into your program.

لائبریری پہلے سے لکھا ہوا کوڈ ہے جسے `import` سے پروگرام میں لایا جاتا ہے۔

3.6.1 Built-in Libraries

Built-in libraries come pre-installed with Python.

- You only import them; you do not run pip.
- Three that every Class XI student must know are math, random and datetime.

```
import math
print(math.pi)
print(math.ceil(4.3))
print(math.floor(4.9))
print(math.pow(2, 4))
print(math.sqrt(81))
```

Remember: ceil always rounds up (ceiling); floor always rounds down. `round()` (built-in, not in math) goes to the nearest integer and is a different function.

math حساب کے لیے، random بے ترتیب عدد کے لیے اور datetime تاریخ وقت کے لیے بنی ہوئی لائبریریاں ہیں۔

3.6.2 Third-Party Libraries

Third-party libraries are written by the community.

- They are not bundled with Python, so you install them from a terminal with `pip install library_name` and then import them as usual.

Library	Primary use
NumPy	Heavy numerical computing and large arrays
Pandas	Data analysis; table-like data (CSV / Excel)
Matplotlib	Data visualisation — graphs and charts
TensorFlow / PyTorch	Artificial intelligence and machine learning

```
pip install pandas
import pandas as pd
```

Trap: pip is a terminal command, not a Python statement.

تھرڈ پارٹی لائبریریاں پائتھن کے ساتھ نہیں آتیں، انہیں `pip install` سے انسٹال کیا جاتا ہے۔

3.7 Debugging Code in Python

Debugging is the process of identifying, tracing and fixing bugs so a program works correctly.

- Every program you write will have bugs.
- The skill is not "never make a mistake"; it is "find the mistake quickly".
- Python errors fall into three families:

Type	When it appears	What you notice	Example
Syntax error	Before the program even starts	Python refuses to run	<code>print("Hello" — missing)</code>
Runtime error (exception)	While the program is running	It crashes with a traceback	<code>x = 10 / 0 → ZeroDivisionError</code>
Logical error	Program finishes "successfully"	The output is wrong	<code>area = length + width</code> instead of <code>*</code>

```
x = int("abc")
```

Remember: Fix syntax errors first (the program cannot start). Then run it. Then check the output against a hand-worked example to catch logical errors.

Exam: Examiners love "identify the error type" questions. Ask: (1) Does it refuse to start? → syntax. (2) Does it crash mid-run? → runtime. (3) Does it run but give a wrong...

ڈیبنگ یعنی غلطی ڈھونڈ کر ٹھیک کرنا۔

Key terms

Term	Meaning	اردو
Program	A set of instructions a computer can follow.	پروگرام
High-level language	A language close to English (Python, Java, C++).	اعلیٰ سطحی زبان
Interpreter	Runs a program line by line without a separate compile step.	انٹرپریٹر
IDE	Software that combines editor, terminal and debugger.	انٹیگریٹڈ ڈویلپمنٹ انوائرنمنٹ
Variable	A named box that holds a value.	متغیر
Data type	The kind of value: int, float, str, bool, list, dict.	ڈیٹا کی قسم
Operator	A symbol that performs an operation on operands.	عامل
Operand	A value an operator works on.	معمول
Sequence	Instructions executed from top to bottom.	تسلسل
Selection	Choosing a path with <code>if / else</code> .	انتخاب
Repetition / loop	Repeating a block with <code>for</code> or <code>while</code> .	تکرار / لوپ
Nested	A structure placed inside another of the same kind.	تو در تو
Library / module	A collection of ready-made functions.	لائبریری / ماڈیول
pip	The tool that installs third-party Python libraries.	پپ
Syntax error	A grammar mistake; the program will not start.	نحوی غلطی
Runtime error	A crash while the program is running (exception).	رن ٹائم غلطی
Logical error	The program runs but the result is wrong.	منطقی غلطی
Traceback	Python's report of where a crash happened.	ٹریس بیک
Flag	A Boolean variable that controls a loop or decision.	فلیگ
Integer division	<code>//</code> division that drops the fraction.	صحیح تقسیم

Chapter in one look

- A program is a set of instructions; programming languages are low-level (close to hardware) or high-level (close to English). Python is a high-level, interpreted, readable language.
- Python is widely used in web development, data science, AI and automation — skills that grow a career in Pakistan and abroad.
- An IDE (such as VS Code plus the Python extension) combines an editor, a terminal, a debugger and syntax highlighting.
- Variables store values; Python types are dynamic. Core types: `int`, `float`, `str`, `bool`, `list`, `dict`.
- `print()` displays output; `input()` always returns a **string** — convert with `int()` or `float()` before arithmetic.
- Operators: arithmetic (`+` `-` `*` `/` `//` `%` `**`), assignment (`=` `+=` `...`), relational, logical (`and` `or` `not`), membership (`in`), bitwise (`&` `|` `^` `~` `<<` `>>`).
- Control structures: sequence (top to bottom), selection (`if` / `if-else` / `if-elif-else` / nested), repetition (`for` / `while`).
- `range(stop)` goes 0 ... stop-1; `range(start, stop, step)` jumps by step. `break` leaves a loop; `continue` skips one iteration.
- Use `for` when the count is known; use `while` when you wait for a condition (and always update that condition).
- Libraries extend Python. Built-in: `math`, `random`, `datetime`. Third-party (need `pip`): NumPy, Pandas, Matplotlib, TensorFlow.
- Errors: syntax (grammar, will not run), runtime (crash, traceback), logical (runs, wrong answer).

CHAPTER 4

Data and Analysis

★ 4 Golden · 15 cards · Keys and Integrity Constraints · Entity Relationship Model (ER-Model) · Referential Integrity · Creating Queries in MS Access

4.1.1 Data, Information and Database

Data are unprocessed facts, figures or observations.

Data → Processing → Information → Decision

- They can be quantitative (numbers: 80, 75, 90) or qualitative (descriptions: the colours of cars in a parking lot).
- Data on its own is not very useful — a list of marks does not yet tell you whether the class is strong.
- When you organise, process and interpret data, it becomes information.

Aspect	Data	Information
Definition	Raw facts and figures	Processed, meaningful data
Example	85, 90, 75	Average marks = 83.3%
Usefulness	Little, on its own	Useful for decisions

Remember: Exam line: Data is raw; information is processed data used for decision-making. Always add one pair of examples.

ڈیٹا خام حقائق ہیں، معلومات وہی حقائق ہیں جب انہیں ترتیب دے کر معنی دے دیا جائے۔

4.1.2 Database Management System (DBMS)

A DBMS is software that lets you create, manage and interact with databases.

- Less redundancy. The department name is stored once, not in every student row.
- Consistency. Change a phone number once; every screen that uses it shows the new number.
- Security. Only authorised users see sensitive fields (an admin sees marks; office staff see names).
- Integrity. Rules stop duplicate roll numbers and empty primary keys.

ڈی بی ایم ایس وہ سافٹ ویئر ہے جو صارف اور ڈیٹا بیس کے درمیان پل کا کام کرتا ہے۔

4.2 Database Components (Tables, Records, Fields)

The basic structure is a table (also called a relation): rows and columns, like a spreadsheet, holding facts about one kind of thing (Students, Teachers, Books).

- A record (row, tuple) is one complete instance — everything about Ali.
- A field (column, attribute) is one kind of fact — the whole Name column, or the whole Grade column.

Roll No	Name	Course	Grade
101	Ali	CS	A
102	Sara	IT	B+
103	Ahmed	CS	A

Remember: Memory: row = record = one person. column = field = one question on the admission form.

ٹیبل قطاروں اور کالموں میں ڈیٹا رکھتا ہے۔

4.3.1 Keys and Integrity Constraints ★ Golden

A key is a field (or a set of fields) that uniquely identifies records and links tables.

- Integrity constraints are the rules that keep those keys honest.
- Any field (or combination) that could uniquely identify a row.
- A table may have several.

Table	Field	Role
Student	RollNo	Primary key
Student	BFormNo	Alternate key
Result	RollNo	Foreign key → Student.RollNo
Result	ExamID + RollNo	Could be a composite primary key

Trap: A phone number is a bad PK: people change numbers, two siblings may share one, and some students have none (NULL).

Exam: Board questions: "Differentiate primary and foreign key" (table of 4 points) and "Why can a PK not be NULL?" Because a NULL would mean "this row has no identity", so the...

کنڈیڈٹ کی وہ فیلڈ ہے جو ریکارڈ کی شناخت کر سکے۔

4.3 Relational Database Management System (RDBMS)

An RDBMS stores data in several tables and uses primary and foreign keys to keep relationships.

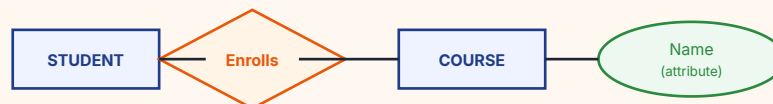
- Reduces redundancy by splitting data into related tables.
- Supports many users at once, with permissions.
- Is queried with SQL (Structured Query Language) — a standard language of SELECT, INSERT, UPDATE, DELETE.

آر ڈی بی ایم ایس کئی ٹیبلز میں ڈیٹا رکھتا ہے اور کیڑ سے ان کے تعلقات برقرار رکھتا ہے۔

4.4 Entity Relationship Model (ER-Model) ★ Golden

The ER model is the architect's blueprint: you draw what will be stored and how it connects before you create tables in Access.

Component	Meaning	Symbol	Example
Entity	A real-world object we store data about	Rectangle	STUDENT, TEACHER, COURSE
Attribute	A property of an entity. PK is underlined	Oval	StudentID, Name, Class
Relationship	How two entities are connected	Diamond	Enrolls, Teaches, Has



Rectangle = entity, diamond = relationship, oval = attribute. Underline the primary-key attribute.

Trap: The M:N rule.

Exam: Always state cardinality on the diagram (1 and M at the ends of the line). For M:N, the long question is "how do you resolve it?" — draw the junction entity with two...

ای آر ماڈل ڈیٹا بیس کا خاکہ ہے۔

4.5 Referential Integrity ★ Golden

Referential integrity means every foreign-key value in a child table must match an existing primary-key value in the parent — or be NULL if the design allows it.

- If Result.StudentID is a FK to Student.StudentID, you cannot type Result for student 999 if 999 is not in Student.
- You also cannot delete student 101 while results for 101 still exist — unless you first handle those child rows.

ID	Name
101	Ali
102	Sara

Trap: Cascade Delete is powerful and dangerous.

Exam: Write: definition, why it matters (no orphans), then one numbered example of Cascade Update and one of Cascade Delete. Mention parent vs child.

ریفرنشل انٹیگریٹی کا مطلب ہے بچہ ٹیبل کی فارن کی ہمیشہ والدین کی موجود پرائمری کی طرف اشارہ کرے۔

4.6 Relational Schema Development

A relational schema is the formal written design of the database: table names, fields, primary keys, foreign keys and how tables relate.

- You write it before clicking around in Access.
- It defines (1) how data is split into tables, (2) which columns each table has, (3) which fields are PK/FK, (4) the relationships.
- Benefits: less redundancy, integrity planned in advance, easier to grow later.

```
Department (DeptID PK, DeptName)
Student (RollNo PK, Name, Class, DeptID FK → Department)
Course (CourseID PK, Title, Credit, DeptID FK → Department)
```

Remember: Skipping the schema is like building a house without a drawing. You get messy, broken structures that are painful to repair.

ریلیشنل اسکیم ڈیٹا بیس کا لکھا ہوا نقشہ ہے: کون سے ٹیبل، کون سے فیلڈ، کون سی کیز، کون سے تعلقات۔

4.7 Case Study — Library Management System

A college library needs a database. Walk the seven steps examiners expect.

- A college library needs a database.
- Walk the seven steps examiners expect.
- A book has ID, title, author, publisher.
- A member (student/teacher) has ID, name, contact.

BorrowID	MemberID	BookID	IssueDate	ReturnDate
5001	102	B-17	2026-10-05	



Final library ER: two 1:M relationships through the BorrowRecord junction table.

```
Member (MemberID PK, Name, ContactInfo)
Book (BookID PK, Title, Author, Publisher)
BorrowRecord (BorrowID PK, MemberID FK, BookID FK, IssueDate, ReturnDate)
```

لائبریری میں ممبر اور کتاب کا تعلق بہت-بہت-سے ہے، اس لیے بورو ریکارڈ نامی پل ٹیبل بناتے ہیں جس میں دونوں کی فارن کیز اور اجراء...

4.8 Database Objects (Tables, Forms, Queries, Reports)

Four objects you will meet in Access. Learn the one-line jobs cold:

- Learn the one-line jobs cold:

Object	Job	You ...
Table	Stores the data (foundation; every other object depends on it)	keep the actual records here
Form	Friendly screen for enter / edit / view (text boxes, drop-downs)	type data in; fewer mistakes than a raw grid
Query	Ask a question: filter, sort, calculate	e.g. "students above 80%"
Report	Formatted output for printing or sharing	read only — never type into a report

Remember: Store → Enter → Search → Print. Tables, Forms, Queries, Reports.

ٹیبل ڈیٹا رکھتی ہے، فارم سے اندراج ہوتا ہے، کوئری سوال پوچھ کر چھانتی ہے، رپورٹ چھپائی کے لیے سنواری ہوئی پیشکش ہے — اس میں ڈیٹا بھرا نہیں جاتا۔

4.9 Creation of Tables in Microsoft Access

Plan first (4.9.1): purpose, field list, data type of each field, which field is the PK.

- Create tab → Table.
- A blank grid appears.
- Double-click Field1, Field2, ... to rename.
- Type data; Access guesses the data type.

Data type	Use it for
Short Text	Names, addresses, phone numbers (you do not add phone numbers). Up to 255 characters.
Number	Values you calculate with: Age, Marks, Quantity.
Date/Time	Date of birth, admission date, appointment time.
Currency	Fees, salary.
AutoNumber	Access fills a unique sequential number — excellent PK.

Trap: Phone numbers look numeric but are Short Text.

پہلے ٹیبل کا منصوبہ بنائیں۔

4.10 Designing Forms for Data Manipulation

A form is the screen people actually type into.

- Create tab → Form Wizard.
- Select the table (e.g. Student).
- Move fields with > (one) or >> (all).
- Choose layout. Columnar = one record at a time, best for data entry.

فارم اندراج کی سادہ اسکرین ہے۔

4.11 Creating Queries in MS Access ★ Golden

A query retrieves, filters and analyses data without changing the original tables.

- Create tab → Query Design.
- Show Table → add Student → Close.
- Double-click the fields you want; they drop into the grid.
- Red Run (!) button executes it.

Goal	Where to type	Example
Single criterion	Criteria row, one column	Department → "Computer Science"
AND (both must match)	Same Criteria row, two columns	Name "Ali" and Department "Computer Science"
OR (either may match)	Criteria row and the or: row	Department "IT" or "Computer Science"

Trap: Text criteria need quotes: "IT".

Exam: Remember: AND = same row in the grid; OR = different rows. Examiners draw a blank grid and ask you to fill Criteria and or:.

کوئری ٹیبل کو بدلے بغیر سوال پوچھتی ہے۔

4.11.4 Data Summarization and Statistical Analysis

Grouping combines rows that share a value (all CS students together) and then an aggregate calculates a number for the group.

- In Access click Totals (Σ) to reveal the Total row in the grid.

Function	What it calculates
COUNT()	How many records (students in IT)
SUM()	Total of a number field (fees collected)
AVG()	Mean
MAX() / MIN()	Highest / lowest in the group

```
SELECT Department, AVG(Marks) AS AvgOfMarks
FROM Student
GROUP BY Department;
```

گروپ بائی ایک جیسے ریکارڈ جوڑتا ہے، پھر COUNT، SUM، AVG، MAX، MIN سے خلاصہ نکلتا ہے۔

4.11.6 Data Visualization in MS Access (Charts)

Charts turn numbers into a picture so a pattern jumps out.

- Pick the chart that matches the question:

Chart	Shape	Best for
Bar (horizontal)	Bars sideways	Categories with long names (English, Mathematics, ...)
Column (vertical)	Bars standing	Side-by-side comparison (2024 / 2025 / 2026 admissions)
Pie	Slices of a circle	Parts of one whole (percent of CS vs IT vs Eng). Few slices.
Line	Dots joined	A trend over time (visitors per month)



Column = compare categories. Pie = share of a total. Line = change over time.

Exam: "Which chart?" — line for time, pie for composition, bar/column for comparing categories. Mention one reason.

بار لمبے ناموں والی اقسام، کالم سادہ موازنہ، پائی حصوں کا تناسب، لائن وقت کے ساتھ رجحان۔

Key terms

Term	Meaning	اردو
Data	Raw facts and figures.	ڈیٹا / خام حقائق
Information	Processed data that supports a decision.	معلومات
Database	Organised collection of related data.	ڈیٹا بیس
DBMS	Software that creates and manages databases.	ڈیٹا بیس مینجمنٹ سسٹم
Table	Rows and columns about one entity.	ٹیبل
Record	One row — one instance.	ریکارڈ
Field	One column — one attribute.	فیلڈ
Primary key	Chosen unique, non-empty identifier.	پرائمری کی
Foreign key	Field that refers to another table's PK.	فارن کی
Candidate key	Any field(s) that could be the PK.	کینڈیڈیٹ کی
Alternate key	A candidate not chosen as PK.	آلٹرنیٹ کی
RDBMS	DBMS that stores related tables linked by keys.	ریلیشنل ڈی بی ایم ایس
SQL	Language to query relational databases.	ساختہ استفساری زبان
Entity	Object we store data about (ER rectangle).	ہستی
Attribute	Property of an entity (ER oval).	صفت
Relationship	Link between entities (ER diamond).	تعلق
Cardinality	1:1, 1:M or M:N.	عددییت
Junction table	Bridge table that splits an M:N into two 1:M.	جنکشن ٹیبل
Referential integrity	Every FK matches a live PK (no orphans).	حوالہ جاتی درستگی
Cascade Update	Parent PK change copied to child FKs.	کیسکیڈ اپ ڈیٹ
Cascade Delete	Deleting the parent deletes related children.	کیسکیڈ ڈیلیٹ
Schema	Written blueprint of tables, keys, relationships.	اسکیمہ / نقشہ
Form	Screen for entering data.	فارم
Query	Saved question that filters/calculates data.	کوئری
Report	Formatted printable output.	رپورٹ
AutoNumber	Access-generated unique integer, often the PK.	آٹو نمبر

Chapter in one look

- Data = raw facts; information = processed data used for decisions; a database stores related data in a structured, updatable way.
- A DBMS is the software layer that provides less redundancy, consistency, security and integrity.
- Table = rows (records) and columns (fields). One table, one entity.
- Candidate keys could identify a row; the primary key is the one you choose (unique, not NULL); leftover candidates are alternate keys; a foreign key points at another table's PK.
- An RDBMS keeps several related tables and is queried with SQL.
- ER symbols: rectangle entity, oval attribute, diamond relationship. Cardinality 1:1, 1:M, M:N. M:N needs a junction table.
- Referential integrity forbids orphan FKs. Cascade Update copies a changed PK into children; Cascade Delete removes children with the parent.
- Write a relational schema before building. Library case: Member and Book are M:N, resolved by BorrowRecord.
- Objects: tables store, forms enter, queries search, reports print.
- Access Design View sets field names, data types and the PK. Phone numbers are Short Text; money is Currency; auto IDs are AutoNumber.
- Forms need a table/query source; Wizard + columnar layout for data entry.
- Query criteria: one value; AND on the same row; OR on two rows. Totals Σ for Group By / Count / Sum / Avg / Max / Min.
- Bar/column compare categories, pie shows parts of a whole, line shows a trend over time.

CHAPTER 5

Application and Impacts of Computing

★ 4 Golden · 13 cards · Artificial Intelligence (AI) · Internet of Things (IoT) & Components · Information Sources (Primary, Secondary, Tertiary) · Assistive Technologies

5.1 Introduction to Computing Impacts

Computing means using computers and computer-based devices to store, process and share information.

- Schools and colleges — online admissions, e-books, online results of the Sindh boards.
- Hospitals — computerised patient records, digital X-rays, online appointment systems.
- Banks — ATMs, mobile banking apps, Raast instant payments, JazzCash and Easypaisa wallets.
- Farms — weather forecasts on mobile phones, soil sensors, tube-well control.

کمپیوٹنگ آج ہماری روزمرہ زندگی کا اہم حصہ بن چکی ہے۔

5.1.1 Artificial Intelligence (AI) ★ Golden

Artificial Intelligence (AI) is the ability of a computer system to perform tasks that normally need human thinking, such as recognising faces, understanding language and making decisions.

IoT (Eyes) — collects data → Analytics — sorts data → AI (Brain) — decides → Action

- Artificial — it is created by humans; it is not natural.
- Intelligence — it can perform tasks that require thinking.
- Face recognition on a mobile phone to unlock the screen.
- YouTube suggesting videos based on your interests.

Remember: Eyes – Sort – Brain – Do. IoT is the eyes, Analytics sorts, AI is the brain, and the actuator does the action. Think of a cricket fielder: his eyes see the...

Exam: Common questions: "Define AI and give two examples" (2–3 marks), and "Explain how AI, IoT and Data Analytics work together" (4–5 marks). To score full marks: (1) write...

مصنوعی ذہانت (AI) کو "مصنوعی" اس لیے کہتے ہیں کہ اسے انسانوں نے بنایا ہے، اور "ذہانت" اس لیے کہ یہ ایسے کام کر سکتی ہے جن کے لیے سوچنا پڑتا ہے۔

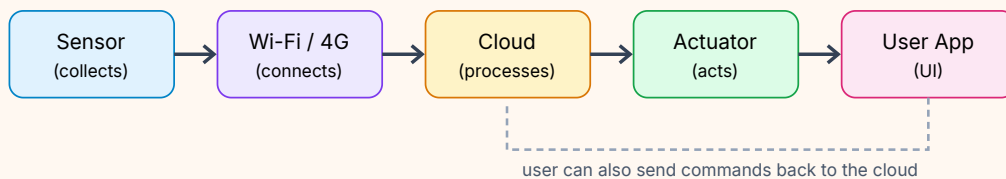
5.1.2 Internet of Things (IoT) & Components ★ Golden

The Internet of Things (IoT) is a system in which physical devices are connected to the Internet and can collect and share data to perform tasks automatically.

Sensor (Collects) → Wi-Fi (Connectivity) → Cloud (Processes) → Actuator (Acts) → User App (UI)

- The "things" are ordinary objects — watches, cameras, cars, fans, water pumps — that have been given sensors and an internet connection so...
- Source example: A smartwatch collects your heart-rate data and sends it to your mobile phone, where it is displayed.

#	Component	What it does	Examples
1	Sensors	Devices that collect information from the physical world. They are the "eyes and ears" of the system.	Temperature sensor, motion sensor, fingerprint sensor, smoke sensor, soil moisture sensor
2	Actuators	Devices that perform physical actions in response to system commands. They convert digital signals into physical movement.	Electric motor, switch, automatic door opener, water valve, fan controller
3	Connectivity	The connection that allows devices to send and receive data . Without it, sensors can't talk to the system.	Wi-Fi, Bluetooth, ZigBee, 4G/5G
4	Data Processing	Changing raw data into useful information. Steps: collecting, organising, analysing and deciding.	Checking if the room temperature is too hot to trigger the AC
5	Cloud Computing	Storing data and using services over the internet instead of local hard drives. Handles massive IoT data.	Google Drive, iCloud, online LMS
6	User Interface (UI)	The component that enables interaction between the user and the device . Helps users see information and give commands.	Mobile apps, touch screens, voice commands



The six IoT components working as one system (data processing happens inside the cloud).

Remember: Sensor senses, Actuator acts. Both words give you the answer: sensor = see/sense (input), actuator = act (output). Remember the six components with "Some Ants..."

Exam: "Define IoT and explain its components" is one of the most frequent long questions. Write the definition with the three key ideas — physical devices, connected to the...

انٹرنیٹ آف تھنگز (IoT) ایک ایسا نظام ہے جس میں مختلف آلات (جیسے گھڑیاں، کیمرے، گاڑیاں) انٹرنیٹ سے جڑے ہوتے ہیں اور خودکار طریقے سے...

5.1.3 Data Analytics & Components

Data Analytics is the process of collecting, organising, cleaning, studying and presenting data to find useful information.

1. Collection → 2. Storage → 3. Cleaning → 4. Analysis → 5. Visualization

- It helps people make better decisions.
- Data means raw facts and figures (for example, "72, 45, 88").
- Information is data that has been processed so it has meaning ("Ali scored 72%; he passed").

Roll No	Name	Marks (raw)
1	Ayesha	42
2	Bilal	18
3	Hina	35
3	Hina	35
4	Kashif	250
5	Sana	(missing)

Remember: "Cows Sit Calmly And Vote" — Collection, Storage, Cleaning, Analysis, Visualization. Or think of cooking daal: buy the lentils (collect), keep them in a jar...

ڈیٹا اینالٹکس ڈیٹا کو جمع کرنے، ترتیب دینے، صاف کرنے، اس کا مطالعہ کرنے اور پیش کرنے کا عمل ہے تاکہ مفید معلومات حاصل ہوں اور بہتر فیصلے کیے جا سکیں۔

Comparison: IoT vs AI vs Data Analytics

These three technologies are often used together, so students mix them up.

- The table below, from the source, compares them on four aspects.

Aspect	IoT	Artificial Intelligence (AI)	Data Analytics
Definition	Network of physical devices connected to the internet.	Technology enabling machines to think and decide.	Process of examining data to find patterns.
Main purpose	Data collection and automation.	Intelligent decision making and predictions.	Extracting meaningful insights.
Type of work	Works with physical devices (hardware).	Works with smart software systems.	Works with data (numbers, records).
Output	Raw data from the environment.	Smart decisions or actions.	Reports, graphs, insights.
Role in a smart system	Eyes and hands	Brain	Organiser / reporter
Pakistani example	Safe City cameras in Lahore/Karachi capturing traffic	Software recognising a car's number plate for an e-challan	Monthly report showing which roads have the most violations

Remember: Ask "What is the output?" Raw readings → IoT. Charts and reports → Data Analytics. A decision or prediction → AI.

IoT جسمانی آلات (ہارڈویئر) کا نیٹ ورک ہے جو ماحول سے خام ڈیٹا جمع کرتا ہے۔

5.2 Uses of IoT in Different Areas

IoT improves efficiency (doing more with less effort), saves time, and enhances the overall quality of life.

Soil Sensor → Cloud AI (Needs water?) → Water Pump

- It plays a significant role in developing modern infrastructure (the basic systems of a country — roads, power, water, hospitals).
- The source lists six areas.

Remember: Remember the six areas with "CHIEF H": Cities, Health, Industry, Education, Farming, Homes.

IoT کارکردگی بہتر بناتا ہے، وقت بچاتا ہے اور زندگی کا معیار بلند کرتا ہے۔

5.2.1 Scope of AI in Education in Pakistan

Artificial Intelligence is gradually becoming an important part of Pakistan's education system, improving teaching, learning and administration (managing the school: admissions, records, timetables).

- Learning Management Systems (LMS) — online platforms (e.g. Google Classroom, Moodle) where teachers post lessons and assignments; AI...
- Automated grading and digital testing — MCQ tests marked instantly by computer.
- AI-based career counselling — software that studies a student's marks and interests and suggests suitable fields.
- Helps students learn at their own speed.

Practical uses	Benefits	Challenges
LMS	Learn at own speed	Limited rural internet
Automated grading & digital testing	Identifies students needing help	Lack of computers/devices
AI career counselling	Supports students with disabilities	Teacher training needed
—	—	Data privacy & security

Trap: Mistake: Writing that AI will replace teachers.

پاکستان کے تعلیمی نظام میں مصنوعی ذہانت آہستہ آہستہ اہم کردار ادا کر رہی ہے، جیسے لرننگ مینجمنٹ سسٹم (LMS)، خودکار جانچ اور کیریئر کونسلنگ۔

5.2.2 Career Connection in IoT and AI

Learning computing skills opens doors to many employers in Pakistan: software houses (e.g.

- IoT Engineer: designs smart systems and connected devices (e.g. a smart irrigation kit).
- Data Analyst: analyses the data generated by IoT systems and produces reports.
- Programming — Python, C++.
- Networking — Wi-Fi, protocols (rules for how devices communicate).

Remember: Skills = "PNEP": Programming, Networking, Electronics, Problem-solving. You start building the first one this year when you learn Python!

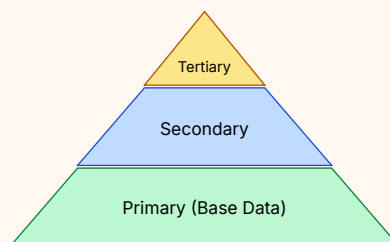
کمپیوٹنگ کی مہارتیں سیکھ کر آپ سافٹ ویئر ہاؤسز، ٹیلی کام کمپنیوں (جیسے جاز، ٹیلی نار)، بینکوں اور سرکاری آئی ٹی محکموں میں کام کر سکتے ہیں۔

5.3 Information Sources (Primary, Secondary, Tertiary) ★ Golden

Information sources are ways to find knowledge and data.

- The internet has vast information, but not all of it is accurate.
- Sources are divided into three main categories.
- Primary sources give original, firsthand information straight from people, events or official records.

Primary (firsthand)	Secondary (explained by others)	Tertiary (summarised, quick reference)
Interview with a flood victim in Dadu	Newspaper report on the 2022 floods	Encyclopedia entry "Floods in Pakistan"
Photograph taken at an event	Textbook chapter explaining the event	Wikipedia article
Personal diary or letter	Biography written by a historian	Dictionary
NADRA official records (CNIC data)	Pakistan Bureau of Statistics report analysing census data	Handbook / almanac of facts
Original survey responses	Research paper analysing the survey	Index or list of research papers
Sensor readings from an IoT weather station	TV news analysis of the weather	A school "fact sheet" on climate
Quaid-e-Azam's original speech recording	Article discussing the speech	Encyclopedia summary of his life



The information pyramid: primary data is the base; each higher level is built from the levels below it and is more summarised.

Remember: Rule of thumb (Author – Date – References): If an online article does not state the author's name, the date of publication, or list its references, it should...

Exam: Typical questions: "Differentiate between primary, secondary and tertiary sources with two examples each" (long/short), "Classify the following sources..." (MCQ/short),...

معلومات کے ذرائع علم اور ڈیٹا تلاش کرنے کے طریقے ہیں، جنہیں تین اقسام میں تقسیم کیا جاتا ہے۔

5.4 Impacts of Computing in Various Fields

Computing has drastically changed society.

- It makes life faster and more connected, but it also brings challenges like privacy risks and technology addiction.
- For every field, think of both sides.

Field	Positive impact (+)	Negative impact (-)
Education	Online classes, e-books and learning apps make studying highly flexible and accessible.	Excessive screen time reduces attention spans and causes physical health issues (eye strain, back pain).
Communication	Emails and video calls allow instant, global communication at almost zero cost (e.g. a WhatsApp video call to a relative in Dubai).	People interact less in person, leading to social isolation and reduced face-to-face skills.
Business	Online shopping, digital banking and automation dramatically improve speed and efficiency.	Traditional manual jobs are lost to machines; risk of severe cyber-attacks on banks.
Healthcare	Digital patient records and telemedicine allow doctors to treat remote patients accurately.	Patient data privacy is at risk if hospital servers are hacked.
Agriculture	Smart irrigation and weather forecasting massively increase crop yields.	Poor internet access in rural areas means many farmers cannot use these tools.

Remember: For the five fields, remember "E-C-B-H-A" — Education, Communication, Business, Healthcare, Agriculture — and always write one + and one – for each.

کمپیوٹنگ نے معاشرے کو بہت بدل دیا ہے۔

5.5 Assistive Technologies ★ Golden

Assistive technologies are devices, software and systems that help people with disabilities or functional limitations (difficulty in seeing, hearing, moving, speaking, remembering, etc.).

Mobility → Sensory → Communication → Cognitive

- They allow more people to participate equally in school, work and society.
- There are four categories:

Category	Purpose	Examples
1. Mobility aids	Help people move from one place to another.	Smart wheelchairs, stair lifts
2. Sensory aids	Support vision and hearing to improve interaction.	Hearing aids, screen readers (VoiceOver), Braille displays
3. Communication aids	Assist individuals with speech difficulties to express themselves.	Speech-generating devices, text-to-speech apps
4. Cognitive / learning aids	Help people with memory, reading and understanding.	Audiobooks, spell-checkers, digital reminder apps

Remember: Four categories = "My Small Cat Cries": Mobility, Sensory, Communication, Cognitive. Link each to a body function: legs (mobility), eyes/ears (sensory), mouth...

Exam: Expect: "What are assistive technologies? Explain their categories with examples" (long), "Name two sensory aids" (short), or "A screen reader is an example of ___"...

معاون ٹیکنالوجیز (Assistive Technologies) وہ آلات، سافٹ ویئر اور نظام ہیں جو معذور افراد کی مدد کرتے ہیں تاکہ وہ اسکول، کام اور مع...

5.5.1 Importance of Assistive Technologies

Assistive technologies are important for three main reasons: Equal access to education: Screen readers allow blind students to read digital textbooks equally with their peers (classmates).

Equal Access → Independence → Inclusion

- Equal access to education: Screen readers allow blind students to read digital textbooks equally with their peers (classmates).
- Independence in daily life: Mobility aids reduce the need for constant human assistance — a person in a smart wheelchair can go to the...
- Social inclusion: They break down barriers, allowing individuals to work, study and socialise confidently.

Remember: Remember "All": Access, Independence, Inclusion.

معاون ٹیکنالوجیز کی اہمیت تین باتوں میں ہے: تعلیم تک برابر رسائی (ناہینا طلبہ اسکرین ریڈر سے کتابیں پڑھ سکتے ہیں)، روزمرہ زندگی می...

5.5.2 Career Connection in Assistive Tech

Designing assistive technology is a highly rewarding career path — you earn a living while improving people's lives.

- CareerWhat they do Assistive Software DeveloperCreates tools such as speech-to-text software (e.g. Urdu voice typing for people who cannot use a keyboard).
- Biomedical EngineerDesigns artificial limbs and hearing aids.

Career	What they do
Assistive Software Developer	Creates tools such as speech-to-text software (e.g. Urdu voice typing for people who cannot use a keyboard).
Biomedical Engineer	Designs artificial limbs and hearing aids.
UX/UI Accessibility Expert	Ensures websites are easy to navigate for visually impaired users (e.g. adding text descriptions to images so screen readers can read them).

معاون ٹیکنالوجی بنانا ایک قابلِ قدر پیشہ ہے۔

Key terms

Term	Meaning	اردو
Computing	Using computers and digital devices to store, process and share information.	کمپیوٹنگ
Artificial Intelligence (AI)	Human-made technology that lets machines think, learn and decide using data.	مصنوعی ذہانت
Internet of Things (IoT)	Physical devices connected to the internet that collect and share data to work automatically.	انٹرنیٹ آف تھنگز، اشیاء کا انٹرنیٹ
Sensor	Device that collects information from the physical world.	حساس آلہ (سینسر)
Actuator	Device that performs a physical action in response to a command.	عمل کار آلہ (ایکچویٹر)
Connectivity	The link (Wi-Fi, Bluetooth, ZigBee, 4G/5G) that lets devices send and receive data.	رابطہ کاری
Data processing	Changing raw data into useful information.	ڈیٹا کی پروسیسنگ
Cloud computing	Storing data and using services over the internet instead of local drives.	کلاؤڈ کمپیوٹنگ
User interface (UI)	The part that lets a user see information and give commands to a device.	صارف انٹرفیس
Data Analytics	Collecting, storing, cleaning, analysing and presenting data to find useful information.	ڈیٹا کا تجزیہ
Data cleaning	Finding and correcting errors such as duplicates and missing values.	ڈیٹا کی صفائی
Data visualization	Showing data as charts, plots or graphs.	ڈیٹا کی بصری پیشکش
Smart city	City using IoT for traffic, lighting, parking and waste.	اسمارٹ شہر
Smart farming	Using sensors, automated irrigation and drones in agriculture.	اسمارٹ کاشتکاری
Learning Management System (LMS)	Online platform for lessons, assignments and tests.	آن لائن تعلیمی نظام
Telemedicine	Treating patients remotely using computing and communication.	دور دراز علاج
Primary source	Original, firsthand information.	بنیادی ذریعہ

Term	Meaning	اردو
Secondary source	Information collected and explained by someone else.	ثانوی ذریعہ
Tertiary source	Highly summarised information for quick reference.	ثالثی ذریعہ
Reliability	How far a source can be trusted.	قابل اعتماد ہونا
Privacy	Keeping personal information safe from others.	رازداری
Cyber-attack	An attempt by hackers to damage or steal from computer systems.	سائبر حملہ
Assistive technology	Devices, software and systems that help people with disabilities.	معاون ٹیکنالوجی
Screen reader	Software that reads screen text aloud for blind users.	اسکرین ریڈر
Braille display	Device with raised pins showing text as Braille dots.	بریل ڈسپلے
Accessibility	Designing so everyone, including people with disabilities, can use a product.	رسائی پذیری

Chapter in one look

- Computing is vital in Pakistan's schools, hospitals, banks, farms and industries; it has both positive and negative impacts.
- **AI** is "artificial" (made by humans) and "intelligent" (does tasks needing thinking); it uses data to learn, recognise patterns and make decisions.
- In smart systems, IoT is the "eyes" that gather data, Analytics sorts it, and AI is the "brain" that decides the action.
- **IoT** is a system of physical devices connected to the internet that collect and share data to perform tasks automatically.
- The six IoT components are sensors (collect), actuators (act), connectivity, data processing, cloud computing and user interface.
- **Data Analytics** is a pipeline: collection → storage → cleaning → analysis → visualization.
- Data cleaning removes duplicates, incorrect values, missing records and improper formats.
- IoT outputs raw data (hardware); AI outputs smart decisions (software); Data Analytics outputs reports, graphs and insights (data).
- IoT is used in smart cities, industrial automation, healthcare, education, home automation and smart farming.
- Smart farming (soil sensor → cloud AI → water pump) saves water and improves yields in Sindh and Punjab.
- AI in Pakistani education: LMS, automated grading, career counselling; challenges are rural internet, lack of devices, teacher training and data privacy.
- IoT/AI careers (IoT engineer, data analyst) need programming, networking, electronics and problem-solving.
- Information sources: primary (firsthand/original), secondary (collected and explained by others), tertiary (summarised for quick reference).
- Do not trust an online article that lacks an author, a publication date or references.
- Computing helps education, communication, business, healthcare and agriculture but risks screen addiction, isolation, job loss, cyber-attacks and privacy loss.
- **Assistive technologies** — mobility, sensory, communication and cognitive aids — give equal access, independence and social inclusion to people with disabilities.

Digital Literacy

★ 11 Golden · 17 cards · Types of Data · Data-Collection Strategies · Interviews and Surveys · Prototypes, Observation and Simulations · Primary and Secondary Data · Case Study: Digital Inquiry Project · Steps 1 – 3: Advanced Search and Methodology · Steps 4 – 6: Survey Preparation and Primary Data Collection · Steps 7 – 9: Secondary Data and Spreadsheet Organisation · Steps 10 – 12: Data Analysis, Conclusion and Recommendations · The Final Digital Artefact

6.1 Introduction to Digital Literacy

Digital literacy is the ability to understand and use digital devices, the internet, and online tools in a safe, effective, and responsible manner.

Locate information → Organise it properly → Turn it into something useful → Share it responsibly

- Safe – you protect yourself and others online (for example, you do not share passwords or private details).
- Effective – you get the job done well (for example, you find the right information quickly).
- Responsible – you use information honestly and fairly (for example, you do not copy someone else's work or spread false news).
- find information,

Remember: Remember the four verbs of a digitally literate student: Locate – Organise – Create – Share ("LOCS": think of a lock that keeps your digital life safe).

ڈیجیٹل خواندگی (Digital Literacy) سے مراد ڈیجیٹل آلات، انٹرنیٹ اور آن لائن ٹولز کو محفوظ، مؤثر اور ذمہ دارانہ طریقے سے سمجھنا اور استعمال کرنا ہے۔

6.2 Digital Literacy in the Information Age

The Information Age is the period we live in now, where vast amounts of data are produced, shared and accessed instantly through the internet, mobile devices and digital platforms (websites, apps, social media).

- Simply having access to technology is not enough.
- Owning a smartphone does not make you good at using information.
- You must develop the skills to understand, manage and use information wisely.

Ability	What it means	Example
Technical	Knowing how to operate devices and software.	Creating a Google Form; using a spreadsheet.
Cognitive (thinking)	Thinking critically about information.	Checking whether a news story is true before believing it.
Social	Communicating and behaving well with others online.	Being polite in a class WhatsApp group; respecting privacy.

Remember: Three abilities = "Tech, Think, Talk": Technical, Cognitive (think), Social (talk with others).

معلوماتی دور (Information Age) میں انٹرنیٹ اور موبائل کے ذریعے بے پناہ معلومات فوراً تیار اور شیئر ہوتی ہیں۔

6.3 Types of Data ★ Golden

Data means facts, numbers, words or observations that we collect.

- The data gathered through a survey, an investigation or a classroom project can be described in two ways: qualitatively or quantitatively.
- Knowing the difference is critical, because each type is analysed in a different way.

Type of data	Nature	Examples	Best used for
Qualitative	Descriptive, non-numerical	Opinions, reasons, behaviour	Understanding why and how something happens
Quantitative	Numerical, countable or measurable	Hours, scores, percentages, totals	Comparing results and making graphs/charts

Remember: QUALitative = QUALity = words and feelings. QUANTitative = QUANTity = numbers and counts. Qualitative answers why/how; quantitative answers how many/how much.

Exam: Common questions: "Differentiate between qualitative and quantitative data" (4–5 marks) or "Classify the following data". Write the answer as a table with four rows:...

کیفیتی ڈیٹا (Qualitative Data) الفاظ پر مشتمل ہوتا ہے اور بتاتا ہے کہ لوگ کیا سوچتے، محسوس کرتے یا تجربہ کرتے ہیں، جیسے آن لائن کل...

6.4 Data-Collection Strategies ★ Golden

Digital literacy is not limited to finding information that already exists on the internet.

Interviews → Surveys → Prototypes → Observation → Simulations

- You should also learn how to gather original (new) information yourself.
- A data-collection strategy is a planned method for gathering this information.
- The five common strategies are: Interviews-Surveys-Prototypes-Observation-Simulations

Strategy	How it works (one line)	Usual data type	Best when...
Interview	Planned conversation with a person or small group	Mostly qualitative	You need deep, detailed answers from a few people
Survey	Same questions asked to many people	Quantitative and qualitative	You need answers from many people quickly
Prototype	Users test an early version and give feedback	Both	You want to improve a product or solution
Observation	Watch and record; no questions asked	Notes (qualitative) or counts (quantitative)	You want to see what people actually do
Simulation	Computer model; change variables and see what happens	Mostly quantitative	Real experiments are costly, risky or difficult

Remember: Mnemonic for the five strategies: "I S P O S" – "I Saw People On Screens": Interviews, Surveys, Prototypes, Observation, Simulations.

Exam: "List and explain the data-collection strategies" is a frequent long question. Name all five, give a one-sentence definition and one example for each, and say what type...

ڈیجیٹل خواندگی صرف موجود معلومات تلاش کرنے تک محدود نہیں، بلکہ نئی معلومات خود جمع کرنا بھی اس کا حصہ ہے۔

6.4.1 – 6.4.2 Interviews and Surveys ★ Golden

An interview is a planned conversation used to gather information directly from a person.

- It may be one-to-one (you and one person) or in a small group.
- Questions are planned in advance, so you do not forget anything important.
- Answers are usually qualitative (opinions, reasons, experiences).

Question type	What it looks like	Data it gives
Closed: Yes/No	Do you study with your phone nearby? (Yes / No)	Countable (quantitative)
Closed: Multiple choice	Which activity do you use your phone for most? (Lectures / Notes / Social media / Games)	Categories you can count
Number / scale	How many hours do you use your phone daily? —	Quantitative
Rating scale	"My phone helps me study." 1 = Strongly disagree ... 5 = Strongly agree	Quantitative (you can average the ratings)
Open-ended	How does your phone help you in studies?	Qualitative (free text)

Remember: Interview = In-depth with few. Survey = Same questions to many. Think of a doctor talking to one patient (interview) vs a census form sent to every house...

Exam: You may be asked to "differentiate between an interview and a survey" or to "identify what is wrong with this survey question and rewrite it". For rewriting, name the...

انٹرویو ایک منصوبہ بند گفتگو ہے جس میں کسی ایک فرد یا چھوٹے گروہ سے براہ راست معلومات حاصل کی جاتی ہیں؛ اس کے جوابات عموماً کیفیتی ہوتے ہیں۔

6.4.3 – 6.4.5 Prototypes, Observation and Simulations ★ Golden

A prototype is a simple early version of a product, model or solution.

- It allows an idea to be tested before time and effort are spent on the final version.
- Users try the prototype and give feedback, and this feedback becomes useful data for improvement.
- Example: Before building a full school app, you draw the screens on paper (or make a few clickable screens in a design tool) and ask five...

Remember: Watch it → Observation. Ask it → Survey/Interview. Try it → Prototype. Model it → Simulation.

Exam: Scenario questions are popular: "Which data-collection method is most suitable for ... ? Give a reason." Always name the method and justify it using its key feature (e.g....

پروٹوٹائپ کسی پروڈکٹ یا حل کا ابتدائی سادہ نمونہ ہے جسے صارفین آزما کر رائے دیتے ہیں تاکہ اسے بہتر بنایا جاسکے۔

6.5 Primary and Secondary Data ★ Golden

Whenever you use data, it is important to know where it came from.

- The source affects how much control the researcher had over collecting it, and how confidently the results can be used.

Aspect	Primary Data	Secondary Data
Source	Collected directly by the researcher.	Already collected by others.
Examples	Survey responses, interviews, observations.	Reports, articles, websites, databases.
Purpose	Prepared for a specific study.	Reused for a new purpose.
Control	High control over questions and collection process.	Limited control because the data already exists.

Primary Data

Fresh, direct collection
(You do the work)

Surveys · Interviews · Observation

Secondary Data

Existing archives
(Someone else did it)

Books · Reports · Databases

Primary data is collected by you; secondary data was collected by others.

Remember: Primary = "1st hand" (you collect it first). Secondary = "2nd hand" (like a second-hand book – someone else used it first).

Exam: "Differentiate between primary and secondary data" is one of the most-asked questions in this chapter. Draw the four-row table: Source, Examples, Purpose, Control. Learn...

بنیادی ڈیٹا (Primary Data) محقق خود کسی خاص مقصد کے لیے براہ راست جمع کرتا ہے، مثلاً سروے، انٹرویو یا مشاہدہ۔

6.6 Designing a Data-Collection Approach

Before collecting data, a researcher must have a plan.

Question → Data type → Method → Sample → Tools → Collect → Store + backup → Ethics

- Define the research question clearly. What exactly do you want to find out?
- Decide the type of data needed: quantitative, qualitative, or both.
- Select suitable methods (survey, interview, observation).
- Choose the sample or target group – the people who will give you data.

Remember: Remember the steps with: "Quiet Teachers Make Students Think, Check, Save Ethically" → Question, Type, Method, Sample, Tools, Collect, Store, Ethics.

ڈیٹا جمع کرنے سے پہلے منصوبہ بندی ضروری ہے: تحقیقی سوال واضح کریں، ڈیٹا کی قسم طے کریں، مناسب طریقہ چنیں، نمونہ (Sample) منتخب کریں...

6.7 Presenting Data Using Digital Tools: 6.7.1 Spreadsheets and 6.7.2 Presentations

Collecting data is only one part of the work.

Spreadsheet data → Chart / Graph → Insert → Slides

- You must also present your findings so that others can understand them quickly and easily.
- Digital tools make this much easier.
- The four main tools are spreadsheets, presentations, infographics and reports.

Formula	What it does
=SUM(B2:B4)	Adds all values in B2 to B4.
=AVERAGE(B2:B4)	Mean: sum ÷ number of values.
=MAX(B2:B4) / =MIN(B2:B4)	Largest / smallest value.
=MEDIAN(B2:B4)	Middle value when sorted.
=COUNT(B2:B4)	Counts cells containing numbers.
=COUNTA(C2:C4)	Counts cells that are not empty (text or numbers).
=COUNTIF(C2:C4, "Yes")	Counts cells that match a condition.

Remember: Presentation rule of thumb: one idea per slide, and no more than about six short bullet points. If you are reading long sentences off the slide, the slide has...

ڈیٹا جمع کرنے کے بعد اسے پیش کرنا بھی ضروری ہے۔

6.7.3 – 6.7.4 Infographics and Reports

An infographic (information + graphic) is a visual way of presenting information, facts or data so that people can understand the main idea quickly and easily.

- Simplify complex concepts.
- Highlight the most important findings and results.
- Use clear visuals, limited text and accurate information.

Situation	Best tool	Why
You must calculate the average marks of 40 students.	Spreadsheet	Formulas calculate instantly.
You will explain results to the class in assembly.	Presentation	Designed for a live audience.
You want a one-page poster for the notice board.	Infographic	Main result understood at a glance.
The principal wants the full study with method and recommendations.	Report	Detailed, structured document.

Remember: Match tool to job: Spreadsheet = Calculate, Presentation = Tell, Infographic = Show quickly, Report = Explain fully.

انفوگرافک معلومات کو تصویری انداز میں پیش کرنے کا طریقہ ہے تاکہ اصل بات جلدی سمجھ آ جائے؛ اس کے لیے کینوا (Canva) اور پاور بی آئی...

6.8 Case Study: Digital Inquiry Project ★ Golden

This case study shows how to apply all the ideas of this chapter to solve a real-world problem from start to finish.

1 Question → 2 Search → 3 Collect → 4 Analyse → 5 Create

- An inquiry is an investigation to find the answer to a question.
- Research question: "How does daily mobile phone use affect the study habits of students?" Scenario: Students notice classmates using phones...
- Some say phones help them learn; others say phones waste time.

Workflow stage	Steps
Question & Search	1 Advanced searches · 2 Identify data types · 3 Select methods
Collect	4 Prepare the survey · 5 Choose the sample · 6 Collect primary data ethically · 7 Use secondary data
Analyse	8 Organise in a spreadsheet · 9 Create charts · 10 Analyse results
Create	11 Write the conclusion · 12 Recommendations → final digital artefact

Remember: Workflow in five words: Ask → Search → Collect → Analyse → Create. Every research project you ever do will follow this pattern.

Exam: Expect a long question: "Describe the steps of a digital inquiry project using the example of mobile phone use." Write the research question first, then the 12 steps...

اس کیس اسٹڈی میں تحقیقی سوال یہ ہے: "روزانہ موبائل فون کا استعمال طلبہ کی پڑھائی کی عادات پر کیسے اثر ڈالتا ہے؟"

6.8 Steps 1 – 3: Advanced Search and Methodology ★ Golden

Students first search for background information using search engines (such as Google or Bing).

- They use advanced search operators – special symbols or words typed into the search box – to narrow down results and find reliable...
- Suggested queryOperator usedWhat it does "mobile phone use" "study habits" students" " (quotes)Finds the exact phrases "mobile phone use"...
- "academic performance" filetype:pdfFinds only PDF documents (often official reports and research papers).

Suggested query	Operator used	What it does
"mobile phone use" "study habits" students	" " (quotes)	Finds the exact phrases "mobile phone use" and "study habits", not the words scattered randomly.
"academic performance" filetype:pdf	filetype:	Finds only PDF documents (often official reports and research papers).
digital distraction site:.edu	site:	Searches only educational websites (addresses ending in .edu).

Remember: Three operators, three jobs: " " = exact, filetype: = format, site: = where. Note: there is no space after the colon (site:.edu, not site: .edu).

Exam: "Explain any three advanced search operators with examples" is a likely short question. For each, write the operator, an example query, and what it does. Also learn the...

پہلے قدم میں طلبہ سرچ انجن پر جدید سرچ آپریٹرز (Advanced Search Operators) استعمال کرتے ہیں: کوٹیشن مارکس (" ") بالکل وہی جملہ تلا...

6.8 Steps 4 – 6: Survey Preparation and Primary Data Collection ★ Golden

Design clear, unbiased questions. Each question is linked to the type of data it produces: QuestionType of data How many hours do you use your phone daily?Quantitative Which activity do you use your phone for the most? Quantitative...

- Design clear, unbiased questions.
- Each question is linked to the type of data it produces: QuestionType of data How many hours do you use your phone daily?Quantitative Which...

Question	Type of data
How many hours do you use your phone daily?	Quantitative
Which activity do you use your phone for the most?	Quantitative (categories)
How does your phone help you in studies?	Qualitative
Suggest one way to use phones responsibly.	Qualitative

Remember: Ethics in five words: Consent, No-Sensitive, Anonymous, Private, Educational – "C-NAPE". Think: "Cover the person's nape (neck) – protect them."

Exam: Questions often ask: "What ethical guidelines should be followed when collecting data?" List all five points. Use the words consent and anonymity – examiners look for...

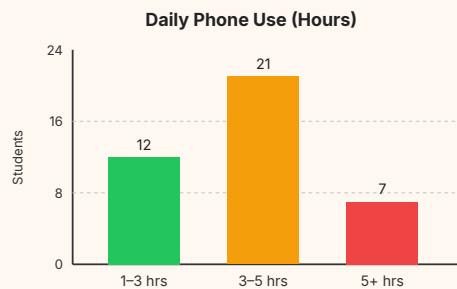
چوتھے قدم میں واضح اور غیر جانبدار سوالات تیار کیے جاتے ہیں۔

6.8 Steps 7 – 9: Secondary Data and Spreadsheet Organisation ★ Golden

Students include a reliable online article stating that "excessive screen time lowers concentration." They will compare this secondary data with their own primary survey results.

- If both agree, the conclusion is stronger.
- If they disagree, the students must think about why (for example, a different age group or country).

Student	Hours	Study Use	Main Activity	Distracted?
1	3	Yes	Lectures	Sometimes
2	5	No	Social Media	Yes
3	2	Yes	Notes	No



Bar chart of daily phone use for 40 students. The counts (12, 21, 7) are estimated from the bar heights in the source chart.

Remember: Chart chooser: Bar = Compare, Pie = Parts of a whole, Line = Change over time.

Exam: You may get the spreadsheet table and be asked to "write formulas to find the total and average hours" or "count students who use the phone for study". Write the full...

ساتویں قدم میں ایک قابل اعتماد آن لائن مضمون کا ثانوی ڈیٹا ("زیادہ اسکرین ٹائم توجہ کم کرتا ہے") اپنے سروے کے بنیادی ڈیٹا سے مواز...

6.8 Steps 10 – 12: Data Analysis, Conclusion and Recommendations ★ Golden

To analyse means to study the data carefully to find patterns and trends.

- Most students (the majority) use phones for 3–5 hours daily.
- Half use it for educational purposes (lectures/notes).
- A clear trend: students who use phones mostly for entertainment report much higher distraction levels than those who use them for study.

Remember: Conclusion formula: "The data shows ... However ... Therefore ..." – state the main finding, mention the other side, then give the overall answer.

Exam: You may be asked to "write a conclusion and two recommendations" for given data. Use the "shows ... however ... therefore" pattern and make sure each recommendation follows...

دسویں قدم میں چارٹس اور اسپریڈشیت سے رجحانات تلاش کیے جاتے ہیں: زیادہ تر طلبہ روزانہ تین سے پانچ گھنٹے فون استعمال کرتے ہیں، نصف ا...

6.8 The Final Digital Artefact ★ Golden

A digital artefact is a final product created using digital technologies.

- Students must create one final artefact that answers the original research question.
- It is not just a decorated file; it must present evidence and conclusions.
- Artefact option Suitable digital tools Main purpose Digital Report MS Word, Google Docs Explain the full study in a structured text document.

Artefact option	Suitable digital tools	Main purpose
Digital Report	MS Word, Google Docs	Explain the full study in a structured text document.
Presentation	PowerPoint, Canva, Slides	Share key findings with a live audience.
Infographic	Canva, Adobe Express	Show the main results visually and quickly on one page.
Dashboard	Power BI, Tableau	Explore data through interactive charts and filters.

Remember: The structure follows the inquiry itself: What (title, question, intro) → How (method) → Found (primary, secondary, charts) → So what (conclusion,...

Exam: Learn the four artefact options with tools and purpose (a table question), and the ten-part structure in order. "What is a digital artefact?" – stress that it must...

ڈیجیٹل آرٹیفیکٹ (Digital Artefact) ڈیجیٹل ٹیکنالوجی سے بنائی گئی آخری پروڈکٹ ہے جو تحقیقی سوال کا جواب ثبوت اور نتائج کے ساتھ پیش...

Careers in Digital Research

The skills in this chapter lead to real jobs: CareerWhat they doChapter skill used Market Research AnalystDesigns surveys to collect primary data to guide business decisions.Surveys, primary data UX (User Experience)...

Career	What they do	Chapter skill used
Market Research Analyst	Designs surveys to collect primary data to guide business decisions.	Surveys, primary data
UX (User Experience) Researcher	Gathers qualitative and quantitative data to improve app and website designs.	Prototypes, interviews, observation
Data Journalist	Uses advanced search skills to research stories and presents them via infographics.	Advanced search, infographics
Digital Marketing Specialist	Collects audience data and presents results through dashboards and reports.	Dashboards, reports

Remember: Link each career to one tool: Market Researcher → Survey; UX Researcher → Prototype; Data Journalist → Search + Infographic; Digital Marketer → Dashboard.

اس باب کی مہارتوں سے کئی کیریئر بنتے ہیں: مارکیٹ ریسرچ اینالسٹ کاروباری فیصلوں کے لیے سروے بناتا ہے، یو ایکس (UX) ریسرچر ایپس اور...

Key terms

Term	Meaning	اردو
Digital literacy	Ability to understand and use digital devices, the internet and online tools safely, effectively and responsibly.	ڈیجیٹل خواندگی
Information Age	Period in which vast amounts of data are produced, shared and accessed instantly.	معلوماتی دور
Data	Facts, numbers, words or observations collected for study.	مواد / ڈیٹا
Qualitative data	Descriptive, non-numerical data about thoughts, feelings and experiences.	کیفیتی ڈیٹا
Quantitative data	Numerical data that can be counted or measured.	مقداری ڈیٹا
Interview	Planned conversation to gather information directly from a person or small group.	انٹرویو / بالمشافہ گفتگو
Survey / Questionnaire	Same set of questions asked to many people.	سروے / سوالنامہ
Prototype	Simple early version of a product used to test an idea and get feedback.	ابتدائی نمونہ
Observation	Watching and recording what happens without asking questions.	مشاہدہ
Simulation	Computer-based model of a real situation in which variables can be changed.	نقلی ماڈل (سمیولیشن)
Variable	A factor that can be changed or measured.	متغیر
Primary data	Data collected first-hand by the researcher for a specific purpose.	بنیادی ڈیٹا
Secondary data	Data already collected by someone else and reused for a new purpose.	ثانوی ڈیٹا
Research question	The clear question an investigation tries to answer.	تحقیقی سوال
Sample	The group of people who provide the data.	نمونہ
Representative sample	A sample that reflects the mix of the whole group.	نمائندہ نمونہ
Consent	Permission given by a person before their data is collected.	رضامندی / اجازت

Term	Meaning	اردو
Anonymity	Not recording people's names so they cannot be identified.	گمنامی
Ethics	Rules of right and fair behaviour when collecting and using data.	اخلاقیات
Spreadsheet	Software that arranges data in rows and columns and performs calculations.	اسپریڈشیٹ / حسابی جدول
Formula	A spreadsheet instruction starting with = that calculates a result.	فارمولا
Infographic	A visual presentation of information so the main idea is understood quickly.	تصویری معلومات (انفوگرافک)
Report	Structured document presenting data, findings, analysis and conclusions.	رپورٹ
Dashboard	Interactive screen of charts and filters for exploring data.	ڈیش بورڈ
Advanced search operator	Special symbol or word (e.g. " ", filetype:, site:) that narrows search results.	جدید تلاش آپریٹر
Methodology	The plan of methods used to answer a research question.	طریقہ کار
Digital artefact	Final product made with digital technology that answers the research question with evidence.	ڈیجیٹل آرٹیفیکٹ / ڈیجیٹل تخلیق
Recommendation	A practical suggestion based on the findings.	سفارش

Chapter in one look

- **Digital literacy** is the ability to understand and use digital devices, the internet and online tools safely, effectively and responsibly – not just using devices, but finding, evaluating and creating information.
- In the **Information Age**, access to technology is not enough; digital literacy needs technical, cognitive and social abilities.
- **Qualitative data** is descriptive and non-numerical (opinions, reasons, behaviour) and explains why/how.
- **Quantitative data** is numerical, countable or measurable (hours, scores, percentages) and is used for comparing and charting.
- The five **data-collection strategies** are interviews, surveys, prototypes, observation and simulations.
- **Interviews** are planned conversations giving mostly qualitative data; **surveys** ask the same questions to many people, on paper or with Google/Microsoft Forms.
- Good survey questions are clear, short, unbiased and ask about one thing; avoid leading, double-barrelled, vague and overlapping questions.
- A **prototype** is an early version tested for feedback; **observation** records what happens without asking questions; a **simulation** is a computer model used when real experiments are costly, risky or difficult.
- **Primary data** is collected first-hand (high control); **secondary data** was collected by others and reused (limited control).
- Designing a data-collection approach has eight steps: question, data type, method, sample, tools, collect, store with backup, and ethics (consent is mandatory).
- **Spreadsheets** organise and calculate (e.g. `=SUM()`, `=AVERAGE()`, `=COUNTIF()`); **presentations** deliver to an audience; **infographics** give quick visual summaries; **reports** give detailed structure.
- Visuals must always be accurate and based strictly on real data, never exaggerated.
- **Advanced search operators** such as `" "`, `filetype:pdf` and `site:.edu` help find reliable sources efficiently; sources must be evaluated for author, date, reliability and bias.
- **Ethics**: get consent, avoid sensitive information, keep anonymity, keep responses private, and use data only for educational purposes.
- The **digital inquiry project** follows Question → Search → Collect → Analyse → Create in 12 steps.
- A **digital artefact** (report, presentation, infographic or dashboard) answers the research question using evidence and visuals, following a ten-part structure ending in references.

NIGHT BEFORE

One-page recaps

Read each recap, close the booklet, say it aloud. Then stop.

Chapter 1 recap — Computer Systems

- Discrete = counted (stairs). Continuous = measured (ramp). Digital systems use bits 0 and 1; they are reliable, accurate, easy to design and programmable.
- Analog signal = smooth sine wave. Digital signal = square wave HIGH/LOW. Digital data resists noise.
- AND · output 1 only if all inputs 1. OR + output 1 if any input 1. NOT ' inverts. NAND = NOT-AND, NOR = NOT-OR (universal). XOR 1 when inputs differ. XNOR 1 when inputs same.
- Truth table rows = 2^n . SOP = sum of minterms where output is 1. POS = product of maxterms where output is 0.
- K-map: group 1, 2, 4, 8 adjacent 1s (power of 2, wrap-around allowed). Each group drops the variables that change.
- SDLC: requirement → design → implementation → testing → deployment → maintenance. Waterfall = sequential; Agile = short sprints and feedback.
- OSI 7: Physical, Data Link, Network, Transport, Session, Presentation, Application. Mnemonic: Please Do Not Throw Sausage Pizza Away. TCP/IP 4: Network Access, Internet, Transport, Application.

Chapter 2 recap — Computational Thinking & Algorithm

- CT pillars: decomposition, pattern recognition, abstraction, algorithms.
- Algorithm = finite, ordered, unambiguous steps. Pseudocode = structured English, not a programming language.
- Bubble Sort: compare neighbours, swap if out of order; largest "bubbles" to the end each pass. $O(n^2)$.
- Selection Sort: find the minimum in the unsorted part and swap it to the front. $O(n^2)$.
- Linear Search: check one by one. Works on unsorted data. $O(n)$.
- Binary Search: list MUST be sorted. Repeatedly test the middle. $O(\log n)$. Trace with low / mid / high.
- Judge an algorithm on correctness, efficiency (time and space) and clarity. Prefer binary search on large sorted lists; linear search on small or unsorted lists.

Chapter 3 recap — Programming Fundamentals

- Python: high-level, interpreted. Variables are names; types int, float, str, bool, list, dict. `input()` returns str.
- Arithmetic `+` `-` `*` `/` `//` `%` `**`. Relational. Logical `and` `or` `not`. Bitwise `&` `|` `^` `~` `<<` `>>`.
- Selection: `if` / `if-else` / `if-elif-else` / nested. Test from the top; first True wins.
- `for` = known count + range. `while` = unknown count; always update the condition. `break` leaves; `continue` skips one.
- Libraries: `import math` / `random` / `datetime`. Third-party via pip: NumPy, Pandas, Matplotlib.
- Syntax = will not run. Runtime = crash + traceback. Logical = runs, wrong answer.

Chapter 4 recap — Data and Analysis

- Data = raw facts. Information = processed data used for decisions. Database = organised related data. DBMS = software that manages it (reduces redundancy, improves security, sharing, integrity).
- Table / relation; row / record / tuple; column / field / attribute. Primary key uniquely identifies a row. Foreign key refers to another table's primary key.
- ER model: entity (rectangle), attribute (oval), relationship (diamond), cardinality 1:1, 1:M, M:N.
- Referential integrity: you cannot orphan a child row. Cascade Update / Cascade Delete copy or remove matching child rows.
- Access objects: tables, forms, queries, reports. Queries filter with criteria, wildcards (`*`, `?`), AND/OR. Totals: Count, Sum, Avg, Group By. Charts visualise the result.

Chapter 5 recap — Application and Impacts of Computing

- IoT = connected devices. Six parts: sensor, actuator, connectivity, processing, cloud, UI. AI = machines that learn and decide. Analytics = collect → store → clean → analyse → visualise.
- Together: IoT (eyes) → Analytics (sort) → AI (brain) → actuator (action).
- Sources: primary (firsthand), secondary (explained by others), tertiary (summaries). No author / date / references → do not trust.
- Assistive tech: mobility, sensory, communication, cognitive. Goal: equal access, independence, inclusion.

Chapter 6 recap — Digital Literacy

- Digital literacy = find, evaluate, create and share information with digital tools, safely.
- Qualitative = words/descriptions. Quantitative = numbers. Strategies: interview, survey, prototype, observation, simulation.
- Primary data you collect; secondary data already exists. Good survey questions are clear, single-idea, unbiased.
- Present with spreadsheets (formulas such as `=AVERAGE()`), slides, infographics and reports. Cite sources. Follow the 12-step inquiry project from question to artefact.

Computer Science XII

Exam revision cards — definitions, tables, mnemonics and ★
Golden topics on one glance each.

Bilingual support: English + اردو

امتحانی چیٹ شیٹ — تعریفیں، جدول، یادداشت اور گولڈن موضوعات

01 Computer Systems (HCI)

02 Computational Thinking & Algorithms

03 Programming Fundamentals

04 Data and Analysis

05 Application and Impacts of Computing

06 Entrepreneurship in the Digital Age

XII

Contents

32 Golden cards are listed below. Start there if you have less than two hours.

Chapter 1: Computer Systems (HCI) — 16 cards

Chapter 2: Computational Thinking & Algorithms — 24 cards

Chapter 3: Programming Fundamentals — 18 cards

Chapter 4: Data and Analysis — 9 cards

Chapter 5: Application and Impacts of Computing — 17 cards

Chapter 6: Entrepreneurship in the Digital Age — 18 cards

Night-before recaps: one-page summaries of every chapter

★ Golden index — if you only have two hours

These are the high-yield topics. Say each definition aloud, then draw the table or diagram from memory.

صرف دو گھنٹے ہوں تو یہ گولڈن کارڈز زبانی دہرائیں، پھر جدول حافظے سے بنائیں۔

- | | |
|--|--|
| 1. ★ Ch 1. Applications of HCI (Domains of Life) | 17. ★ Ch 3. The with Statement (Context Manager) |
| 2. ★ Ch 1. Importance of HCI | 18. ★ Ch 4. Data Sources and Database Connection |
| 3. ★ Ch 1. Human-Computer Interaction Problems | 19. ★ Ch 4. Pandas and DataFrames |
| 4. ★ Ch 1. User Interface Design (UI vs UX) | 20. ★ Ch 4. Handling Missing Values (NaN) |
| 5. ★ Ch 1. Testing Methods (Usability, A/B, Automated) | 21. ★ Ch 4. Data Visualization and Types of Graphs |
| 6. ★ Ch 2. Trace Tables | 22. ★ Ch 4. Descriptive Statistics |
| 7. ★ Ch 2. Comparing Trace Tables and Stepwise Reasoning | 23. ★ Ch 5. Neural Networks (NN) |
| 8. ★ Ch 2. Number of Steps (Big O Notation) | 24. ★ Ch 5. Components of Neural Networks |
| 9. ★ Ch 2. Concept of Data Structure | 25. ★ Ch 5. Secure Collaboration (Authentication, MFA, Access Control) |
| 10. ★ Ch 2. Linear Data Structures: Stack | 26. ★ Ch 5. Common Security Threats (Malware, Phishing, DoS) |
| 11. ★ Ch 2. Linear Data Structures: Queue | 27. ★ Ch 5. Functions of Equity and Equal Access |
| 12. ★ Ch 3. Lists (Ordered, Mutable, Methods) | 28. ★ Ch 6. Entrepreneurship in the Digital Age |
| 13. ★ Ch 3. Tuples (Ordered, Immutable) | 29. ★ Ch 6. Prototype |
| 14. ★ Ch 3. Dictionaries (Key-Value Pairs) | 30. ★ Ch 6. Prototype Development Cycle |
| 15. ★ Ch 3. Functions in Python | 31. ★ Ch 6. Minimum Viable Product (MVP) |
| 16. ★ Ch 3. & 3.4 File Handling and Operations (Modes r, w, a) | 32. ★ Ch 6. Beachhead Market |

How to use these Cheat Sheets

This booklet is the **night-before and last-week** revision pack. Each topic is one card: the definition you must write, the table or formula, a memory trick, the usual exam wording, and one Urdu line. It is not a textbook. Open the Student's Edition when you need a worked example.

یہ چیٹ شیٹ امتحان سے پہلے رات اور آخری ہفتے کے لیے ہے۔ ہر موضوع ایک کارڈ ہے۔ مثالیں اصل کتاب میں ہیں۔

How to revise with this booklet

1. **Two hours left.** Read only the ★ Golden cards and the Golden index. Say each definition aloud.
2. **One day left.** One chapter at a sitting. Cover the card, write the table from memory, then check.
3. **The night before.** Read the night-before recaps. Do not start a new chapter.
4. **In the hall.** Remember the recipe: *Define. Explain. Example. Diagram. Working.*

What each card holds

- **Definition** — the 1-mark sentence. Write it exactly.
- **Points / table / snippet** — the 3-mark and 5-mark body.
- **Remember** — a mnemonic or trap to avoid.
- **Exam** — how the board asks it.
- **اردو** — the same idea in one spoken line.

★ **Golden** cards are high-yield. If time is short, skip non-Golden cards first, not these.

CHAPTER 1

Computer Systems (HCI)

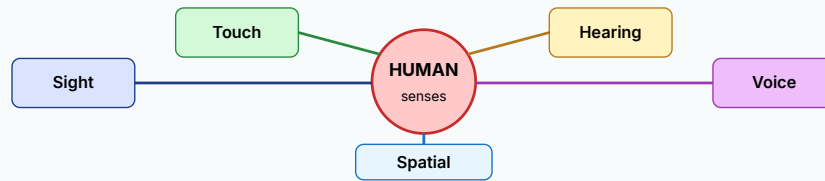
★ 5 Golden · 16 cards · Applications of HCI (Domains of Life) · Importance of HCI · Human-Computer Interaction Problems · User Interface Design (UI vs UX) · Testing Methods (Usability, A/B, Automated)

1.1 Fundamentals of Human-Computer Interaction

Human-Computer Interaction (HCI) is the study and design of technology so that it serves people.

- An HCI system is easy to use, efficient, and responsive to human needs.
- HCI is also known as CHI (Computer-Human Interface) or MMI (Man-Machine Interaction).
- The three names mean the same idea: the meeting point of a person and a machine.

Channel	How humans use it	How computers respond
Sight (بصارت)	Look at screens, graphics or videos.	Displays visuals on monitors and screens.
Touch (چھونا)	Tap, type or swipe.	Responds with clicks or vibrations (haptics).
Hearing (سننا)	Listen to sounds or voices.	Provides audio feedback via speakers.
Voice (آواز)	Speak commands (for example, "Alexa").	Processes speech and replies.
Spatial (مکانی)	Move in physical space.	Tracks position or body movement (for example, VR).



Five sensory channels meet at the human. HCI design must support the channels the user actually needs.

Remember: Remember the three names: HCI = CHI = MMI. Think "Human meets Machine." Five channels: S-T-H-V-S — Sight, Touch, Hearing, Voice, Spatial. "See The Handy Voice..."

بیومن کمپیوٹر انٹرایکشن (HCI) کا مطلب ہے ٹیکنالوجی کو اس طرح ڈیزائن کرنا کہ وہ انسانوں کے لیے فائدہ مند، آسان اور مؤثر ہو۔

1.1.1 Traditional Interaction and 1.1.2 Natural Interaction

1.1.1 Traditional interaction is the basic way people use everyday devices.

- Computer: moving a mouse and typing on a keyboard; output via monitor and speakers.
- Smartphone (traditional use): tapping and swiping on a touchscreen; output via screen lighting and vibrations.
- Voice: speaking commands into a device (for example, "Alexa, play a song").
- Body and gesture: facial recognition to unlock a phone, or playing VR games by waving hands.

Feature	Traditional	Natural
Main input	Mouse, keyboard, keypad, simple tap/swipe	Voice, face, hand/body gesture, movement
Feels like	Operating a machine	Talking or moving as you do with people
Examples	TV remote, ATM keypad, school-portal login typed on a PC	FaceID, Siri, Kinect games, "Hey Google"
Typical output	Screen, speakers, vibration	Spoken reply, unlock, tracking of the body in VR

Remember: Test sentence: If you must press keys or tap labelled buttons, it is traditional. If the machine understands speech, face or body movement, it is natural.

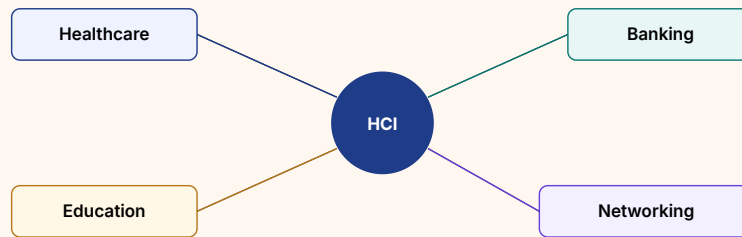
روایتی تعامل (Traditional Interaction) میں ماؤس، کی بورڈ، کی پیڈ یا عام ٹیپ اور سوائپ استعمال ہوتے ہیں، جیسے اے ٹی ایم کی پیڈ۔

1.2 Applications of HCI (Domains of Life) ★ Golden

HCI is used to design and develop interactive systems such as desktop applications, websites and mobile apps.

- These are human-centred solutions built to meet user needs across many domains of life.
- A domain is an area of daily activity — health, money, school, talking to people.
- When you design for a domain, you design for that domain's users, tasks and risks.

Domain	Application examples	User benefit
Health care	Patient portals, telemedicine apps, assistive technologies	Improves access to medical services remotely
Banking	Mobile banking apps, secure ATM interfaces	Enables quick transactions and secure financial management
Education	E-learning platforms (LMS), online exam portals	Enhances teaching, learning and student engagement
Networking and communication	Social media, messaging apps (WhatsApp), Zoom	Connects people globally through collaboration tools



HCI sits at the centre of four everyday domains: healthcare, banking, education and networking.

Remember: Exam mnemonic H-BEN: Health, Banking, Education, Networking. For each, write: one application + one user benefit. That structure scores full marks on...

Exam: Very common 5–8 mark question: "Explain applications of HCI in different domains of life with examples." Draw a four-row table (Health care, Banking, Education,...

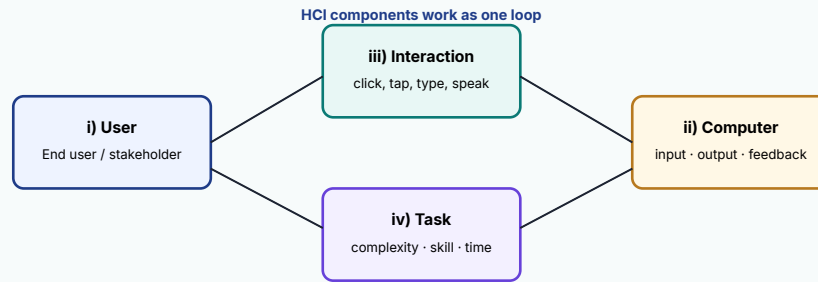
ایچ سی آئی کی ایپلیکیشنز صحت، بینکنگ، تعلیم اور نیٹ ورکنگ کے شعبوں میں انسانی ضروریات پوری کرتی ہیں۔

1.3 Components of HCI

HCI relies on several primary components that work together.

- If one component is weak, the whole interaction feels broken.
- The first four are the people, the machine, the conversation, and the job to be done.

	Component	Meaning
i	User (صارف)	Any individual or group that interacts with a computer system to achieve defined goals. Two categories: end users (people who directly operate the system, from beginners to experts) and stakeholders (people with an interest in the outcome, such as managers and clients).
ii	Computer system (کمپیوٹر سسٹم)	The platform where humans and computers communicate through input, output and feedback using an interface. It connects human actions with machine responses.
iii	User interaction (صارف کا تعامل)	The process through which a user communicates with a system by performing actions (clicking, tapping, typing, speaking). It transforms user intent into system responses.
iv	Task (کام)	The specific action users perform (for example, booking a ticket). It has three characteristics: task complexity (filling a form vs designing a 3D model), user knowledge (basic typing vs advanced programming), and time requirement (ATM withdrawal vs online passport application).



The user performs a task through interaction with a computer system. All four parts must fit together.

Remember: U-C-I-T: User, Computer system, Interaction, Task. "U Can Interact with Tasks." Then add three task questions: How hard? What skill? How long?

ایچ سی آئی کے چار بنیادی اجزاء ہیں: صارف (اینڈ یوزر جو خود سسٹم چلاتا ہے اور اسٹیک ہولڈر جس کا نتیجے میں مفاد ہے)، کمپیوٹر سسٹم (ا۔۔۔)

Types of User Interaction

User interaction can happen in many different ways depending on the technology used.

- Learn every type with one clean example — board questions often ask you to "list and explain" five or more.

Remember: Group them: hands on screen (direct, gesture, haptic) · words (command, conversational, voice) · lists and blanks (menu, form) · automatic/body (context-aware,...

تعامل کی اقسام میں براہ راست بھرا پھیری، کمانڈ، مینو، فارم، گفتگو (چیٹ بوٹ)، اشارہ، آواز، سیاق و سباق سے آگاہ نظام، بایومیٹرک، بی۔۔۔

Interface Types, Environment and Feedback

Components v–vii complete the HCI picture: the look of the controls, the place where you use them, and the signals that tell you "it worked" or "try again".

- Graphical User Interface (GUI): visual elements such as windows, icons and menus (Windows OS, Android).
- Command-Line Interface (CLI): text-based typing (MS-DOS, Linux terminal).
- Touch interface: tapping, swiping, pinching (smartphones, tablets).
- Voice User Interface (VUI): spoken commands (smart speakers).

Remember: GUI vs CLI vs NUI: GUI = pictures you click. CLI = words you type. NUI = body you move. VUI = voice you speak. Touch = fingers on glass. Feedback rule: system...

یوزر انٹرفیس وہ واسطہ ہے جس سے انسان سسٹم سے بات کرتا ہے: GUI تصاویر اور مینو، CLI لکھی کمانڈ، ٹچ، آواز (VUI) اور قدرتی اشارے (NUI)۔

1.4 Importance of HCI ★ Golden

HCI ensures that machines are approachable and usable, turning complex technology into everyday tools.

- Good design makes technology simple and enjoyable (touchscreens).
- Poor design causes confusion and risks (unclear buttons on medical devices).
- This is why HCI is not optional decoration.

Implication	Description
1. Enhance productivity	Well-designed interfaces help users work faster, make fewer mistakes, and achieve goals efficiently.
2. Promotes inclusivity	Accessibility features (screen readers, voice commands) empower marginalized groups and ensure equal use of technology for everyone.
3. Drives innovation	Emerging tech such as AR/VR, AI and IoT rely strictly on solid HCI principles for people to actually adopt and use them.
4. Economic impact	User-friendly products gain higher adoption rates, customer loyalty and market success, driving profits.
5. Ethical responsibility	Responsible design ensures technology respects user privacy, reduces bias and supports human well-being.

Remember: P-I-I-E-E: Productivity, Inclusivity, Innovation, Economic impact, Ethics. Sentence: "Please Include Innovative, Ethical Earnings."

Exam: Typical wording: "Why is HCI important? Discuss its implications." (8 marks) Write a short opening (good design vs poor design) then five numbered points with one...

اچھا ایچ سی آئی پیچیدہ ٹیکنالوجی کو روزمرہ کا آسان اوزار بناتا ہے، برا ڈیزائن الجھن اور خطرہ پیدا کرتا ہے، جیسے طبی آلے پر غیر واضح بٹن۔

1.5 Accessibility Principles

Accessibility principles are HCI guidelines that ensure digital systems are usable by all individuals, including people with diverse physical, sensory and cognitive abilities.

- They remove barriers to create equal access.
- Accessibility is not a "special extra".
- It is how you make sure a student with low vision, a shopkeeper who cannot use a mouse, or a colour-blind driver can still complete the...

Principle	What it means	Example
1. High contrast (واضح رنگ)	Make text and pictures stand out clearly against the background.	Black text on a white background in a railway app, readable in bright sunlight.
2. Readable fonts (پڑھنے میں آسان تحریر)	Clear, simple fonts with proper size and spacing.	A school website using large Calibri so homework instructions are easy to read.
3. Colour-blind friendly design (رنگوں کی پہچان)	Do not depend <i>only</i> on colours; also use symbols or text labels.	A traffic app showing a red circle with a "STOP" label, not just a red colour.
4. Inclusive navigation (آسان نیویگیشن)	Give users different ways to move around a system.	A website that lets you use the Tab key to jump between buttons if you cannot use a mouse.
5. Multimodal detection (متعدد حسی ذرائع)	Present information in different formats (sight, sound, touch).	A video with text captions, or a phone vibrating to alert a user.

Remember: C-F-C-N-M: Contrast, Fonts, Colour-blind, Navigation, Multimodal. "Can Friends Catch New Messages?"

رسائی پذیری (Accessibility) کے اصول یقینی بناتے ہیں کہ ہر انسان سسٹم استعمال کر سکے، بشمول مختلف جسمانی، حسی اور ذہنی صلاحیتوں والے افراد۔

1.6 Need Analysis for Interface Design

Need analysis is the process of identifying the gap between the existing situation and the desired outcome.

1. Identify the users → 2. Collect information → 3. Analyse needs → 4. Prioritise → 5. Develop solutions

- In HCI it is done before designing a UI, so that you understand users' goals, tasks, abilities and expectations.
- If you skip need analysis, you guess.
- Guessing is how a school portal ends up with 40 menus and no "Download date sheet" button on the home screen.

Remember: Need analysis = gap finding. Write: Existing = what hurts now. Desired = what "done" looks like. The UI must close that gap, not decorate it.

نیڈ اینالیسس موجودہ صورتحال اور مطلوبہ نتیجے کے فرق کو یو آئی بنانے سے پہلے تلاش کرتا ہے۔

1.7 Human-Computer Interaction Problems ★ Golden

HCI problems are common difficulties that stop users from using technology smoothly.

- They happen when systems fail to meet human needs or cognitive limits.
- Cognitive limits means the mind can only hold and process so much at once — if an app asks you to remember three SMS codes at the same...
- Board answers score well when you name the problem, explain it in one line, and give a concrete example (the source examples are safe; a...

Problem	Explanation and example
Interface complexity	Too many options or a confusing layout. Example: a shopping app with so many menus you cannot find the checkout button.
Poor feedback	The system does not show if an action worked or failed. Example: clicking Submit and the screen freezes with no confirmation.
Inconsistency	Different parts of the app look or act differently. Example: a magnifying-glass icon for Search on one page, a binocular icon on another.
Accessibility gaps	Design ignores people with disabilities. Example: a video tutorial platform without captions for deaf users.
Compatibility issues	Interface fails on different devices or browsers. Example: a website that works on a laptop but crashes on a mobile browser.
Contextual neglect	The system does not adjust to the user's situation. Example: a map app with no dark mode, blinding the driver at night.
High cognitive load	Users must memorize too much or follow overly complex steps. Example: a registration system forcing you to remember three different SMS codes at once.
Trust deficiency	The system feels unsafe or hides security cues. Example: an online store asking for credit cards but lacking an SSL/HTTPS lock icon.

Remember: Eight problems in two families: (1) Screen problems — complexity, inconsistency, accessibility, compatibility, context. (2) Mind-and-trust problems — poor...

Exam: Common 8-mark question: "Describe problems of HCI with examples." Aim for at least six of the eight, each in the pattern Name → one-line meaning → example. Memorise the...

ایچ سی آئی مسائل وہ رکاوٹیں ہیں جو سسٹم کو انسانی ضرورت یا ذہنی حد کے مطابق نہ ہونے پر پیدا ہوتی ہیں۔

1.8 Methods for Improving HCI

Once you can name an HCI problem, you fix it with a matching method.

- Enhance usability: simplify navigation (a clear "Buy Now" button).
- Strengthen accessibility: support disabilities (captions on videos).
- Ensure consistency: keep layouts uniform (the same "Save" button everywhere).
- Provide clear feedback: show action results ("Payment Successful" popup).

Problem (1.7)	Method (1.8)
Cannot find "Send Money"	Enhance usability — one clear button
No captions on help video	Strengthen accessibility
Search icon changes from page to page	Ensure consistency
Submit freezes with no message	Clear feedback — "Payment Successful"
Portal crashes on mobile	Improve adaptability — responsive layout
Built features nobody asked for	UCD — interview, then pilot-test
Elderly user cannot type a long number	Multimodal — voice + touch
Complaints ignored after launch	Continuous testing and updates

Remember: Think "problem → method" as a pair. Complexity ↔ usability. No captions ↔ accessibility. Different icons ↔ consistency. Freeze ↔ feedback. Phone crash ↔...

ایچ سی آئی بہتر بنانے کے طریقے: نیویگیشن آسان کرنا، رسائی بڑھانا (کیپشن)، ایک جیسا لے آؤٹ، واضح فیڈبیک ("ادائیگی کامیاب")، ریسپانس...

1.9 User Interface Design (UI vs UX) ★ Golden

UI design is the process of creating and organizing the visual elements of a software application so users can interact with it easily and efficiently.

- 1.9.1 Components of a user interface ComponentRole ButtonsClickable elements that perform actions (Submit, Cancel).
- MenusLists of options for navigating the system.
- IconsSmall symbols representing tools or actions.

Component	Role
Buttons	Clickable elements that perform actions (Submit, Cancel).
Menus	Lists of options for navigating the system.
Icons	Small symbols representing tools or actions.
Text fields	Boxes where users type information (passwords, search).
Checkboxes / radio buttons	Selecting options in a form. Checkboxes allow several choices; radio buttons allow only one in a group.
Sliders	Adjusting values along a bar (volume, brightness).

Remember: Clothes vs day out. UI is the outfit (colours, buttons, icons). UX is the whole day (Did you arrive on time? Did you get wet? Were you stressed?). Examiners...

Exam: Highest-yield short question in the chapter: "Differentiate between UI and UX with an example." Write: (1) the two definitions, (2) the sentence "UI is what you see; UX..."

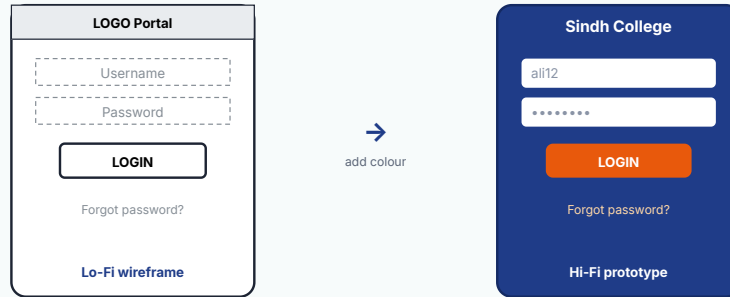
یو آئی ڈیزائن بصری عناصر — بٹن، مینو، آئیکن، ٹیکسٹ فیلڈ، چیک باکس/ریڈیو، سلائیڈر — کو ترتیب دینا ہے۔

1.9.3–1.9.5 Wireframing, Figma and Prototypes

1.9.3 Wireframing is creating a basic visual outline or blueprint of a user interface before the final design.

Lo-Fi sketch → Add details → Hi-Fi prototype

- It plans the structure without focusing on colours or detailed graphics.
- Low-fidelity wireframes (Lo-Fi) A simple black-and-white sketch showing basic layout (boxes and lines).
- Used early to test and share ideas quickly.



A school-portal login: Lo-Fi boxes and labels on the left; a coloured, realistic Hi-Fi prototype on the right.

Remember: Lo-Fi = boxes. Hi-Fi = almost the real app. Prototype = clickable. Figma can hold all three stages. Trigger = what the user does. Action = where the fake app...

وائر فریم یو آئی کا بنیادی خاکہ ہے، رنگوں سے پہلے۔

1.11 Test and Evaluate HCI (Evaluation Methods)

Testing checks if the product works properly (finding bugs).

- Evaluation determines whether it meets users' needs (usability).
- Together they improve the product before release.
- A school portal can be "bug-free" (testing passed) and still unusable if students cannot find the date sheet (evaluation failed).

Method	Description
SUS (System Usability Scale)	A standard 10-item questionnaire used to calculate a score for how usable a system is.
NPS (Net Promoter Score)	Asks users: "How likely are you to recommend this app to a friend?" Measures user loyalty.
Formative evaluation	Done during development to collect early feedback and improve the design continuously.
Summative evaluation	Done at the end of development to measure overall success against the original goals.
Expert reviews	Usability specialists review the app against established design standards (heuristics).

Remember: Formative = forming it (during). Summative = summing it up (end). SUS = 10 questions on usability. NPS = "Would you recommend it?" Expert = specialist vs...

ٹیسٹنگ یہ دیکھتی ہے کہ پروڈکٹ ٹھیک چلتا ہے (بگز)؛ ایوبلیو ایشن یہ دیکھتی ہے کہ صارف کی ضرورت پوری ہوئی۔

1.11.2 Testing Methods (Usability, A/B, Automated) ★ Golden

Testing involves observing how the system behaves under specific conditions.

- Usability testing: observing real users while they perform realistic tasks, to measure efficiency and satisfaction. If the user gets stuck,...
- Task-based testing: assigning a specific task (for example, "Find a product and add it to cart") and strictly measuring completion time,...
- Automated audits: software bots scan websites to detect technical issues such as broken links, slow loading speeds or low colour contrast.
- A/B testing: comparing two versions of an interface (A and B). Half the users see A, half see B. The one with more clicks or sign-ups wins.

Task-based "Add to cart" Time: 12 s Success: Yes	A/B test Version A Version B half / half users more clicks wins	Automated audit Bot scanning... 3 broken links Low contrast
--	---	---

Three golden testing pictures: timed tasks, A versus B, and a bot that finds broken links and weak contrast.

Remember: Task-based three numbers: Time, Success, Errors. A/B: two versions, split users, the higher clicks/sign-ups win. Bot: links, speed, contrast. Load: many users...

Exam: Golden 8-mark question: "Explain testing methods used in HCI." Write seven short paragraphs or a seven-row table. For full marks, define usability testing (observe real...

سات ٹیسٹنگ طریقے: یوزیبلٹی (حقیقی صارف کو کام کرتے دیکھنا)، ٹاسک بیسڈ (وقت، کامیابی، غلطیاں ناپنا)، خودکار آڈٹ (بوت ٹوٹے لنک، مسست...

Career Opportunities in UI/UX

The same HCI skills you practised on ATMs, JazzCash and school portals are real jobs.

- Each role uses a different slice of this chapter.

Career	What the person does
UX designer	Researches user needs to ensure the app is intuitive and satisfying.
UI designer	Designs the visual layout (colours, typography, buttons) using tools such as Figma.
Interaction designer	Focuses on the flow between screens, animations and gestures.
Usability researcher	Runs SUS/NPS tests and A/B tests with real users to find flaws.
Accessibility specialist	Ensures products meet inclusive standards for people with disabilities.

Remember: UI looks; UX asks; interaction connects screens; research measures; accessibility includes everyone.

اس باب سے ملازمتیں: یو ایکس ڈیزائنر صارف کی ضرورت تحقیق کرتا ہے، یو آئی ڈیزائنر فگما میں رنگ اور بٹن بناتا ہے، انٹریکشن ڈیزائنر اس...

Key terms

Term	Meaning	اردو
HCI	Human-Computer Interaction; designing technology that is easy, efficient and responsive to people. Also called CHI or MMI.	انسان اور کمپیوٹر کا باہمی تعامل
Haptics	Physical feedback you can feel, such as a vibration.	لمسی / ارتعاشی جواب
Traditional interaction	Using standard input/output such as mouse, keyboard, keypad or simple tap and swipe.	روایتی تعامل
Natural interaction	Interacting without mouse or keyboard, using voice, face or body gesture.	قدرتی تعامل
End user	A person who directly operates the system.	حتمی صارف
Stakeholder	A person or group with an interest in the system's outcome (manager, client).	فریق مفاد
GUI	Graphical User Interface using windows, icons and menus.	تصویری صارف انٹرفیس
CLI	Command-Line Interface using typed text commands.	کمانڈ لائن انٹرفیس
NUI	Natural User Interface using gestures, body movement or immersive environments (VR, Kinect).	قدرتی صارف انٹرفیس
System feedback	The computer's response that confirms or guides an action.	نظام کا جواب
User feedback	Input the user sends to the system (Submit click, star rating).	صارف کی رائے / ان پٹ
Accessibility	Making digital systems usable by all, including people with diverse abilities.	رسائی پذیری
Need analysis	Finding the gap between the current situation and the desired outcome before designing a UI.	ضروریات کا تجزیہ
Cognitive load	How much the user must remember or process at once.	ذہنی بوجھ
UCD	User-Centred Design: focus on real user goals and pilot-test.	صارف مرکز ڈیزائن
UI	The visual elements of an application (buttons, menus, icons, fields, sliders).	صارف انٹرفیس

Term	Meaning	اردو
UX	The overall smoothness, comfort and satisfaction of using a system.	صارف کا تجربہ
Wireframe	A basic outline of a UI that plans structure without colours or detailed graphics.	انٹرفیس کا خاکہ
Lo-Fi	Low-fidelity sketch of boxes and lines used early to test ideas.	کم تفصیل والا خاکہ
Hi-Fi	High-fidelity prototype that looks almost like the final product.	زیادہ تفصیل والا نمونہ
Prototype	An early interactive model used to test design ideas before writing code.	ابتدائی چلنے والا نمونہ
Figma	A free, web-based tool for UI design and prototypes.	فگما (ویب پر یو آئی ٹول)
SUS	System Usability Scale: a 10-item questionnaire that scores how usable a system is.	نظام کی استعمال پذیری کا پیمانہ
NPS	Net Promoter Score: how likely a user is to recommend the app to a friend.	تجویز کا درجہ (وفاداری)
Formative evaluation	Evaluation during development to improve the design continuously.	دورانِ تعمیر جانچ
Summative evaluation	Evaluation at the end, measured against original goals.	اختتامی جانچ
A/B testing	Showing version A to half the users and version B to the other half; the better-performing version wins.	اے/بی موازنہ ٹیسٹ
Automated audit	Bots scanning for broken links, slow pages or low contrast.	خودکار جانچ
Heuristic	An established design standard used in expert reviews.	مستند ڈیزائن اصول

Chapter in one look

- **HCI** (also CHI, MMI) designs technology to be easy, efficient and responsive to people, using five sensory channels: sight, touch (haptics), hearing, voice and spatial.
- **Traditional interaction** uses mouse, keyboard, keypad or simple tap/swipe; **natural interaction** uses voice, face or body gesture without a mouse or keyboard.
- HCI applications cover **health care, banking, education, and networking/communication**, each with a clear user benefit (remote care, quick secure money, better learning, global connection).
- Four core components: **user** (end user vs stakeholder), **computer system**, **user interaction**, and **task** (complexity, knowledge, time).
- Interaction types include direct manipulation, command-based, menu-driven, form fill, conversational, gesture, voice, context-aware, biometric, haptic and multimodal.
- Interfaces: GUI, CLI, touch, VUI, NUI. **Environment** is physical, social and technical. **System feedback** is the machine's reply; **user feedback** is what the person sends in.
- HCI matters for productivity, inclusivity, innovation, economic impact and ethical responsibility. Poor design (unclear medical buttons) creates risk.
- Five accessibility principles: high contrast, readable fonts, colour-blind friendly labels, inclusive navigation, multimodal presentation.
- **Need analysis** finds the gap before UI design: identify users, collect data, analyse, prioritise, develop solutions (banking case: FaceID, large buttons, local language).
- Eight HCI problems: complexity, poor feedback, inconsistency, accessibility gaps, compatibility, contextual neglect, high cognitive load, trust deficiency.
- Improve HCI with usability, accessibility, consistency, clear feedback, adaptability (responsive design), UCD and pilot tests, multimodal input, and continuous testing.
- **UI** is the visual components (buttons, menus, icons, fields, checkboxes, sliders); **UX** is how the whole experience feels. UI is what you see; UX is how it feels.
- Lo-Fi wireframes plan structure; Hi-Fi prototypes look real; **Figma** is a free browser tool; prototypes use Trigger (On Tap) and Action (navigate) before code.
- Evaluation (SUS, NPS, formative, summative, expert/heuristic reviews) checks needs; testing (usability, task-based, automated, A/B, compatibility, load, keyboard/screen-reader) checks behaviour under conditions.

CHAPTER 2

Computational Thinking & Algorithms

★ 6 Golden · 24 cards · Trace Tables · Comparing Trace Tables and Stepwise Reasoning · Number of Steps (Big O Notation) · Concept of Data Structure · Linear Data Structures: Stack · Linear Data Structures: Queue

2.1 Analyze algorithms for correctness

An algorithm is correct if its output matches the expected results for a given input.

Trace tables → **paper execution**, **line by line** → **Stepwise reasoning** → **logic blocks**, **overall state**

- Analysing correctness means showing that the algorithm produces the intended output for all valid inputs, not just one lucky example.
- Two methods are used in this course:

Remember: Remember: runs ≠ correct. A logical error still produces output — just the wrong output. That is why we analyse correctness on paper before we trust the code.

الگورتھم تب درست (Correct) سمجھا جاتا ہے جب دیے گئے ان پٹ پر اس کا آؤٹ پٹ وہی ہو جو ہونا چاہیے۔

2.1.1 Trace Tables ★ Golden

A trace table is a technique used to monitor how the values of variables change during the execution of an algorithm.

- Dry running: execute the algorithm line by line by hand.
- Debugging: find logical errors where the algorithm runs but the output is wrong.
- Validation: check that the algorithm does what it is intended to do.
- Line / step number — which line is being executed.

```
AllHeights = [1.91, 2.01, 1.87, 1.57, 1.48]
Total = 0
for height in AllHeights:
    if height ≥ 1.57 and height ≤ 1.91:
        Total = Total + 1
print(Total)
```

Remember: How to fill a trace table without getting lost. Write a new row every time a variable changes or a condition is tested. Copy unchanged values into the new row...

Exam: Very common long question: "Dry-run the given algorithm using a trace table and state the output." Draw columns for every variable plus a condition column. Show the loop...

ٹریس ٹیبل الگورتھم کا کاغذی اجرا (Dry Run) ہے۔

2.1.2 Stepwise Reasoning

Stepwise reasoning (also called logic verification) breaks an algorithm into logical sections and checks the state of the algorithm at each stage.

- Analyse each step of the algorithm.
- Check: does it start in a valid state?
- Does it handle all possible cases?
- Are the conditions correct?

Check	What you verify
Initial state	Sum = 0 before the loop. Valid: nothing has been added yet.
Loop boundary	The loop runs exactly from $i = 1$ to $i = N$. No missing number, no extra number.
Iteration	Each i is added once: after k iterations, $\text{Sum} = 1+2+\dots+k$.
Final state	When the loop ends, Sum holds $1+2+\dots+N$, which is the required total.
Edge case $N = 0$	The loop body never runs, Sum stays 0. Correct (the sum of no numbers is 0).
Check case $N = 5$	Sum becomes 15, which matches $5 \times 6 / 2 = 15$.

```
N = 5
Sum = 0
for i in range(1, N + 1):
    Sum = Sum + i
print(Sum)
```

Remember: Think of stepwise reasoning as a teacher checking a proof, not as a cashier counting coins one by one. You ask "is this block always true?" instead of "what is..."

مرحلہ وار استدلال میں الگورتھم کو منطقی حصوں میں توڑ کر ہر مرحلے پر اس کی حالت (State) دیکھی جاتی ہے: کیا شروع درست ہے، کیا تمام ص...

2.1.3 Comparing Trace Tables and Stepwise Reasoning ★ Golden

Both methods verify correctness, but they fit different situations depending on the complexity and length of the algorithm.

Feature	Trace table	Stepwise reasoning
Best for	Small algorithm blocks, complex loops, and beginners learning to track variables.	Large systems, architectural logic, and abstract mathematical proofs.
Effort required	High. Very tedious and impractical if a loop runs 1,000 times.	Moderate. Needs strong logical deduction and abstract thinking.
Primary goal	Catching syntax misinterpretations or specific arithmetic errors line by line.	Catching conceptual or structural flaws in the overall design.
Level	Micro-level (line by line). Checks exact values.	Macro-level (logic blocks). Checks overall state.

Trace table

Micro-level
Line by line
Exact values

Stepwise reasoning

Macro-level
Logic blocks
Overall state

Trace tables zoom in on values; stepwise reasoning zooms out on design.

Remember: Micro vs macro. Trace table = microscope (every cell). Stepwise reasoning = map (whole city). Beginners start with the microscope; designers need the map.

Exam: Differentiate questions are almost guaranteed: "Compare a trace table with stepwise reasoning." Write a four-row table (best for, effort, primary goal, micro/macro). Add...

ٹریس ٹیبل خرد بینی (Micro) سطح پر کام کرتا ہے: ہر لائن اور ہر قیمت۔

2.2 Evaluating the Clarity of an Algorithm

Clarity means an algorithm is easy to read, understand and modify.

Clarity → Modularity (parts) → Readability (text and flow)

- In many cases, evaluating clarity is even more important than making the algorithm slightly faster.
- A clear algorithm is easier to maintain, debug and extend over time.
- The two key aspects of clarity are:

Remember: Ask two questions: "Can I split this into parts?" (modularity) and "Can another student read it aloud and understand it?" (readability).

وضاحت (Clarity) کا مطلب ہے الگورتھم پڑھنا، سمجھنا اور بدلنا آسان ہو۔

2.2.1 Modularity

Modularity means breaking an algorithm into smaller, independent parts such as functions, classes or modules.

- Single Responsibility Principle (SRP). Each module should focus on one specific task. If a function calculates a sum, prints a report and...
- Coupling. Modules should be reusable in different contexts. Good modularity means loose coupling: a change in one part does not smash the...
- Abstraction. A well-designed algorithm hides complex implementation details. You can call a sorting function without knowing whether it...

Remember: SRP in one line: one function, one job. If you cannot name the function with a short verb-phrase (CalculateSum, FindAverage), it is doing too much.

ماڈیولریتی کا مطلب الگورتھم کو آزاد حصوں (فنکشن، کلاس، ماڈیول) میں توڑنا ہے۔

2.2.2 Readability

Readability is how easily another person can understand the purpose and logic of an algorithm without running it.

Names → Self-doc → Simple flow → Consistent

- Naming conventions — consistent rules for names. Good names make an algorithm easier to search and understand. Bad: $v = d / t$ Good:...
- Self-documentation — the algorithm explains itself, so you need fewer comments. Meaningful names and clean formatting make the purpose...
- Control-flow simplicity — keep the logic straightforward. Avoid deep nesting (IF inside IF inside IF), limit extra conditions, and keep...
- Consistency — one style throughout: indentation, spacing, and casing (camelCase or snake_case). Then the logic stands out instead of the...

```
# Hard to read
d = 100
t = 4
s = 20
v = d / t
if v > s:
    if t > 0:
        print(v)
```

Remember: Remember readability as N-S-S-C: Names, Self-documentation, Simple flow, Consistency.

پڑھائی پن یہ ہے کہ دوسرا شخص الگورتھم چلائے بغیر اس کا مقصد سمجھ لے۔

2.3 Assess Algorithm Efficiency

Algorithm efficiency is measured in terms of time and space — how fast it runs and how much memory it uses.

Time (speed) → Space (memory) → Efficiency

- Number of steps (described with Big O notation).
- Number of conditions (IF decisions).
- Number of repetitions (loops).

Remember: Efficiency is always “as the input grows”. An algorithm that feels instant on 10 items can freeze on 10 million items if it is $O(n^2)$.

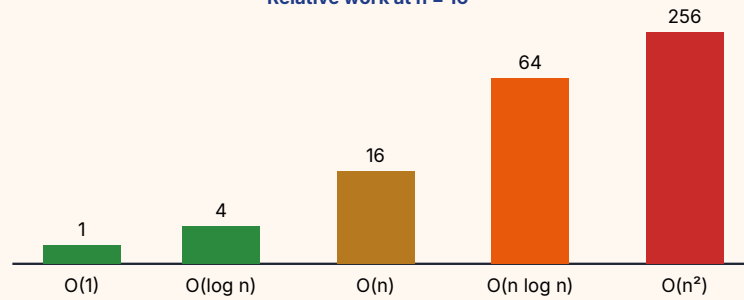
کارکردگی (Efficiency) وقت (رفتار) اور جگہ (میموری) سے ناپی جاتی ہے۔

2.3.1 Number of Steps (Big O Notation) ★ Golden

When we discuss efficiency we ask how performance changes as the input size n increases.

- Big O notation is a mathematical method used to describe how the running time (or memory) grows as n grows.
- We ignore small constants and keep the most important term: if an algorithm does $3n + 5$ steps we write $O(n)$.
- In computer science, $\log n$ almost always means log base 2 (how many times you can halve n until you reach 1).

Class	Name	What happens when n grows	Typical example	Grade
$O(1)$	Constant	Time stays the same for 10 items or 10 million.	Access <code>arr[0]</code> (or any known index).	Excellent
$O(\log n)$	Logarithmic	Time grows very slowly; the problem is halved each step.	Binary search on a sorted list.	Great
$O(n)$	Linear	If n doubles, time roughly doubles.	Linear search of an unsorted list.	Fair / Good
$O(n \log n)$	Linearithmic	A little worse than linear; much better than n^2 . Common in good sorting.	Merge Sort (split in half, then merge).	Good
$O(n^2)$	Quadratic	Time grows with the square of n . Nested loops. Painful for large n .	Bubble Sort; compare every item with every other item.	Poor

Relative work at $n = 16$ 

Same $n = 16$. Quadratic work is already four times $n \log n$ and sixteen times linear.

```
arr = [23, 76, 12, 90, 34, 68]
print(arr[0])
```

Remember: Order from fastest to slowest (for large n): $O(1) \rightarrow O(\log n) \rightarrow O(n) \rightarrow O(n \log n) \rightarrow O(n^2)$. Mnemonic: "Constant Logs Never Need Squares" — Constant, Logarithmic,...

Exam: Memorise the five rows of the class table (name, growth, one example, grade). Typical MCQ: "Accessing `arr[0]` is ...?" $\rightarrow O(1)$. "Binary search is ...?" $\rightarrow O(\log n)$. "Nested..."

بگ او (Big O) بتاتا ہے کہ ان پٹ n بڑھنے پر وقت یا میموری کیسے بڑھتی ہے۔

2.3.2 Number of Conditions

Conditions (IF statements) are decision points.

- Computational cost. Each IF is extra work, even if the rest of the algorithm is simple.
- Nested conditions. IF inside IF often means a brute-force approach. It hurts both readability and speed.
- Short-circuiting (the optimisation goal). Stop evaluating as soon as the result is known. In a logical AND, if the first part is False the...

```
x = 0
y = 10
if x != 0 and (y / x) > 2:
    print("ratio is large")
else:
    print("skipped safely")
```

Remember: Put the cheapest / most likely False test first in an AND, and the most likely True test first in an OR. That is short-circuiting used on purpose.

ہر IF پروسیسر سے True/False پوچھتی ہے اس لیے زیادہ شاخیں کارکردگی گھٹاتی ہیں۔

2.3.3 Number of Repetitions

Repetitions (loops) are the most significant factor in algorithm efficiency.

- The structure of the loop decides how time grows with n .

Type of repetition	Conceptual impact	Efficiency grade
Linear loop	Processes each element exactly once. Time grows in proportion to size.	Good — $O(n)$
Logarithmic repetition	Reduces the problem size by half each iteration (binary search). Huge time savings.	Excellent — $O(\log n)$
Nested loops	For each element, process all other elements. Operations explode.	Poor — $O(n^2)$

Remember: Find the deepest loop. One loop $\approx O(n)$. Loop inside a loop $\approx O(n^2)$. A loop that halves $n \approx O(\log n)$.

دہرائی (لوپ) کارکردگی کا سب سے بڑا عامل ہے۔

2.3.4 The Efficiency Evaluation Framework

To evaluate an algorithm at a conceptual level, ask three questions: Does it scale?

Does it scale? → Where is the bottleneck? → Is the extra complexity worth it?

- Does it scale? If n grows 10 times, does time jump 10 times (linear) or 100 times (quadratic)? Quadratic scaling fails for large datasets.
- What is the bottleneck? Focus on the most time-consuming section (usually the deepest loop) and try to improve that part.
- Is the cost worth it? A theoretically faster $O(\log n)$ algorithm can be too complex to code and maintain. For a list of just 50 items, a...

Situation	Scale	Bottleneck	Worth the cost?	Choice
50 unsorted class names	Tiny n	A single linear scan	Binary search needs a sort first, which is extra work and extra code	Linear search $O(n)$
1,000,000 sorted account numbers, searched all day	$n \times 10$ compared with 100,000 is already huge; quadratic would be impossible	The search loop	Yes — binary search is standard and well understood	Binary search $O(\log n)$

Remember: Framework mnemonic: S-B-W — Scale, Bottleneck, Worth it.

کارکردگی کے تین سوال: کیا الگورتھم بڑے n پر چل سکتا ہے (Scale)?

2.4 Refinements to improve clarity and efficiency

Improving an algorithm means reviewing it step by step: simplifying logic, removing redundant work, and picking smart data structures.

- Use meaningful names. Replace vague variables with descriptive ones. Bad: $a = b / c$. Good: $\text{average} = \text{totalSum} / \text{totalItems}$.
- Break into modules. Divide a massive block into functions, for example separating `CalculateSum()` from the printing logic.
- Improve structure and indentation. Packed loops and IF statements confuse the reader. Spacing reduces mental effort.
- Add comments. Keep them short. Explain why something is done, not just what is happening, especially for tricky logic.

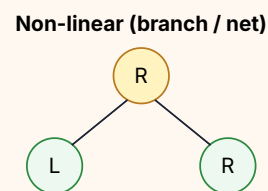
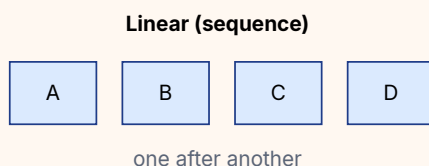
Remember: Clarity: N-M-I-C (Names, Modules, Indentation, Comments). Efficiency: S-A-C-D (Shrink steps, Algorithm choice, Cache, Data structure).

وضاحت بہتر کرنے کے لیے معنی خیز نام، ماڈیول، درست indentation اور مختصر تبصرے استعمال کریں۔

2.5 Concept of Data Structure ★ Golden

A data structure is a systematic way of organising, storing and managing data so that an algorithm can access, modify and process it efficiently.

- Books on a shelf are arranged in a set order, so you can find a title quickly.
- Files in folders are grouped on purpose.
- Data structures do the same job inside computer memory.



Linear: a line of neighbours. Non-linear: a node with more than one connection.

Remember: Ask: "Does each item have at most one next item?" Yes → linear. "Can one item link to many, or can links form a net?" → non-linear.

Exam: Define data structure in full (organise, store, manage; efficient access and processing). Then split linear vs non-linear with the "one predecessor and one successor"...

ڈیٹا سٹرکچر ڈیٹا کو منظم طریقے سے رکھنے کا نظام ہے تاکہ الگورتھم اسے جلدی استعمال کر سکے۔

2.5.1 Linear Data Structures: Array

An array holds a collection of similar items (the same data type) placed next to each other in contiguous memory (one continuous block).

- Each item has a position number called an index, usually starting from 0.
- Because the address of any cell can be calculated from the index, you can jump straight to it.
- That is direct access (also called random access), and it is $O(1)$.

Array name: A

23	76	12	90	34	68
0	1	2	3	4	5

A[3] = 90 (index starts at 0)

Six integers in contiguous cells. Direct access: A[3] jumps to 90 without walking 23, 76 and 12.

```
A = [23, 76, 12, 90, 34, 68]
print(A[3])
print(A[0])
```

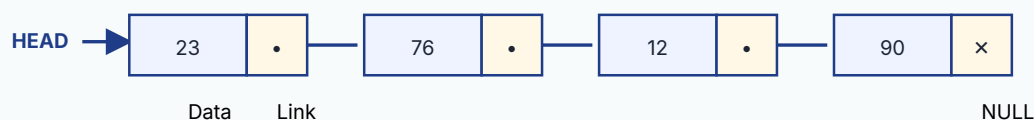
Remember: Index 0 is the first shelf on the row. The third item that humans call "number 4" is index 3. Formula in your head: position from the left, minus one.

ایرے ایک جیسے عناصر کو میموری کے لگاتار بلاک میں رکھتا ہے۔

Linear Data Structures: Linked List

A linked list stores data as a chain. Unlike an array, the items are not one continuous block.

- the actual data;
- a link (pointer) holding the memory address of the next node.



Follow HEAD → 23 → 76 → 12 → 90 → NULL

Four nodes. The last link is NULL, which ends the chain.

```
nodes = [23, 76, 12, 90]
for data in nodes:
    print(data, end=" → ")
print("NULL")
```

Remember: Array = row of lockers with numbers on the doors. Linked list = treasure hunt: each clue (link) tells you where the next box is.

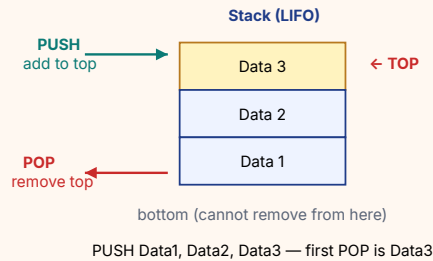
لنکڈ لسٹ زنجیر کی طرح نوڈز پر مشتمل ہے جو میموری میں بکھرے ہو سکتے ہیں۔

Linear Data Structures: Stack ★ Golden

A stack is a linear data structure that works on the LIFO principle: Last-In, First-Out.

- PUSH — add an item to the top.
- POP — remove the item from the top.

Step	Operation	Stack (bottom → top)	TOP	Popped value
0	(empty)	[]	none	
1	PUSH Data1	[Data1]	Data1	
2	PUSH Data2	[Data1, Data2]	Data2	
3	PUSH Data3	[Data1, Data2, Data3]	Data3	
4	POP	[Data1, Data2]	Data2	Data3
5	POP	[Data1]	Data1	Data2



Three PUSH operations. The TOP is Data 3, so the first POP returns Data 3.

```
stack = []
stack.append("Data1")
stack.append("Data2")
stack.append("Data3")
print("after 3 PUSHes:", stack)
print("POP:", stack.pop())
print("POP:", stack.pop())
print("left:", stack)
```

Remember: LIFO = Last In First Out = "last plate washed is first plate used" on a canteen pile. Undo, Redo, Back button and function calls are all stacks.

Exam: Define stack + LIFO, define PUSH and POP, mention TOP, give the Undo example, and draw the vertical boxes with TOP labelled. Trace questions: "Show the stack after PUSH..."

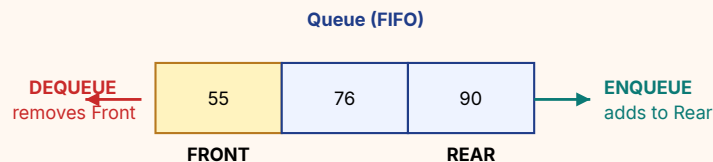
سٹیک LIFO پر چلتا ہے: جو سب سے بعد میں آیا وہ سب سے پہلے نکلے گا۔

Linear Data Structures: Queue ★ Golden

A queue organises data on the FIFO principle: First-In, First-Out.

- ENQUEUE — add an element at the Rear (back).
- DEQUEUE — remove an element from the Front.

Step	Operation	Queue (Front → Rear)	Front	Rear	Removed
0	(empty)	[]	—	—	
1	ENQUEUE 55	[55]	55	55	
2	ENQUEUE 76	[55, 76]	55	76	
3	ENQUEUE 90	[55, 76, 90]	55	90	
4	ENQUEUE D	[55, 76, 90, D]	55	D	
5	DEQUEUE	[76, 90, D]	76	D	55



Next ENQUEUE joins after 90. Next DEQUEUE removes 55.

Three jobs waiting. New jobs join at the rear; service always takes the front.

Remember: FIFO = First In First Out = a canteen line. Front = the person being served. Rear = the person who just joined.

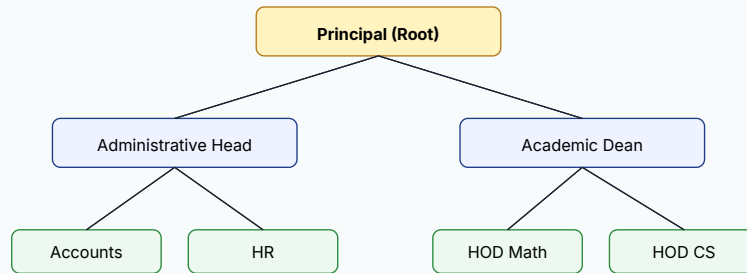
Exam: Define queue + FIFO, define ENQUEUE (rear) and DEQUEUE (front), give the printer example, draw the labelled Front/Rear diagram, and trace a short sequence. Examiners mix...

کیو FIFO پر چلتی ہے: جو پہلے آیا وہ پہلے نکلے۔

2.5.2 Non-Linear Data Structures: Tree

A tree organises data in a hierarchy. Everything branches out from a single starting point called the root.

- Everything branches out from a single starting point called the root.
- It creates a strict one-to-many (parent-child) relationship.
- That is an excellent way to show how roles flow in an organisation.



College tree: one root (Principal), then parent-child branches. No child has two parents in this model.

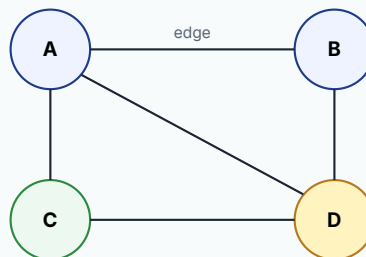
Remember: Tree = family / folders / organogram. One root. Branches down. Parent-child, not a web of equals.

ٹری ڈیٹا کو درجہ بندی (Hierarchy) میں رکھتی ہے جو ایک جڑ (Root) سے پھوٹتی ہے۔

Non-Linear Data Structures: Graph

A graph organises data when things connect in a flexible, complex way.

- Nodes (vertices) — the items (cities, people).
- Edges — the connections (roads, friendships).



Four vertices A, B, C, D. Edges can form a cycle (A-B-D-C-A). A tree cannot do that.

Remember: Tree $\subset$ Graph in shape-talk: a tree is a special graph with one root and no cycles. If you need cycles or many-to-many links, say graph.

گراف لچکدار روابط کے لیے ہے: کوئی سخت درجہ بندی نہیں، کوئی بھی نوڈ کسی سے جڑ سکتا ہے۔

2.5.3 Operations on Data Structures

Whatever the shape, you keep meeting the same five operations:

Operation	Meaning	Quick picture
Insertion	Adding a new item into the structure.	PUSH on a stack; ENQUEUE on a queue; putting 40 into an array.
Deletion	Removing an item that is no longer needed.	POP; DEQUEUE; deleting a node from a list.
Traversal	Going through each item one by one.	Walking HEAD $\rightarrow \dots \rightarrow$ NULL; visiting every tree node.
Searching	Finding a specific item.	Linear search in an array; following links until 87 is found.
Sorting	Arranging items in a proper order (ascending or descending).	Bubble Sort on an array of marks.

Remember: Memorise I-D-T-S-S: Insert, Delete, Traverse, Search, Sort.

ہر ڈیٹا سٹرکچر پر پانچ عمل ہوتے ہیں: داخل کرنا (Insertion)، حذف کرنا (Deletion)، ایک ایک کر کے گزرنا (Traversal)، تلاش (Searching)...

2.6 Identify data structures in problem scenarios

To pick a structure, first see how the data is organised.

- Storing student grades for a class → Array. You usually know how many students there are, and all grades are the same type. Indices give...
- Music playlist / browser history → Linked list. Every song knows what comes next. When one finishes, follow the pointer. Frequent...
- Undo or Redo → Stack (LIFO). Each typing or formatting action is PUSHed. Undo POPs the most recent action.
- Printer task scheduling → Queue (FIFO). Documents print in the exact order they were received. First-come, first-served.

If the data...	Then it is...	Recommended structure
Follows "next" or "previous" logic	Linear	Array, Linked list
Follows "parent" and "child" logic	Non-linear	Tree
Follows "first-come, first-served" logic	Linear	Queue
Follows "connections / relationships" logic	Non-linear	Graph
Needs to be accessed in the reverse order of entry	Linear	Stack

Remember: Decision order: reverse-of-entry? → stack. First-come? → queue. Parent-child? → tree. Many-to-many routes? → graph. Known list with indexes? → array. Lots of...

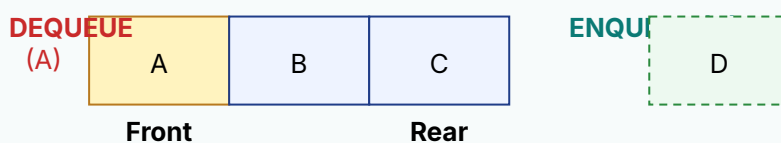
اگر ہر عنصر کا واضح اگلا/پچھلا ہے تو ڈھانچہ لکیری ہے؛ اگر شاخیں یا کئی روابط ہیں تو غیر لکیری۔

2.7 Trace data retrieval in Array, List and Queue

Tracing data retrieval means following the step-by-step path of how an algorithm finds or accesses data.

- Array, linked list and queue are all linear, but the effort to find an item is very different because of how they are organised inside.

Loop iteration	Index	Array access	Value
1	0	Array[0]	10
2	1	Array[1]	50
3	2	Array[2]	20



You cannot pull B until A has left.

Queue A, B, C. ENQUEUE D joins the rear. DEQUEUE must remove A, not C or B.

```
Array = [10, 50, 20, 70, 90, 30]
print("Direct Array[2] =", Array[2])
for i in range(3):
    print(i + 1, i, f"Array[{i}]", Array[i])
```

Remember: Array jumps, list walks, queue waits. If the exam asks "why is array retrieval different?", write direct access by index versus sequential links versus Front...

ایرے میں اگر انڈیکس معلوم ہو تو سیدھی چھلانگ لگتی ہے، جیسے $\text{Array}[2] = 20$ ۔

Career Opportunities in Algorithms

The skills in this chapter — proving correctness, writing clear modules, judging Big O, and choosing data structures — are the daily tools of several computing careers.

Career	What the person does
Software Engineer	Builds and maintains software that relies on data structures and efficient algorithms to solve logic problems.
Backend Developer	Builds server-side logic where choosing the right structure (array, queue) directly affects web performance.
Algorithm Engineer	Specialises in optimising algorithms for search engines, finance or logistics.
Systems Analyst	Evaluates and improves the efficiency and clarity of existing computer systems in an organisation.
Competitive Programmer	Solves complex algorithmic problems under time limits, a common path into top tech roles.

Remember: Every career in the table uses the same three tests you learned: correct, clear, efficient — plus the right data structure.

اس باب کی مہارتیں سافٹ ویئر انجینئر، بیک اینڈ ڈیولپر، الگورتھم انجینئر، سسٹمز اینالسٹ اور کمپیٹیٹو پروگرامر کے کام میں روز استعمال...

Key terms

Term	Meaning	اردو
Correctness	The algorithm's output matches the expected results for valid inputs.	درستگی
Trace table	A paper dry-run table of steps, variables, I/O and conditions.	ٹریس ٹیبل
Dry run	Executing an algorithm by hand, line by line.	دستی اجرا
Stepwise reasoning	Checking each logic block's state; also called logic verification.	مرحلہ وار استدلال
Clarity	How easy an algorithm is to read, understand and modify.	وضاحت
Modularity	Breaking work into independent functions, classes or modules.	ماڈیولریتی
Single Responsibility Principle	Each module should do one specific task.	واحد ذمہ داری کا اصول
Coupling	How tightly modules depend on each other; loose coupling is the goal.	ماڈیولز کی جڑت
Abstraction	Hiding implementation details so a caller need not know the inner method.	تجربہ
Readability	How easily another person understands the logic without running it.	پڑھائی پن
Efficiency	How well an algorithm uses time (speed) and space (memory).	کارکردگی
Big O notation	A description of how running time or memory grows as n grows.	بگ او نوٹیشن
$O(1)$	Constant time; e.g. accessing arr[0].	مستقل وقت
$O(\log n)$	Logarithmic time; the problem is halved each step (binary search).	لا جارتھمک وقت
$O(n)$	Linear time; work grows in proportion to n (linear search).	لکیری وقت
$O(n \log n)$	Work about n times log n (Merge Sort).	این لاگ این وقت
$O(n^2)$	Quadratic time; nested loops (Bubble Sort).	مربع وقت
Bottleneck	The slowest, most time-consuming section, often the deepest loop.	رکاوٹ، سب سے سست حصہ

Term	Meaning	اردو
Short-circuiting	Stopping a boolean evaluation as soon as the result is known.	جلدی ختم ہونے والی شرط
Memoization / caching	Computing a fixed value once and reusing it instead of recalculating.	ایک بار حساب، بار بار استعمال
Data structure	A systematic way to organise, store and manage data for efficient use.	ڈیٹا سٹرکچر
Linear structure	A sequence; each middle item has one predecessor and one successor.	لکیری ڈھانچہ
Array	Same-type items in contiguous memory, accessed by index.	ایرے
Linked list	Nodes of data + pointer, starting at HEAD and ending at NULL.	لنکڈ لسٹ
Stack	LIFO structure: PUSH and POP at the TOP.	سٹیک
Queue	FIFO structure: ENQUEUE at Rear, DEQUEUE at Front.	کیو
Tree	Hierarchical non-linear structure with one root and parent-child links.	ٹری
Graph	Nodes (vertices) joined by edges, with no strict hierarchy.	گراف
Traversal	Visiting each item in a structure one by one.	گزرنا، طواف
Direct access	Jumping to an array index without walking earlier items.	براہ راست رسائی

Chapter in one look

- **Correctness:** output matches expected results for valid inputs. Test with a trace table (manual dry run) or stepwise reasoning (logic verification).
- A **trace table** is micro-level: line, variables, I/O, conditions. Best for small loops. The soldier heights [1.91, 2.01, 1.87, 1.57, 1.48] print Total = 3.
- **Stepwise reasoning** is macro-level: valid start, all cases, correct conditions, loop stays correct. Sum 1..N starts at 0; N = 0 stays 0; N = 5 becomes 15.
- **Clarity** (read, understand, modify) often beats a tiny speed gain. It has two parts: modularity and readability.
- **Modularity:** SRP (one job), loose coupling, abstraction (call Sort without knowing Bubble vs Selection). Split CalculateSum and FindAverage.
- **Readability:** names (`velocity = distance / time`), self-documentation, simple flow, consistent style.
- **Efficiency** = time + space. Drivers: number of steps (Big O), conditions, repetitions.
- **Big O:** $O(1)$ constant (arr[0]), $O(\log n)$ binary search, $O(n)$ linear search, $O(n \log n)$ Merge Sort, $O(n^2)$ nested loops / Bubble Sort. Quadratic fails as n grows.
- **Framework:** Does it scale? What is the bottleneck (deepest loop)? Is the extra complexity worth it? For n = 50, simple $O(n)$ can beat fragile $O(\log n)$.
- **Refinements:** meaningful names, modules, indentation, why-comments; shrink Bubble Sort's inner loop; better algorithms; cache constants; pick the right structure.
- **Data structure:** systematic organising of data. Linear (one after another): array, linked list, stack, queue. Non-linear: tree, graph.
- **Array:** contiguous, index from 0, direct access $O(1)$, shifting on mid-insert. **Linked list:** nodes of data+link, HEAD to NULL, sequential access.
- **Stack LIFO:** PUSH/POP at TOP (Undo). **Queue FIFO:** ENQUEUE rear, DEQUEUE front (printer).
- **Tree:** one root, parent-child (college chart). **Graph:** vertices + edges, no strict hierarchy (GPS, friends).
- Five operations: insert, delete, traverse, search, sort. Choose by next/previous, parent/child, FIFO, relationships, or reverse-of-entry.
- **Retrieval:** array can jump to an index; a list walks from HEAD; a queue only removes the Front.

Programming Fundamentals

★ 6 Golden · 18 cards · Lists (Ordered, Mutable, Methods) · Tuples (Ordered, Immutable) · Dictionaries (Key-Value Pairs) · Functions in Python · & 3.4 File Handling and Operations (Modes r, w, a) · The with Statement (Context Manager)

3.1 Data Structures in Python

A data structure is a way of organising values in a program so you can store them, find them and change them.

- Primitive data structures — the basic types that hold one value. Examples: integers (int), floating-point numbers (float), and booleans...
- Abstract data structures — higher-level containers built from primitive types so you can hold many values and run richer operations....

Category	Description	Examples
Primitive	Basic types that represent a single value.	int, float, bool
Abstract	Higher-level structures for collections and complex operations.	Lists, tuples, sets, dictionaries

```
age = 17
height = 5.8
passed = True
marks = [80, 75, 90]
print(type(age))
print(type(height))
print(type(passed))
print(type(marks))
```

Remember: Primitive = one piece. Abstract = a box of pieces. Think of a single rupee coin (primitive) versus a wallet that can hold many coins (list, tuple, set or...

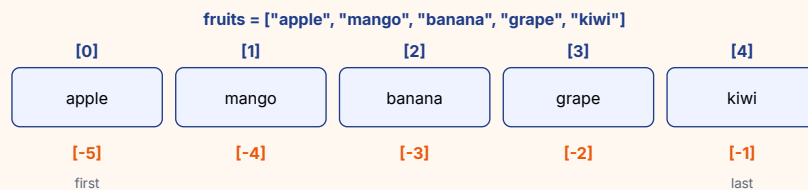
ڈیٹا سٹرکچر (Data Structure) پروگرام میں ڈیٹا کو ترتیب دینے کا طریقہ ہے۔

3.1.1 Lists (Ordered, Mutable, Methods) ★ Golden

A list is Python's most versatile built-in data structure.

- It stores an ordered collection of items in a single variable.
- Lists are mutable (you can change them after you create them) and they are written with square brackets [].
- One variable can hold thousands of values.

After this line	List in memory
create	['apple', 'banana', 'apple']
.append('grape')	['apple', 'banana', 'apple', 'grape']
.remove('apple')	['banana', 'apple', 'grape'] — only the first apple goes
.sort()	['apple', 'banana', 'grape']



Positive indices count left to right from 0. Negative indices count right to left from -1.

```
numbers = [10, 2, 15, 14, 3, 33]      # list of integers
names = ["Hassan", "Qasim", "Malika"] # list of strings
mixed = [1, "Python", 3.14, True]    # mixed data types
print(numbers)
print(names)
print(mixed)
```

Remember: Always remember [-1]. If a list has 10,000 items and you do not want to compute len(list) - 1, write list[-1]. It grabs the last item in one step.

Exam: Lists appear every year: "What is a list?", "List four methods with one line each", "Differentiate list and tuple", "Write a program to search an element", "Predict the..."

فہرست (List) مربع بریکٹ [] میں لکھی جاتی ہے، ترتیب برقرار رکھتی ہے اور تبدیل کی جا سکتی ہے (Mutable)۔

3.1.2 Tuples (Ordered, Immutable) ★ Golden

A tuple is an ordered collection that is immutable — once created, you cannot add, remove or replace items.

- Think of it as a read-only list.
- Tuples use parentheses ().
- Like lists, they allow duplicates, mixed types, and indexing from 0 (including negative indices).

Method	Meaning
<code>.count(value)</code>	How many times the value appears.
<code>.index(value)</code>	Index of the first occurrence.

List []

Can add / remove / replace

Tuple ()

LOCKED — read-only

A list is an open box you can refill. A tuple is a sealed box you may only look into.

```
t = (10, 20, 30, 40, 50)
print(t[0])
print(t[-1])
print(t[1:4])
print(t[::-1])
```

Remember: Tuple = locked lunchbox. You can see what is inside; you cannot swap the sandwich. Memory: "Tuple is Tight (immutable), written with round brackets that look..."

Exam: Expect "Differentiate list and tuple" (at least four points: brackets, mutability, methods, speed/protection), "Why use a tuple?", and "What happens if you write `t[0] = ...`"

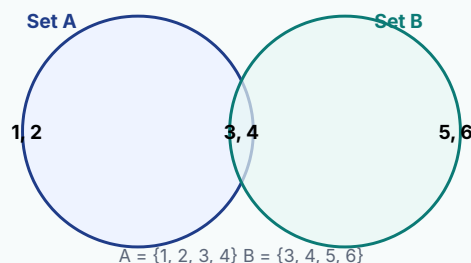
ٹیپل (Tuple) گول بریکٹ () میں لکھا جاتا ہے، ترتیب رکھتا ہے مگر تبدیل نہیں ہو سکتا (Immutable).

3.1.3 Sets (Unordered, Unique, Math Operations)

A set stores an unordered, unindexed collection of unique elements.

- It is inspired by mathematical sets.
- You write a set with curly braces {}.
- Duplicates are removed as soon as the set is created.

Method	Description
<code>.add(value)</code>	Adds one element.
<code>.update(values)</code>	Adds many elements from a list, tuple or set.
<code>.remove(value)</code>	Removes the value. Raises <code>KeyError</code> if it is missing.
<code>.discard(value)</code>	Removes the value. Does not crash if it is missing.
<code>.clear()</code>	Removes every element. Result is <code>set()</code> .



Overlap {3, 4} is the intersection. Left-only {1, 2} is $A - B$. Right-only {5, 6} is $B - A$.

```
nums = {1, 4, 3, 3, 2, 2}
print(nums)
```

Remember: SET = Same-item Eaten Twice-never. Empty set is `set()`, never `{}`. Safe delete = `discard`; angry delete = `remove`.

سیٹ (Set) منفرد عناصر کا بغیر ترتیب اور بغیر انڈیکس مجموعہ ہے، جو { } میں لکھا جاتا ہے۔

3.1.4 Dictionaries (Key-Value Pairs) ★ Golden

A dictionary stores data as key-value pairs.

- Ordered (Python 3.7+): insertion order is kept.
- Mutable: you can add, change and delete pairs.
- Keys are unique: if you repeat a key, the last value silently overwrites the earlier one. Values may repeat.

Method	Returns
<code>.keys()</code>	All keys.
<code>.values()</code>	All values.
<code>.items()</code>	Key-value pairs as tuples.



You never search a dictionary by position. You search by key.

```
d = {"name": "Alice", "age": 30}
print(d)
d["age"] = 31
print(d)
d["city"] = "NYC"
print(d)
del d["city"]
print(d)
```

Remember: Keys are unique, like roll numbers. Values are the student records and may repeat (two students can both score 80). Memory: "get is gentle, brackets can break."

Exam: High-yield: definition of a dictionary, difference between `d[key]` and `d.get(key)`, "what if a key is repeated?", traversal with `.items()`, and a full comparison table of...

ڈکشنری (Dictionary) کلید اور قدر (Key-Value) کے جوڑوں میں ڈیٹا رکھتی ہے۔

3.1.5 Common Built-in Functions (min, max, len, sum)

Several built-in functions work across lists, tuples, sets and dictionaries.

- You do not call them with a dot on the object (except when a type also has its own method, such as `list.sort()`).

Function	Purpose	Works on
<code>min()</code>	Smallest element (numeric or alphabetical)	List, tuple, set, dictionary keys
<code>max()</code>	Largest element	List, tuple, set, dictionary keys
<code>len()</code>	Number of elements (for a dict: number of keys)	List, tuple, set, dictionary
<code>sum()</code>	Adds numeric elements	List, tuple, set, dictionary values via <code>sum(d.values())</code>
<code>sorted()</code>	Returns a new sorted list ; original unchanged	List, tuple, set, dictionary keys

```
s = {5, 1, 9, 3, 7}
print(len(s))
print(min(s))
print(max(s))
print(sum(s))
print(sorted(s))
```

Remember: LEN-MIN-MAX-SUM — count, smallest, largest, total. For a dictionary: `len/min/max/sorted` use keys; `sum` needs `.values()`.

`len` () عناصر گنتا ہے، `min` () سب سے چھوٹا، `max` () سب سے بڑا، `sum` () اعداد کا مجموعہ، اور `sorted` () نئی ترتیب شدہ فہرست دیتا ہے بغیر اصل کو بدلے ہیں۔

3.1.6 Activity: Student Attendance System

Teachers need a quick way to mark who is in class.

Store names in a list → Ask present? for each → Count yes → Print totals

- This program stores names in a list, asks about each student, counts how many said yes, and prints a summary with len().

Student	Typed answer	After <code>.lower()</code>	<code>present_count</code>
Ali	yes	yes → add 1	1
Sara	no	not yes	1
Ayesha	yes	yes → add 1	2

Remember: Counter pattern: start at 0, loop, add 1 when a condition is true. You will reuse this in the word-frequency program (add 1 per word) and the grade tracker...

حاضری کا پروگرام طلبہ کے نام ایک فہرست میں رکھتا ہے، ہر نام کے لیے پوچھتا ہے، اور yes پر شمار کار (Counter) ایک بڑھاتا ہے۔

3.1.7 Activity: Word Frequency Counter

Count how many times each word appears in a sentence.

- This is a classic use of a dictionary plus `.get()`: the word is the key, the count is the value.

After seeing	<code>word_freq</code>
python	<code>{'python': 1}</code>
is	<code>{'python': 1, 'is': 1}</code>
easy	<code>{'python': 1, 'is': 1, 'easy': 1}</code>
python (again)	<code>{'python': 2, 'is': 1, 'easy': 1}</code>
... end of sentence	<code>{'python': 3, 'is': 3, 'easy': 1, 'fun': 1, 'powerful': 1}</code>

Remember: The heart of every counter is one line: `d[item] = d.get(item, 0) + 1`. You can count letters, roll numbers or votes with the same pattern.

جملے کو `.split()` سے الفاظ کی فہرست بناتے ہیں، پھر ہر لفظ ڈکشنری میں کلید ہوتا ہے اور اس کی گنتی قدر۔

3.2 Functions in Python ★ Golden

When the same instructions are written again and again, that is duplicate code.

def → name → parameters → indented body → return (optional)

- Duplicate code makes a program longer, harder to read, and painful to update (you must fix every copy).
- A function lets you write the instructions once and run them whenever you call the function's name.

Benefit	Meaning
Code reusability	Write once, use anywhere.
Modularity	Break a big problem into small named tasks.
Readability	Shorter, cleaner files with meaningful names.
Maintainability	Fix a bug in one place and every call is fixed.

```
def greet():  
    print("Hello, World!")  
  
greet()  
greet()
```

Remember: DRY — Don't Repeat Yourself. If you are about to copy-paste a block, wrap it in `def`. Name functions with verbs: `greet`, `add`, `save_grades`.

Exam: Write the four benefits (reusability, modularity, readability, maintainability) as separate bullets — that is a standard short question. For long questions, define a...

فنکشن (Function) ہدایات کا وہ بلاک ہے جو ایک بار لکھا جاتا ہے اور بار بار بلایا جا سکتا ہے۔

3.2.1 Types of Functions (Built-in vs User-defined)

There are two main types of functions in Python.

- a) Built-in functions are already defined by Python.
- Examples you have already used: `len()`, `range()`, `sum()`, `print()`, `input()`, `min()`, `max()`, `sorted()`, `int()`, `float()`, `type()`.
- b) User-defined functions are functions you create with `def` for tasks that no built-in covers in one step — greeting a user, dividing with...

	Built-in	User-defined
Who writes it?	The Python language	The programmer
How you start	Just call the name	First write <code>def name(...):</code>
Examples	<code>len("Ali")</code> → 3	<code>def add(a, b): return a + b</code>
Purpose	Everyday operations needed by everyone	Your program's specific jobs

```
print(len("Sindh"))

def greet_student(name):
    print("Hello,", name)

greet_student("Hassan")
greet_student("Malika")
```

Remember: If Python already has it (`len`, `sum`, `max`), do not rewrite it. Write a user-defined function when you are packaging your steps.

بلٹ ان فنکشن پائنتھن میں پہلے سے موجود ہوتے ہیں، جیسے `len()` اور `print()`.

3.2.2 Functions with Return Values

A function can either print a result (show it on the screen) or return a result (send it back to the caller).

- A returned value can be stored in a variable, printed later, or used in a further calculation.
- The call is replaced by the returned value, as if you had written that value in its place.

```
def multiply(a, b):
    return a * b

result = multiply(5, 3)
print("The answer is:", result)
```

Remember: `return` = send back; `print` = show now. `return` also exits the function immediately. Lines after `return` in the same function do not run.

`return` نتیجہ کال کرنے والے پروگرام کو واپس بھیجتا ہے تاکہ وہ متغیر میں محفوظ ہو سکے یا آگے حساب میں لگے۔

3.2.3 Activity: Arithmetic Calculator Using Functions

Each arithmetic operation is a separate function.

- That is modularity: addition is one small task, division is another (with a guard against dividing by zero).
- The main program only asks for two numbers and calls the functions.

Remember: Four functions, four names: `add`, `subtract`, `multiply`, `divide`. The main program should look like a contents page — a sequence of calls — not a pile of formulae.

کیلکولیٹر میں ہر عمل الگ فنکشن ہے: جمع، تفریق، ضرب، تقسیم۔

3.2.4 Scope of Local and Global Variables

Scope is the region of a program where a variable can be seen and used.

- Python has two scopes you must know for this chapter.
- A local variable is created inside a function.
- It exists only while that function is running.

```
def calc():
    secret = 50
    print(secret)

calc()
print(secret)
```

Remember: Local = inside the room; global = in the corridor. Everyone can see the corridor. Things left in a room are gone when you leave. Prefer locals. Too many...

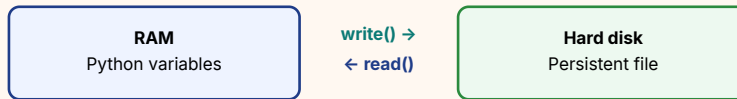
دائرہ کار (Scope) بتاتا ہے کہ متغیر کہاں دکھائی دیتا ہے۔

3.3 & 3.4 File Handling and Operations (Modes r, w, a) ★ Golden

File handling is a program's ability to create, read, write, update and delete files on disk.

- Variables live in RAM and vanish when the program ends.
- Files live on a hard disk or SSD and stay there — they are persistent.

Type	What it stores	Examples
Text file	Human-readable characters. You can open it in a text editor.	.txt, .csv, .py
Binary file	Raw bytes, not meant to be read as text.	.jpg, .mp3, .dat



Read copies data from disk into RAM. Write copies data from RAM onto disk.

```
new_file = open('myfile.txt', 'w')
new_file.write("The quick brown fox jumps over the lazy dog!!!\n")
new_file.close()
print("File has been created successfully.")
```

Remember: r = Read existing, w = Wipe then write, a = Add at end. Remember: write does not add a newline unless you type \n.

Exam: Define file handling, contrast RAM with disk, list modes r / w / a with one property each, and warn that w overwrites. Short programs that write one line and close the...

فائل ہینڈلنگ سے ڈیٹا ڈسک پر مستقل (Persistent) رہتا ہے، برخلاف RAM کے متغیرات کے جو پروگرام بند ہوتے ہی مٹ جاتے ہیں۔

3.6 The with Statement (Context Manager) ★ Golden

It is easy to forget .close(). If the program crashes before .close() runs, the file may be incomplete or corrupted.

with open(...) as f → use f → block ends → file closed automatically

- It is easy to forget .close().
- If the program crashes before .close() runs, the file may be incomplete or corrupted.
- Python's with statement is a context manager: it opens the file, runs the indented block, and always closes the file when the block ends —...

```
with open('myfile.txt', 'w') as new_file:
    new_file.write("Context managers are safe!\n")
# no .close() here — with already closed the file
```

Remember: with = "take, use, put back." Prefer with open(...) as f: in every new program. You still choose the mode: 'r', 'w' or 'a'.

Exam: Define the with statement as a context manager that guarantees close. Contrast it with manual open/close. Be ready to write a four-line program that writes with with and...

with بیان ایک کانٹیکسٹ مینیجر (Context Manager) ہے جو فائل کو بلاک کے اختتام پر خود بند کر دیتا ہے، چاہے اندر غلطی ہو جائے۔

3.8 Errors and Exceptions (Try-Except)

An exception is an error raised while the program is running — for example opening a file that is not there.

- Without handling, Python prints a traceback and stops.
- A try/except block catches the error and lets you show a clear message instead of crashing.

Error type	Meaning
FileNotFoundError	The path does not exist.
PermissionError	The operating system will not let you read or write that file.
IOError	A general input/output problem (disk full, disk disconnected).

```
try:
    with open('missing_data.txt', 'r') as f:
        content = f.read()
        print(content)
except FileNotFoundError:
    print("Error: The file does not exist. Please check the name.")
except Exception as e:
    print(f"An unexpected error occurred: {e}")
```

Remember: try = attempt; except = backup plan. Catch the specific error first (FileNotFoundError), then a general Exception if you need it. Order matters: a bare except...

ایکسپشن (Exception) چلتے پروگرام کی غلطی ہے، جیسے غیر موجود فائل کھولنا۔

3.9 Mini File Handling Project: Grade Tracker

This project uses 'w' to create a grades file, 'a' to add a student later, and 'r' to read every line and compute an average.

Write names and marks ('w') → Append a new student ('a') → Read lines ('r') → Split, total, average

- It combines file handling, string methods and arithmetic.

Remember: Same file, three modes, three with blocks. Never open the same file in 'w' when you only meant to add David — that would delete Alice, Bob and Carol.

گریڈ ٹریکر پہلے 'w' سے فائل بناتا ہے، پھر 'a' سے نیا طالب علم جوڑتا ہے، پھر 'r' سے تمام سطریں پڑھ کر اوسط نکالتا ہے۔

Career Opportunities

The skills in this chapter — data structures, functions and files — are the same skills used in many jobs.

- Python / software developer — builds applications, scripts and tools in Python, one of the most widely used languages in industry.
- Data analyst — uses lists and dictionaries, then file handling, to process, visualise and interpret data for decisions.
- Automation engineer — writes scripts that repeat file and testing tasks so humans do not have to.
- Game developer — uses functions, data structures and logic to build interactive games.

Remember: Four careers: P-D-A-G — Python developer, Data analyst, Automation engineer, Game developer.

اس باب کی مہارتیں پائنتھن ڈویلپر، ڈیٹا اینالسٹ، آٹومیشن انجینئر اور گیم ڈویلپر کے کام میں روز استعمال ہوتی ہیں۔

Key terms

Term	Meaning	اردو
Data structure	A way of organising values so a program can store and use them.	ڈیٹا سٹرکچر
Primitive type	A basic single value: <code>int</code> , <code>float</code> , <code>bool</code> .	بنیادی قسم
List	Ordered, mutable sequence written with <code>[]</code> .	فہرست
Index	The position of an item, starting at 0 (or -1 from the end).	انڈیکس
Slicing	Copying a portion of a sequence with <code>start:stop:step</code> .	سلائسنگ
Traversal	Visiting each element in turn, usually with a loop.	گشت / تراورسل
Mutable	Can be changed after creation.	قابل تبدیلی
Immutable	Cannot be changed after creation.	ناقابل تبدیلی
Tuple	Ordered, immutable sequence written with <code>()</code> .	ٹیپل
Set	Unordered collection of unique items written with <code>{}</code> .	سیٹ
Union	All unique items from two sets (<code> </code>).	اجتماع
Intersection	Items common to both sets (<code>&</code>).	اشتراک
Dictionary	Mutable map of unique keys to values.	ڈکشنری
Key	The unique lookup name in a dictionary.	کلید
Value	The data stored under a key.	قدر
Function	A named, reusable block of code.	فنکشن
Parameter	An input name in a function definition.	پیرامیٹر
Return value	The result sent back with <code>return</code> .	واپسی کی قدر
Local variable	A name that exists only inside a function.	مقامی متغیر
Global variable	A name defined outside all functions.	عالمی متغیر
File handling	Creating, reading, writing and closing disk files.	فائل ہینڈلنگ
File mode	How a file is opened: <code>r</code> , <code>w</code> , <code>a</code> (or binary).	فائل موڈ
Context manager	A <code>with</code> block that sets up a resource and cleans it up.	کانٹیکسٹ مینیجر
Exception	A runtime error object, such as <code>FileNotFoundError</code> .	ایکسپشن
try / except	A pair of blocks that attempt code and handle errors.	ٹرائی / ایکسپٹ

Chapter in one look

- Python programs rest on primitive types (`int`, `float`, `bool`) and abstract structures: lists, tuples, sets and dictionaries.
- A **list** is ordered and mutable, written with `[]`. Index from 0; `[-1]` is last. Slice stop is exclusive. Methods include `append`, `insert`, `remove`, `sort`, `reverse`, `clear`.
- A **tuple** is ordered and immutable, written with `()`. Item assignment raises `TypeError`. Only `count` and `index` exist as methods.
- A **set** is unordered and unique, written with `{}`. Use `|` & `-` & `^` for union, intersection, difference and symmetric difference. Prefer `discard` over `remove`.
- A **dictionary** maps unique keys to values. `d[key]` can raise `KeyError`; `d.get(key, default)` is safe. Duplicate keys keep the last value.
- `min`, `max`, `len`, `sum` and `sorted` work across structures. On a dict, `min/max/sorted` use keys; total values with `sum(d.values())`. `sorted()` returns a new list; `.sort()` changes a list in place.
- Functions written with `def` give reusability, modularity, readability and maintainability. Built-in functions already exist; user-defined functions solve your tasks.
- `return` sends a value to the caller. Without `return`, the result is `None`. Local variables live inside a function; global variables live outside all functions.
- Files persist on disk. Modes: `r` read (must exist), `w` write (overwrites), `a` append. Methods: `read`, `readlines`, `write`, `close`.
- The `with` statement is a context manager that closes the file automatically. Loop line by line to read large files safely.
- `try/except FileNotFoundError` stops a missing file from crashing the program.
- The grade tracker writes with `w`, appends with `a`, reads with `r`, splits each `Name: mark` line, and computes the mean.

CHAPTER 4

Data and Analysis

★ 5 Golden · 9 cards · Data Sources and Database Connection · Pandas and DataFrames · Handling Missing Values (NaN) · Data Visualization and Types of Graphs · Descriptive Statistics

4.1 Concept of Data Analysis

Data analysis is examining, organising and interpreting data so you can extract useful information.

- Schools study results, shops study sales, hospitals study patient records — all to make a decision, spot a trend or solve a problem.
- Python is popular for this work because it is readable, talks to files and databases, does statistics, and draws graphs.
- Five lines of Pandas often replace an hour of spreadsheet clicking.

ڈیٹا کا تجزیہ یعنی اعداد کو ترتیب دے کر ان سے فیصلے کے قابل معلومات نکالنا۔

4.1.1 – 4.1.2 Data Sources and Database Connection ★ Golden

Data sources are where information comes from: school records, websites, sensors, databases.

Python → Driver → Connection → Cursor → SQL → Database

- Connectivity is the conversation between your Python program and that source so you can read and update data.
- Five components of a database connection, in the order they actually work:

#	Component	Role
1	Database	Organised store (SQLite = one .db file; MySQL = a server for big apps)
2	Driver	Translator between Python and that DBMS (sqlite3 is built in)
3	Connection object	Opens, manages and closes the pipeline
4	Cursor object	Runs SQL and walks through result rows
5	SQL queries	CREATE, INSERT, SELECT, UPDATE, DELETE

Exam: Write the five names in order and one line each. A common trap is swapping cursor and connection: the connection is the pipe; the cursor is the messenger that travels...

خمسہ اجزاء: ڈیٹا بیس، ڈرائیور، کنکشن، کرسر، ایس کیو ایل۔

4.2 Creating a SQLite Database in Python

SQLite is built into Python. The whole database is one .db file next to your script — no password, no port, no server.

- SQLite is built into Python.
- The whole database is one .db file next to your script — no password, no port, no server.
- Nine steps you must be able to write from memory:

Step	Why it matters
connect	Creates the file if it does not exist
cursor	Vehicle that carries SQL
execute	Runs one SQL command
commit	Writes INSERT/UPDATE/DELETE to disk
fetchall	Brings SELECT results into a Python list of tuples
close	Frees memory and unlocks the file

Remember: Strings inside SQL use single quotes: 'Ali'. Python strings around the SQL use double quotes (or a triple-quoted string).

ایس کیو لائٹ پائتھن کے ساتھ آتی ہے، پورا ڈیٹا بیس ایک فائل ہے۔

4.3 Pandas and DataFrames ★ Golden

Pandas is a third-party library for tables.

- Install once: pip install pandas in the terminal (not inside the .py file).
- Convention: import pandas as pd.
- A DataFrame is a 2-D table of rows and columns, like Excel or an SQL table.

```
import pandas as pd
data = {
    "Name": ["Ali", "Sara", "Ahmed"],
    "Math": [85, 90, 78],
    "Science": [80, 88, 70],
}
df = pd.DataFrame(data)
print(df)
```

پینڈا ڈیٹا فریم دیتا ہے۔

4.3.3 Handling Missing Values (NaN) ★ Golden

Missing cells are NaN ("Not a Number"). Statistics crash or lie if you ignore them.

- Statistics crash or lie if you ignore them.
- Three tools: isnull() (True where empty), fillna(value) (replace), dropna() (delete the row).

```
import pandas as pd
df = pd.DataFrame({
    "Name": ["Ali", "Sara", "Ahmed", "Ayesha"],
    "Marks": [85.0, None, 78.0, 92.0],
    "Grade": ["A", "A", None, "A"],
})
print(df.isnull().sum())
```

Trap: Filling a salary gap with the mean of a company that includes the CEO can hide poverty (the same outlier problem as the mean).

Exam: State: identify with isnull/sum, fill numeric gaps with the mean, fill categories with a label, or drop rows if filling would invent a lie. Never analyse first and clean...

خالی خانے NaN ہیں۔

4.4 Data Organization and Representation

Organised data is easier to read. Three everyday forms: Table — exact values side by side (Ali 85 Math, 80 Science).

- Table — exact values side by side (Ali 85 Math, 80 Science).
- Frequency distribution — how many items fall in each range (bin).
- Graph — a picture of a trend or comparison (next topics).

Range of marks	Frequency (students)
50–59	5
60–69	7
70–79	9
80–89	4
90–100	2

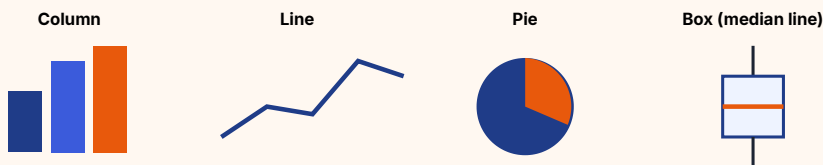
ٹیبل درست عدد دکھاتی ہے، تعدد کی جدول بتاتی ہے کتنے طلبہ کس رینج میں ہیں، گراف رجحان دکھاتا ہے۔

4.5 Data Visualization and Types of Graphs ★ Golden

Pick the graph that matches the question. Then label axes and give a title — an unlabelled chart scores zero in an exam.

- Then label axes and give a title — an unlabelled chart scores zero in an exam.

Chart	Shows	Watch out
Bar / column	Compare discrete categories (subjects, departments)	Column = vertical; bar often = horizontal
Line	Change over time (temperature, visitors)	Needs an ordered x-axis (months, years)
Pie	Parts of one whole as percentages	Bad with many or almost-equal slices
Histogram	Distribution of one numeric variable in bins	Bars touch (continuous range), unlike a bar chart
Scatter	Relationship of two numeric variables	Each dot is one pair (x, y)
Box plot	Min, Q1, median, Q3, max, and outliers	Best tool for spotting extreme values



Column compares; line shows a trend; pie shows share; the box summarises spread around the median.

```
import matplotlib.pyplot as plt
subjects = ["English", "Math", "Science", "History"]
marks = [64, 82, 71, 49]
plt.bar(subjects, marks, color="skyblue", edgecolor="black")
plt.title("Marks vs Subjects")
plt.xlabel("Subjects")
plt.ylabel("Marks out of 100")
```

Exam: "Which graph and why?" is a 3-mark gift. Write the name, what it shows, and one reason the others are worse.

بار/کالم اقسام کا موازنہ، لائن وقت کا رجحان، پائی حصے، ہسٹوگرام مسلسل تقسیم (بار آپس میں لگے)، اسکیٹر دو متغیرات کا تعلق، باکس پلا...

4.6 Descriptive Statistics ★ Golden

Descriptive statistics squeeze a long list into a few honest numbers.

- Central tendency asks "what is typical?"
- Spread asks "how mixed is the list?"

Measure	How	Use when
Mean	Sum ÷ n. Uses every value.	Values sit close together. Example: ordinary test scores.
Median	Middle after sorting. Odd n: centre. Even n: mean of the two centre values.	Outliers would wreck the mean (CEO salary).
Mode	Most frequent value. May be none, one, or several.	Most popular choice (shoe size).

Trap: Do not report variance as if it were in "marks".

Exam: Always state whether you used n or n-1. Show the sorted list for the median. For SD, show: mean → deviations → squares → divide → square root. A box plot pictures min,...

اوسط سب کا مجموعہ تقسیم تعداد، میڈین ترتیب کے بعد بیچ، موڈ سب سے زیادہ بار آنے والا۔

Career Opportunities

The skills in this chapter map onto real jobs: data analyst (Pandas, Excel), database administrator (SQLite/MySQL), data scientist (statistics + modelling), visualisation specialist (Matplotlib, Power BI),...

یہ باب ڈیٹا اینالسٹ، ڈیٹا بیس ایڈمن، ڈیٹا سائنسدان، ویژولائزیشن اور بیک اینڈ ڈویلپر جیسے کیریئر کی بنیاد ہے۔

Key terms

Term	Meaning	اردو
Data analysis	Examining data to extract useful information.	ڈیٹا کا تجزیہ
Connectivity	Link between a program and a data source.	رابطہ کاری
Driver	Translator between Python and a DBMS.	ڈرائیور
Connection	Open pipeline to the database.	کنکشن
Cursor	Object that executes SQL and fetches rows.	کرسر
SQLite	Built-in file-based database.	ایس کیو لائٹ
commit	Save INSERT/UPDATE/DELETE to disk.	کمیٹ / محفوظ کرنا
Pandas	Library for table-like data.	پینڈاز
DataFrame	2-D labelled table of rows and columns.	ڈیٹا فریم
NaN	Missing value ("Not a Number").	خالی / غیر موجود قدر
Frequency	How many items fall in a class/bin.	تعدد
Histogram	Distribution chart whose bars touch.	ہسٹوگرام
Scatter plot	Dots showing two numeric variables.	اسکیٹر پلاٹ
Box plot	Min, Q1, median, Q3, max (and outliers).	باکس پلاٹ
Mean	Arithmetic average.	اوسط
Median	Middle value of a sorted list.	میڈین / درمیانی قدر
Mode	Most frequent value.	موڈ / کثرت
Range	Max – min.	حد / رینج
Variance	Mean of squared deviations from the mean.	ویریانس / تغیر
Standard deviation	Square root of variance (original units).	معیاری انحراف
Outlier	An extreme unusual value.	آؤٹ لائر

Chapter in one look

- Data analysis turns raw facts into decisions. Python reads files and databases, calculates and plots.
- Connection chain: database → driver → connection → cursor → SQL.
- SQLite: connect, cursor, execute, commit, fetchall, close. No commit means no save.
- A DataFrame is a 2-D table. Load with a dictionary, `read_csv` or `read_excel`. New columns are vectorised arithmetic.
- NaN: `find ( isnull )`, `fill ( fillna )`, often with the mean, or `drop ( dropna )`.
- Tables give exact values; frequency bins show how many in each range.
- Bar/column = categories; line = time; pie = parts of a whole; histogram = numeric distribution (touching bars); scatter = two variables; box = five-number summary + outliers.
- Mean uses every value and is sensitive to outliers; median is the middle; mode is the most common.
- Range = max-min. Variance = mean squared deviation ($\div n$ or $\div n-1$). SD = $\sqrt{\text{variance}}$. pandas `.std()` uses $n-1$.

CHAPTER 5

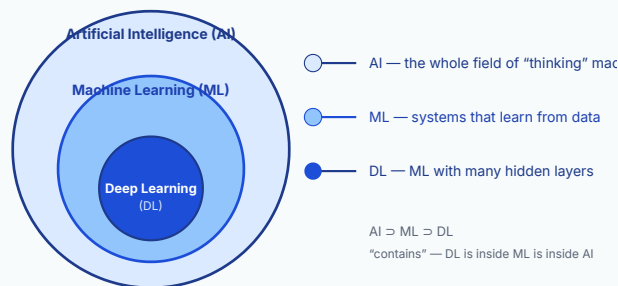
Application and Impacts of Computing

★ 5 Golden · 17 cards · Neural Networks (NN) · Components of Neural Networks · Secure Collaboration (Authentication, MFA, Access Control) · Common Security Threats (Malware, Phishing, DoS) · Functions of Equity and Equal Access

5.1 Machine Learning, Neural Networks and Deep Learning

Machine learning (ML), neural networks (NN) and deep learning (DL) are important areas of artificial intelligence (AI).

- Face unlock on smartphones.
- Speech recognition (for example Siri, Alexa, or Urdu voice typing on a phone).
- Photo search categorisation (typing “car” in a gallery and seeing car photos).
- Medical diagnosis from X-rays and scans.



The set relationship $AI \supset ML \supset DL$. Deep learning is a method of machine learning; machine learning is a part of artificial intelligence.

Remember: Russian-doll memory trick. Picture three boxes. The outer box is AI. Inside it is a smaller box ML. Inside that is the smallest box DL. You cannot take DL out...

مصنوعی ذہانت (AI) سب سے بڑا میدان ہے۔

5.1.1 Machine Learning (ML)

A machine learning model is the system that learns from data and gives an output.

1. Data input → 2. Learning (find patterns) → 3. Prediction on new data

- Think of it as the trained brain of a computer program.
- In ordinary programming you write every rule yourself: if marks ≥ 40 then “Pass”.
- In machine learning you do not list every rule.

Remember: D-L-P. Machine learning is Data in, Learn patterns, Predict. “Data Learns to Predict.”

مشین لرننگ ماڈل وہ نظام ہے جو ڈیٹا سے سیکھ کر نتیجہ دیتا ہے — کمپیوٹر پروگرام کا تربیت یافتہ دماغ۔

5.1.2 Neural Networks (NN) ★ Golden

A neural network is a computer model that learns patterns from data.

Input layer → Hidden layer(s) → Output layer

- Input layer — receives the raw data (pixels of a face, samples of a sound, marks of a student).
- Hidden layer(s) — the “thinking” in the middle. Nodes here combine the inputs and learn features (eye distance, nose shape, an edge in a...
- Output layer — gives the final answer (unlock / keep locked; “this is a cat”; “pass / fail”).

Remember: I-H-O. Input receives, Hidden thinks, Output decides. “In Hidden Out.” A neural network is a brain-inspired maths machine, not a copy of a living brain.

Exam: Very common questions: “What is a neural network?” (2–3 marks), “Name its three layers and explain each with the face-unlock example” (5 marks), and “How is a neural...”

نیورل نیٹ ورک ایک کمپیوٹر ماڈل ہے جو ڈیٹا سے پیٹرن سیکھتا ہے۔

5.1.3 Deep Learning (DL)

Deep learning is a machine-learning method in which computers use multi-layer neural networks to learn complex patterns from data.

- Early layers learn simple features — edges, shapes, colours.
- Deeper layers combine those simple features to understand complex patterns — a whole face, or a specific car model (a Cultus versus a...

	Neural network (the model)	Deep learning (the method)
What it is	A system/model that has layers, nodes and connections to help it learn.	A specific method that trains neural networks containing many hidden layers .
Depth	Can be simple (shallow) with just one or two hidden layers.	Always “deep,” requiring huge amounts of data and processing power to learn complex patterns.
Role	The architecture (the “machine”).	The way we train a deep version of that machine.

Remember: Deep = many hidden layers. Shallow NN: 1–2 hidden layers. Deep NN: many hidden layers, huge data, strong computers. Simple features first, complex meaning...

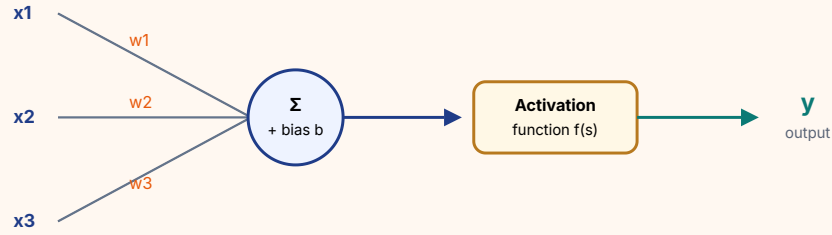
ڈیپ لرننگ مشین لرننگ کا وہ طریقہ ہے جس میں کئی پوشیدہ تہوں والے نیورل نیٹ ورک استعمال ہوتے ہیں۔

5.1.4 Components of Neural Networks ★ Golden

The components of a neural network can be understood through its layered architecture.

- Data enters, gets processed, and exits, with mathematical adjustments happening inside.

#	Component	Description
1	Input layer	Receives the initial raw data (for example image pixels, sound waves, text).
2	Neurons / nodes	Small processing units that receive, calculate, and pass information forward.
3	Hidden layer(s)	The layers between input and output. They process the data and learn patterns from it.
4	Weights	Numbers that show the importance or strength of connections between neurons.
5	Bias	An extra constant value added to help the model adjust its result and improve accuracy.
6	Activation function	A mathematical gate that decides whether the information is important enough to move to the next layer.
7	Output layer	Gives the final computed result, decision or prediction (for example “This is a cat”).
8	Loss function	Checks the error (how much the model's answer is wrong compared with reality) so the network can correct itself.



One neuron: inputs $\times$ weights are summed, bias is added, then an activation function decides the output y .

```

x1, x2 = 0.8, 0.5
w1, w2 = 0.4, 0.6
b = 0.1
s = x1 * w1 + x2 * w2 + b
y = 1 if s ≥ 0.5 else 0
print(round(s, 2))
print(y)
  
```

Remember: Weights = importance. Bias = extra nudge. Activation = yes/no gate. Loss = how wrong. A large weight means that input matters more. Bias shifts the sum up or...

Exam: Golden questions: "List the components of a neural network" (often 8 names + one line each), "What are weights and bias?", and a numerical neuron: given x, w, b and $a...$

نیورل نیٹ ورک کے اجزاء ہیں: ان پٹ تہ، نیورون، پوشیدہ تہ، وزن (Weights)، بائیس (Bias)، ایکٹیویشن فنکشن، آؤٹ پٹ تہ اور لاس فنکشن۔

5.1.5 Components of Deep Learning

Deep learning relies on specific advanced components to achieve its high accuracy.

- These components differentiate it from basic machine learning.

#	Component	What it means
1	Large input data	Deep learning is " data hungry ." It needs massive datasets (thousands or millions of records) for better learning and accuracy.
2	Many hidden layers	Uses dozens or hundreds of hidden layers to learn deeper, more intricate patterns.
3	Automatic feature learning	Unlike older models where humans had to manually highlight important features, deep learning extracts important features by itself .
4	Learning algorithm	Math routines (like backpropagation) that help the system reduce mistakes and improve learning continuously.
5	High computing power	Processing millions of data points across deep layers needs specialised, strong hardware, such as GPUs (Graphics Processing Units).
6	Training process	The model is trained again and again in loops called epochs to gradually improve its final performance.

Remember: D-L-A-L-G-T. Data (lots), Layers (many), Automatic features, Learning algorithm, GPU power, Training epochs. "Deep Learners Always Love Good Training."

ڈیپ لرننگ کو عام مشین لرننگ سے یہ چیزیں الگ کرتی ہیں: بہت بڑا ڈیٹا، بہت سی پوشیدہ تہیں، خود بخود خصوصیات نکالنا، سیکھنے کا الگورتھم...

Future of AI, NN and DL in Pakistan

Machine learning, neural networks and deep learning will play a key role in Pakistan's future.

- The National AI Policy 2025 highlights how artificial intelligence is now part of national planning, focusing on innovation,...

Sector	Potential impact in Pakistan
Education	Smart learning systems, online tutoring, automatic assessments, and finding where students need specific help.
Healthcare	Disease detection, medical image analysis, and managing hospital data. Crucial for rural areas lacking specialist doctors.
Agriculture	Monitoring crops, predicting weather, managing irrigation, and estimating yields — highly important for Pakistan's economy.
Business and banking	Fraud detection, customer support (chatbots), market forecasting, and intelligent risk analysis.
Local languages	Speech recognition, translation and text-to-speech tools for Urdu, Sindhi and other regional languages.

Remember: Five sectors: E-H-A-B-L. Education, Healthcare, Agriculture, Business/banking, Local languages. Policy words: innovation, skills, ethics, digital growth.

قومی اے آئی پالیسی 2025 کے مطابق مصنوعی ذہانت پاکستان کی منصوبہ بندی کا حصہ ہے: جدت، مہارت، اخلاقی استعمال اور ڈیجیٹل ترقی۔

5.2 Applications of Neural Networks and Deep Learning

Neural networks and deep learning are widely used in modern life.

- Many intelligent systems use them in the background to recognise information, make predictions and support users.

#	Application	What it does	Example
1	Image recognition	Identifies objects, faces or patterns in images. Used in security cameras, photo search and biometric locks.	Unlocking a smartphone using face recognition; a camera at a Karachi plaza matching a face with a staff list.
2	Speech recognition	Converts spoken words into digital text or commands. Deep learning helps with accents, pronunciation and voice patterns.	A student speaks to Google Assistant or Alexa to set a morning alarm; Urdu voice search on a phone.
3	Natural Language Processing (NLP)	Helps computers understand, process and respond to human language grammatically and in context. Core of chatbots, translators and AI writers.	Using Google Translate to turn a difficult English paragraph into Urdu; a bank chatbot answering in Urdu.
4	Healthcare systems	Analyse complex medical images (X-rays, CT scans, MRIs) to detect diseases early and accurately.	A hospital system highlighting abnormal patterns in a lung X-ray that might indicate pneumonia or tumours — useful at a DHQ hospital that has few radiologists.
5	Recommendation systems	Analyse past behaviour and preferences to suggest content, products or services.	A student watches computer-science videos on YouTube; the platform recommends more programming tutorials.
6	Autonomous vehicles	Self-driving cars use neural networks to process real-time camera and radar data: roads, signals, pedestrians, obstacles.	A self-driving car detects a pedestrian crossing and automatically applies the brakes.
7	Fraud detection	Banks compare current activity with previous spending patterns to catch unusual transactions.	An HBL or Meezan app flags a sudden Rs 80,000 payment from a new city at 3 a.m.
8	Assistive technologies	AI supports people with disabilities.	Speech-to-text tools for users who have difficulty typing; screen readers that speak Urdu text aloud.

Remember: Eight applications: I-S-N-H-R-A-F-A. Image, Speech, NLP, Healthcare, Recommendations, Autonomous vehicles, Fraud, Assistive tech. "I See New Hospitals..."

نیورل نیٹ ورکس اور ڈیپ لرننگ کی مشہور ایپلی کیشنز یہ ہیں: تصویر کی پہچان، آواز کی پہچان، قدرتی زبان کی پروسیسنگ (NLP)، صحت کے نظام...

5.3.1 Secure Collaboration (Authentication, MFA, Access Control) ★ Golden

Digital tools allow people to share files, attend online classes, edit documents and work together from different locations.

- Collaboration is useful only when data is protected and communication remains safe.
- Secure collaboration means using digital tools to work together in a safe, controlled and responsible way.
- It depends on several key elements.

Remember: A-A-A-P-S. Authentication, Access control, Accountability, Privacy, Safe transmission. MFA = at least two different factor types (know / have / are). 2FA is...

Exam: Typical items: "Define secure collaboration", "Differentiate authentication and access control", "What is MFA/2FA?", "Explain five elements of secure collaboration with..."

محفوظ تعاون کا مطلب ہے ڈیجیٹل ٹولز سے محفوظ، کنٹرول شدہ اور ذمہ دارانہ انداز میں ساتھ کام کرنا۔

5.3.2 Data Protection (Encryption, Passwords, Backup, Firewall)

Data protection means keeping digital information safe from unauthorised access, misuse, loss, damage or accidental deletion.

3 copies → 2 media types → 1 copy off-site

Plain	S	I	N	D	H
+3	V	L	Q	G	K

Remember: Cipher = lock the message. Password = lock the door. 3-2-1 = spare keys in three places. Firewall = gate. Antivirus = indoor search. Caesar: shift letters;...

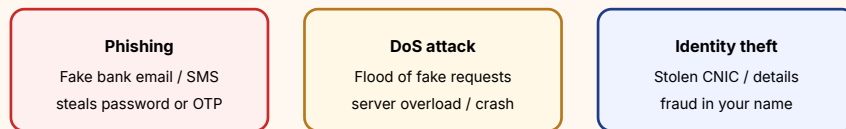
ڈیٹا پروٹیکشن کا مطلب ہے معلومات کو غیر مجاز رسائی، غلط استعمال، نقصان یا حادثاتی حذف سے بچانا۔

5.4.1 Common Security Threats (Malware, Phishing, DoS) ★ Golden

A security threat is any action, event or weakness that can harm a computer system or its data.

- Mitigation techniques are protective measures used to prevent threats or reduce their impact.
- This section names the common threats; later sections show how to spot them and how to reduce the damage.

#	Threat	What it is	Example
1	Malware	Harmful software designed to damage a computer, steal data, or gain unauthorised access. Includes viruses, worms, spyware, ransomware and Trojans .	Installing cracked software that secretly corrupts files, or a "free game" USB that encrypts a college lab's coursework (ransomware).
2	Phishing	A social-engineering trick that deceives users into sharing personal information (passwords, bank details, OTPs) using fake emails or websites that look genuine.	An SMS that looks like JazzCash or HBL asking you to "verify" by typing your OTP on a fake page.
3	Unauthorised access	Entering a system, account or file without permission. Often caused by weak passwords, shared passwords, or leaving devices unlocked.	A classmate using an unlocked lab PC to open someone else's Google Classroom.
4	Data theft / identity theft	Data theft is stealing or copying data without permission. Identity theft is using another person's stolen details (name, CNIC, bank details) to pretend to be them online and commit fraud.	A leaked list of CNICs used to open a SIM or a wallet in someone else's name.
5	Denial of Service (DoS)	A website or system is flooded with thousands of fake requests, causing it to slow down or crash, so real users cannot access the service.	Hackers flooding an online exam portal so students cannot log in on BISE result day or during an e-test.
6	Human error	Mistakes by legitimate users: accidentally deleting files, sending confidential data to the wrong email, or falling for a simple scam.	A teacher emailing the whole marksheet to the wrong parent; a student forwarding an OTP "to help the bank."



Three high-yield threats: phishing (fake messages), DoS (server overload), identity theft (stolen CNIC or bank details).

Remember: Malware = bad program. Phishing = fake message. DoS = flood. Identity theft = stolen 'you'. Human error = honest mistake. Cracked software and unknown USBs are...

Exam: Expect "Define a security threat", "Explain malware / phishing / DoS with examples", and "Differentiate data theft and identity theft." List all six threats if the...

سیکیورٹی خطرہ وہ عمل یا کمزوری ہے جو نظام یا ڈیٹا کو نقصان پہنچاے۔

5.4.2 Identifying Security Threats

Identifying security threats means noticing signs that data or systems may be at risk.

- Early identification helps you act before serious damage occurs.

Sign / indicator	What it likely means
Slow performance, missing files, or random pop-ups.	Malware or background unauthorised activity.
Failed login attempts or strange login locations.	Attempted unauthorised access .
Emails urging quick action, asking for an OTP, or containing bad grammar / strange links.	Classic phishing .
Unexpected deletion, copying or movement of records.	Data theft or insider misuse.
Sudden massive spike in server requests or a very slow website.	An ongoing DoS attack.

Remember: Slow/pop-ups → malware. Odd logins → unauthorised access. Urgent OTP email → phishing. Files moving → theft. Traffic spike → DoS.

خطرے کی جلد پہچان نقصان سے پہلے اقدام ممکن بناتی ہے۔

5.4.3 Apply Mitigation Techniques

Mitigation techniques are actions taken to prevent security threats or reduce their harmful effects once identified.

- Technology helps, but it cannot stop all threats if the user is careless.

#	Technique	How it helps
1	Antivirus and firewalls	Antivirus scans and removes harmful software (for example from USB drives). A firewall monitors network traffic and blocks suspicious access from the internet.
2	Strong passwords and 2FA	Complex passwords slow brute-force guessing. 2FA adds an extra layer: an OTP on your phone as well as the password.
3	Data encryption	Converts data into unreadable code during transmission. Protects passwords and sensitive data if they are intercepted.
4	Regular backups	Copies on cloud or external drives allow quick recovery after a crash, human error or ransomware.
5	User awareness (the human factor)	Training users to avoid unknown links, suspicious attachments and password sharing. The human is often the weakest link — and the strongest defence when trained.

Remember: Five shields: A-F, P-2, E, B, U. Antivirus+Firewall, Passwords+2FA, Encryption, Backups, User awareness.

Ransomware is why backups matter: if files are locked,...

تحفظ کی تدابیر خطرات کو روکتی یا ان کا نقصان کم کرتی ہیں: اینٹی وائرس اور فائر وال، مضبوط پاس ورڈ اور دو عنصری تصدیق (2FA)، انکرپشن...

5.5 Equity and Equal Access in Digital Collaboration

Collaboration tools are vital for online learning and teamwork.

- To use these tools fairly, all users must participate without barriers.
- You must understand the difference between equal access and equity.

Remember: Equal = same tools. Equity = extra support for the same outcome. Same shoes for all vs a ramp for the wheelchair user.

مساوی رسائی (Equal Access) سب کو ایک جیسے اوزار دیتی ہے، جیسے ایک ہی زوم لاگ ان۔

5.5.1 Functions of Equity and Equal Access ★ Golden

Equity and equal access are not slogans. They do work in a digital classroom.

- They do work in a digital classroom.
- The source lists four functions.

Function	What it does	Pakistani example
1. Providing need-based support (equity)	Help tailored to individuals.	Audio descriptions for visually impaired students; a screen reader so a blind student in Karachi can hear a digital textbook.
2. Removing barriers	Help students in remote areas attend online classes.	Mobile-friendly, low-data apps so a student in Tharparkar or the katcha areas of Sindh can join without a fibre connection.
3. Enabling active participation	Students who miss live class can still take part.	Recorded lectures so load-shedding or an internet outage in Hyderabad does not make a student fall behind.
4. Supporting different learning needs	Offer more than one format.	Videos, text notes and audio of the same Chapter 5 topic so each student learns in the way that suits them best.

Remember: Four functions: Support, Remove, Participate, Formats. "Some Rural Pupils Fail" without them — Support (need-based), Remove barriers, Participate via...

Exam: Golden question: "Explain the functions of equity and equal access in digital collaboration" (5–8 marks). Write the four headings in order, one example each, and add the...

انصاف اور مساوی رسائی کے چار کام ہیں: ضرورت کے مطابق مدد (جیسے نابینا طالب علم کے لیے آڈیو وضاحت)، رکاوٹیں ہٹانا (دور دراز علاقوں...)

5.5.2 Collaboration Tools

Digital collaboration tools are online platforms that support communication, file sharing, learning and teamwork across different locations.

Tool	Primary function and educational use
Google Docs	Write, edit, comment and share documents in real time. Excellent for group assignments.
Zoom	Online meetings, virtual classes, screen sharing and video-based communication.
Microsoft Teams	A unified workspace for online meetings, file sharing, class groups and structured teamwork.
Google Classroom / LMS	Learning Management Systems help teachers manage lectures, distribute assignments, conduct quizzes and grade student activities.
WhatsApp	Quick class announcements, urgent reminders and instant group communication. Must be used responsibly.
Email	Formal communication, official notices, submitting large attachments, and keeping a professional written record.

Remember: Docs = write together. Zoom/Teams = meet. Classroom = official class hub. WhatsApp = fast but informal. Email = formal record.

ڈیجیٹل تعاون کے اوزار مختلف مقامات سے بات چیت، فائل شیئرنگ اور ٹیم ورک چلاتے ہیں: گوگل ڈاکس مشترکہ تحریر کے لیے، زوم اور مائیکروسا...

Career Opportunities

This unit covers neural networks, deep learning, secure collaboration, data protection and digital equity.

- Mastery of these ideas opens high-demand careers in Artificial Intelligence and Cybersecurity — including in Pakistan's banks, software...

Career	What the person does
Machine Learning Engineer	Builds and trains machine-learning and neural-network models used in face recognition, NLP and recommendation systems.
Cybersecurity Analyst	Monitors systems for threats such as malware and phishing, and applies mitigation (firewalls, 2FA) to protect organisational data.
AI Researcher	Studies and develops new techniques in artificial intelligence and deep learning, often in university labs or high-tech R&D centres.
Data Privacy Officer	Ensures the organisation's data handling complies with legal privacy principles, protecting sensitive user information and secure collaboration.
Ethical Hacker	Legally tests systems and networks for vulnerabilities (penetration testing) before malicious attackers can exploit them via DoS or malware.

Remember: AI careers lean on Python, maths and ethics. Cyber careers lean on threats, 2FA, firewalls and careful testing. Both need clear communication.

اس باب سے جڑے پیشے ہیں: مشین لرننگ انجینئر، سائبر سیکیورٹی اینالسٹ، اے آئی محقق، ڈیٹا پرائیویسی آفیسر اور اخلاقی ہیکر۔

Key terms

Term	Meaning	اردو
Artificial intelligence (AI)	The broad field in which machines perform tasks that normally need human thinking.	مصنوعی ذہانت
Machine learning (ML)	A model that learns patterns from data and then predicts on new data.	مشین لرننگ
Neural network (NN)	A brain-inspired mathematical model of connected neurons in layers.	نیورل نیٹ ورک
Deep learning (DL)	An ML method that trains neural networks with many hidden layers.	ڈیپ لرننگ
Weight	A number showing the strength or importance of a connection between neurons.	وزن (وزنِ تعلق)
Bias	An extra constant added to a neuron's sum to improve accuracy.	بائیس
Activation function	A gate that decides whether a signal is important enough to pass on.	ایکٹیویشن فنکشن
Loss function	A measure of how wrong the model's answer is compared with reality.	لاس فنکشن
Epoch	One full training pass through the dataset.	ایپوک (تربیتی چکر)
GPU	Graphics Processing Unit — specialised hardware used to train deep networks.	جی پی یو
NLP	Natural Language Processing — computers understanding and producing human language.	قدرتی زبان کی پروسیسنگ
Authentication	Confirming a user's identity before access (password, PIN, OTP, biometrics).	تصدیقِ شناخت
MFA / 2FA	Multi-/two-factor authentication: two or more different proofs of identity.	کثیر / دو عنصری تصدیق
Access control	Rules for what an authenticated user may do (view, edit, share, ...).	رسائی پر قابو
Encryption	Turning plaintext into ciphertext so unauthorised people cannot read it.	رمز کاری (انکریپشن)
Caesar cipher	A symmetric cipher that shifts letters by a fixed number of places.	سیزر سائفر

Term	Meaning	اردو
3-2-1 backup	Three copies, two media types, one copy stored off-site.	3-2-1 بیک اپ
Firewall	A barrier that monitors network traffic and blocks suspicious access.	فائر وال
Malware	Harmful software such as viruses, worms, spyware, ransomware and Trojans.	میلویئر
Phishing	Fake messages or sites that trick people into giving passwords or OTPs.	فشنگ
Denial of Service (DoS)	Flooding a service with fake requests so real users cannot use it.	سروس سے انکار
Equal access	Giving everyone the same tools and opportunity.	مساوی رسائی
Equity	Giving extra, specific support so people facing barriers reach the same outcome.	انصاف (ایکویٹی)
LMS	Learning Management System — platform for lectures, assignments, quizzes and grades.	تعلیمی نظم کا نظام

Chapter in one look

- **ML** is a model (a trained brain) that learns from data in three steps: input, learning, prediction on new unseen data — without a hand-written rule for every case.
- **AI** $\supset$ **ML** $\supset$ **DL**: deep learning sits inside machine learning, which sits inside artificial intelligence.
- A **neural network** is a brain-inspired mathematical model of connected neurons in an input layer, hidden layer(s) and output layer (face unlock: camera $\rightarrow$ features $\rightarrow$ unlock).
- **Deep learning** trains neural networks with many hidden layers; early layers learn simple features, deeper layers learn complex patterns. It needs huge data and computing power.
- NN components: input layer, neurons, hidden layers, weights, bias, activation function, output layer, loss function. A neuron computes $s = \sum(x \times w) + b$, then an activation gate.
- DL components: large data, many hidden layers, automatic feature learning, learning algorithm (backpropagation), GPUs, training epochs. Pakistan's National AI Policy 2025 covers education, health, agriculture, banking and local languages.
- Applications: image and speech recognition, NLP, healthcare imaging, recommendations, autonomous vehicles, fraud detection, assistive tech.
- **Secure collaboration**: authentication (who), MFA/2FA (two or more factors), access control (what you may do), accountability, privacy, safe transmission.
- **Data protection**: encryption (plaintext $\rightarrow$ ciphertext; symmetric vs asymmetric; Caesar shift), strong passwords, 3-2-1 backup, firewall and antivirus.
- Threats: malware, phishing, unauthorised access, data/identity theft, DoS, human error. Spot them by pop-ups, odd logins, urgent OTP emails, missing files, traffic spikes.
- Mitigation: antivirus + firewall, strong passwords + 2FA, encryption, regular backups, user awareness.
- **Equal access** = same tools for all; **equity** = extra support for the same outcome. Functions: need-based support, removing barriers, recorded participation, multiple formats. Tools: Docs, Zoom, Teams, Classroom/LMS, WhatsApp, email.

Entrepreneurship in the Digital Age

★ 5 Golden · 18 cards · Entrepreneurship in the Digital Age · Prototype · Prototype Development Cycle · Minimum Viable Product (MVP) · Beachhead Market

6.1 Entrepreneur and Entrepreneurship

An entrepreneur is a person who identifies a problem, thinks of a solution, and tries to change that solution into a product or service.

1. Problem → 2. Solution → 3. Product

- They observe people and listen to their needs. They notice what annoys people again and again.
- They take calculated risks. A calculated risk is a risk you take after thinking about what you could lose and how to keep that loss small....
- They improve their idea step by step. They do not expect the first attempt to be perfect.

Remember: Simple meaning. An entrepreneur asks: "What problem can I solve?" Entrepreneurship is the process of testing and improving that solution until it becomes...

کاروباری شخص (Entrepreneur) وہ ہوتا ہے جو کسی مسئلے کو پہچانتا ہے، اس کا حل سوچتا ہے اور اس حل کو کسی پراڈکٹ یا سروس میں بدلنے کی کوشش کرتا ہے۔

6.2 Entrepreneurship in the Digital Age ★ Golden

Entrepreneurship in the digital age means using digital tools to start, test, improve, and grow a business idea.

- These tools may include mobile phones, social media, websites, online forms, digital payments, maps, videos, and simple apps.
- In Pakistan you already know many of them: WhatsApp, Facebook, Instagram, TikTok, YouTube, Google Forms, Google Maps, and digital payments...

Remember: Remember "S-T-I-G": digital tools help you Start, Test, Improve and Grow. Think: "Start The Idea Growing."

Exam: Common questions: "Define entrepreneurship in the digital age", "How have digital tools changed entrepreneurship?" and "Give examples of digital entrepreneurship without..."

ڈیجیٹل دور میں انٹرپرائیورسپ کا مطلب ہے موبائل فون، سوشل میڈیا، ویب سائٹ، آن لائن فارم اور ڈیجیٹل ادائیگی جیسے ٹولز کے ذریعے کاروبار...

Local Examples of Digital Entrepreneurship (Table 6.1)

Digital entrepreneurship can begin from ordinary problems.

- The examples below are close to the context of Sindh and Pakistan, where many students, parents, and small service providers already use...

Problem	Digital Idea	Possible First Users	Urdu Context
Parents miss college notices.	WhatsApp notice and reminder service.	Parents of first-year students.	والدین تک کالج کے نوٹس پہنچانے کے لیے واٹس ایپ سروس۔
Small embroidery shop has few customers.	Simple catalogue with order form on Facebook or WhatsApp.	Customers in the same city.	کڑھائی کی دکان کے لیے واٹس ایپ کیٹلاگ۔
Students forget test dates.	Google Calendar guide or simple reminder app.	Grade XII students.	ٹیسٹ کی تاریخیں یاد دلانے والی ایپ یا کیلنڈر۔
Canteen has long queues during break.	Pre-order form for snacks.	Students and teachers.	کینٹین کے لیے پری آرڈر فارم۔

Remember: Think "Problem first, tool second." Start with the pain, not with the technology.

جدول 6.1 بتاتی ہے کہ ڈیجیٹل کاروبار عام مسائل سے شروع ہو سکتا ہے، جیسے والدین کا کالج نوٹس نہ دیکھ پانا، کڑھائی کی دکان پر کم گاہک...

6.3 From Problem to Business Idea

A good business idea usually starts from observing daily life.

Problem → Idea → Prototype → Feedback → MVP

- What problem do people face?
- Who faces this problem?
- How often does it happen?
- What do people currently do to solve it?

Step	What Students Do	Example
Observe	Notice repeated problems around home, college, or market.	Students forget practical file dates.
Ask Users	Talk to people who face the problem.	Ask five classmates how they remember deadlines.
Select One Problem	Choose one clear problem only.	Practical file reminder.
Suggest Solution	Think of a simple digital or paper-based solution.	Reminder form or app screen.
Test Early	Show the idea to users before final development.	Ask classmates if they will use it.

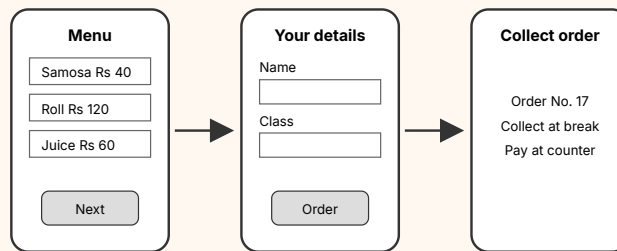
Remember: Mnemonic for Table 6.2: "Only Ask Simple Solutions Today": Observe, Ask users, Select one problem, Suggest solution, Test early.

اچھا کاروباری خیال روزمرہ زندگی کے مشاہدے سے شروع ہوتا ہے۔

6.4 Prototype ★ Golden

A prototype is an early model of a product, service, website, or app.

- It shows how the idea may look or work.
- It is NOT the final product.
- It is made to test the idea and collect feedback.



A paper prototype of a canteen pre-order system: three hand-drawn screens joined by arrows (select item → enter name → collect order).

Remember: Prototype = "practice model". It is like a rough draft of an essay: you show it, get corrections, and rewrite it before the final copy.

Exam: Expect "Define prototype and write its benefits" (3–5 marks). Write: (1) definition with the words early model, not the final product, test the idea and collect...

پروٹوٹائپ (Prototype) کسی پراڈکٹ، سروس، ویب سائٹ یا ایپ کا ابتدائی نمونہ ہوتا ہے جو دکھاتا ہے کہ خیال کیسا نظر آنے گا یا کیسے کام کرے گا۔

6.5 Types of Prototypes

Prototypes are commonly divided into three types according to their level of detail, called fidelity.

Low-Fi (Paper) → Mid-Fi (PowerPoint) → High-Fi (Figma/HTML)

- Fidelity means how close the prototype is to the final product.
- Low fidelity = far from final; high fidelity = very close to final.

Type	Meaning	Best Use	Example
Low-Fidelity	A quick and rough model of the idea.	Very early testing.	Hand-drawn app screens for homework reminder.
Mid-Fidelity	A clearer model that shows layout and flow.	Testing navigation.	Clickable screens in PowerPoint or Figma.
High-Fidelity	A detailed model close to the final product.	Testing realistic experience.	Working demo of a small order form website.

Remember: Fidelity = "faithfulness" to the final product. Remember Paper → PowerPoint → Polished (low → mid → high).

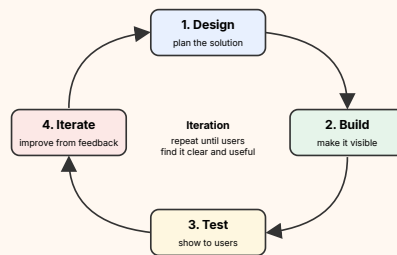
فِڈیلیٹی (Fidelity) کا مطلب ہے کہ پروٹوٹائپ حتمی پراڈکٹ سے کتنا ملتا جلتا ہے۔

6.6 Prototype Development Cycle ★ Golden

Prototype development is not a one-time activity.

- A prototype is created, tested, improved, and tested again.
- This repeating process is called iteration.
- The main stages are Design, Build, Test, and Iterate.

Stage	What happens
1. Design	Understand the problem and plan the solution. Decide what the prototype should show (e.g. screen flow, features).
2. Build	Create the first version (physical or digital). The aim is not beauty; the aim is to make the idea visible.
3. Test	Show the prototype to users. Observe what they understand and what they find difficult.
4. Iterate	Improve the prototype based on user feedback. Remove unnecessary parts and add missing features.



The prototype development cycle: Design → Build → Test → Iterate, then back to Design.

Remember: Mnemonic: "Do Big Things Iteratively" = Design, Build, Test, Iterate. And remember the arrow at the end goes back to Design: it is a cycle, not a line.

Exam: A favourite long question: "Explain the prototype development cycle with a diagram." Draw the four boxes in a circle with arrows, write one or two lines for each stage...

پروٹوٹائپ ایک ہی بار میں نہیں بنتا؛ اسے بنایا، آزمایا، بہتر کیا اور دوبارہ آزمایا جاتا ہے۔

6.7 Class Activity: Create a Prototype

Business idea scenario. Students often forget homework deadlines, test dates, and required class items.

1. Understand → 2. Draw low-fi → 3. Test → 4. Analyse feedback → 5. Improve

- Students often forget homework deadlines, test dates, and required class items.
- A group of students wants to design a simple "Homework Helper" or "ClassBuddy" reminder system.
- The team will not immediately build a complete mobile app.
- They will first understand the problem, draw a low-fidelity prototype, test it with classmates, and improve it.

Finding	Meaning	Design Decision
7 out of 10 forget tasks.	The problem is common.	App should show a daily task list.
Reminders needed before leaving home.	Reminder time is important.	Add morning notification option.
Students want simple apps.	Design must be simple.	Use large buttons and clear labels.

Remember: Feedback → Problem → Improvement. Every comment should be turned into a problem statement and then one clear change, exactly like the three columns of Table...

اس سرگرمی میں طلبہ "کلاس بڈی" (ClassBuddy) نامی یاد دہانی نظام بناتے ہیں۔

6.8 Minimum Viable Product (MVP) ★ Golden

A Minimum Viable Product (MVP) is the simplest working version of a product.

- Minimum: it has only the essential features. No extras.
- Viable: it is usable enough to actually solve the main problem. ("Viable" means able to work successfully.)

Complete canteen app (final product)	Canteen MVP
Login, online payment, delivery tracking, discount offers...	A simple Google Form where students select snacks before break.

Remember: Simple meaning. An MVP answers: "Will people actually use this solution?" It tests real use and behaviour, not only the visual design. Analogy: before opening...

Exam: Typical questions: "What is an MVP? Explain the words Minimum and Viable" and "Give an example of an MVP." Write the full form, the definition (simplest working version,...

ایم وی پی (Minimum Viable Product) کسی پراڈکٹ کا سب سے سادہ مگر کام کرنے والا ورژن ہے جس میں صرف ضروری خصوصیات ہوتی ہیں۔

6.9 Difference between Prototype and MVP

Prototype and MVP are related, but not the same.

- A prototype is mainly used to show and test the design of an idea.
- An MVP is used to test real use and market demand.
- Market demand means whether enough people want the product and will use (or pay for) it.

Point	Prototype	MVP
Main purpose	To show how the idea may look or work.	To test whether users will actually use the solution.
Working status	May not work; can be a sketch or model.	Must work at least in a basic way.
Users	Small group gives feedback on design.	Early real users test the product in daily life.
Cost	Very low cost.	Low cost but may need some basic development.
Example	Paper screens for a canteen order system.	Google Form used for real canteen orders for three days.

Remember: Key takeaway Prototype = "Does it look right and make sense?" MVP = "Will people actually use it and find value in it?"

Memory hook: Prototype → Picture of the...

پروٹوٹائپ اور ایم وی پی آپس میں جڑے ہیں مگر ایک جیسے نہیں۔

6.10 Identifying the Riskiest Assumption

Every business idea is based on assumptions.

- An assumption is something we believe to be true, but have not tested yet.
- The riskiest assumption is the belief that can make the entire business fail if it turns out to be wrong.
- Entrepreneurs should identify and test the riskiest assumption early, because it saves time and money.

Business Idea	Possible Risky Assumption	How to Test It (MVP)
Online tuition finder	Students will pay for online tutoring.	Ask 20 students and offer a trial class to see if they sign up.
Canteen pre-order form	Students will order snacks before break.	Test a Google Form for one week in real time.
Ajrak online catalogue	Customers will place orders through WhatsApp.	Share catalogue with 30 customers and record actual orders.
College van update system	Parents and students will check van updates daily.	Create a trial WhatsApp broadcast group for one route.

Remember: Ask: "What would kill this idea?" The answer is usually the riskiest assumption. Test it first, cheaply, with real behaviour.

ہر کاروباری خیال کچھ مفروضوں (Assumptions) پر قائم ہوتا ہے، یعنی وہ باتیں جنہیں ہم سچ سمجھتے ہیں مگر ابھی آزمایا نہیں۔

6.11 Developing an MVP

To develop an MVP, the team must select only the essential features.

- Extra features should be avoided at the beginning.
- A common mistake is trying to build everything at once.
- This makes the product difficult, costly, and slow.

MVP Element	Question to Ask	Example Answer
Problem	What problem are we solving?	Students wait too long at canteen.
Target users	Who will use it first?	Grade XII students in one college.
Main feature	What is the ONE feature that solves it?	Pre-order snacks before break.
Basic design	What steps are needed?	Menu, name, class, order time.
Cost	What free/low-cost tools to use?	Google Form or printed slips.
Marketing	How will early users hear about it?	WhatsApp message to class group.
Feedback	How will users respond?	Short feedback form after 3 days.

Remember: Mnemonic for the 8 elements: "People Try My Bakery; Cheap Muffins Find Improvement": Problem, Target users, Main feature, Basic design, Cost, Marketing,...

ایم وی پی بناتے وقت صرف ضروری خصوصیات چنیں؛ سب کچھ ایک ساتھ بنانے سے کام مشکل، مہنگا اور سست ہو جاتا ہے۔

6.12 Case Study: MVP for a Canteen Pre-Order System

Problem: In a college, students wait in long lines at the canteen.

Paper prototype → Feedback: short menu, pay at collection → Google Form MVP → 3-day test → Improve

- Sometimes snacks finish before students reach the counter.
- A group wants to test a canteen pre-order idea.
- Goal: Their aim is NOT to create a complete app immediately.

MVP Element	Decision
Features	Menu list, student name, class, item, and quantity.
Cost & Tools	Free Google Form or printed order slips.
Marketing	Class announcement and WhatsApp group message.
Testing period	Three days.
Feedback method	Ask students if they saved time and would use it again.
Improvement	Keep only 5 items; close orders 20 minutes before break.

Remember: Remember the case study as "5 items, 3 days, 20 minutes": five menu items, a three-day test, orders closing twenty minutes before break.

ایک کالج میں طلبہ کینٹین کی لمبی قطاروں میں انتظار کرتے تھے۔

6.13 Testing an MVP

MVP testing means giving the basic working version to real users and studying their response.

- Numbers (quantitative data) show how many users tried the MVP and how many may use it again.
- Comments (qualitative data) explain what users liked, disliked, or found confusing.
- How many users will test it.
- How long the test will continue.

Type of Feedback	Question	Example Result
Number-based	How many users tried it?	40 students used the form.
Number-based	How many would use it again?	30 out of 40 said yes.
Comment-based	What was confusing?	Did not know order closing time.
Observation	Where did users face difficulty?	Forgot to enter class name.

Remember: Data skill link. Feedback is useful only when it is studied: Count the responses → Look for repeated problems → Decide which improvement comes first. Remember...

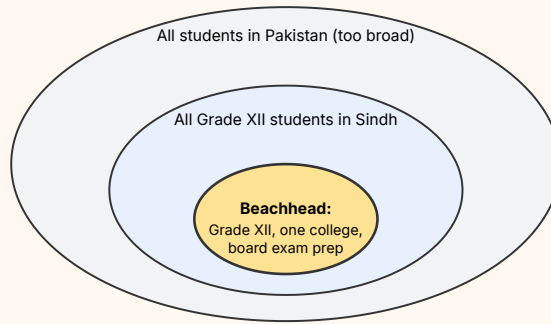
ایم وی پی کی جانچ کا مطلب ہے کہ بنیادی کام کرنے والا ورژن اصل صارفین کو دیا جائے اور ان کے ردعمل کا مطالعہ کیا جائے۔

6.14 Beachhead Market ★ Golden

A beachhead market is a small and focused group of first users.

- These users have a clear need and are likely to try the product early.
- Instead of trying to serve everyone, entrepreneurs first focus on one small group.
- This helps them test the MVP properly and improve it before expanding.

Business Idea	Weak First Market	Better Beachhead Market
Study app	All students in Pakistan.	Grade XII students preparing for exams in one college.
Canteen pre-order	All canteens in the city.	One college canteen during lunch break.
Ajrak catalogue	All online shoppers.	Existing customers of one local shop.
Van update system	All transport users.	Students and parents of ONE specific college van route.



Start with the small inner circle (beachhead), learn and improve, then expand outward.

Remember: Remember S-R-U: Specific, Reachable, Urgent need. If you cannot meet your first users within a week, your market is too big.

Exam: Asked as "Define beachhead market and give its characteristics" or "Suggest a beachhead market for ...". Write the definition (small, focused group of first users with a...

بیج ہیڈ مارکیٹ (Beachhead Market) پہلے صارفین کا ایک چھوٹا اور مخصوص گروہ ہے جسے حل کی واضح اور فوری ضرورت ہو اور جو پراڈکٹ جلد آزمانے پر تیار ہو۔

6.15 Complete Student Project

You will work in groups and complete one small entrepreneurship project.

- The project should be simple, local, and possible in the classroom.
- The purpose is to practise the complete journey: problem → prototype → feedback → MVP plan → risky assumption → beachhead market.
- Suggested project ideas: homework and test reminder system, college notice organizer, canteen pre-order system, small shop WhatsApp...

Criteria	Excellent	Satisfactory	Needs Improvement
Problem ID	Clear, local, and important.	Understandable but needs detail.	Vague or too broad.
Prototype	Shows clear flow and features.	Shows the basic idea.	Incomplete or unclear.
Feedback & Fixes	Collected properly; changes clearly based on feedback.	Collected but analysis is weak; some changes made.	No clear feedback collected; changes not explained.
MVP Plan	Essential features and clear testing plan.	Basic features but needs testing clarity.	Confused with the final product.

Remember: Use the rubric before you submit: read your project sheet row by row and ask, "Is this Excellent yet?" For "Feedback & Fixes", always write which feedback...

طلبہ گروپوں میں ایک چھوٹا، مقامی اور آسان منصوبہ مکمل کرتے ہیں جس میں مسئلے سے پروٹوٹائپ، رائے، ایم وی پی منصوبہ، سب سے خطرناک مفر...

6.16 Ethical and Safe Use of Digital Tools

Digital entrepreneurship should be responsible.

- Privacy: people's personal information belongs to them.
- Truthfulness: report results honestly; do not make up numbers or hide bad feedback.
- Fairness: treat all users equally.

Remember: Ask yourself "P-T-F-E": Is it Private, Truthful, Fair, and Equal-access?

ڈیجیٹل کاروبار ذمہ داری سے کرنا چاہیے۔

Career Opportunities

This unit covered entrepreneurship in the digital age, turning problems into business ideas, prototyping, building MVPs, and identifying beachhead markets.

- Mastery of these concepts opens the door to careers in startups and digital innovation.
- Pakistan has many incubators (for example, National Incubation Centres, NICs, in major cities) that support young founders.

Career	What the person does
Tech Entrepreneur / Startup Founder	Identifies a real-world problem and builds a digital product or service to solve it, guiding it from prototype through to MVP and market launch.
Product Manager	Guides a product's direction by identifying user needs, prioritising features, and testing ideas through prototypes and MVPs.
Business Analyst	Analyses business problems and processes to recommend digital solutions, often testing assumptions before full investment.
Innovation / Startup Consultant	Advises businesses or founders on validating ideas, building MVPs, and identifying beachhead markets.
Digital Marketing Specialist	Plans how a new product or MVP will reach its first (beachhead) market and grow its user base.
Venture / Incubator Analyst	Evaluates early-stage startup ideas and prototypes for potential investment or incubation support.

Remember: Every career in the table uses the same chapter skills: understand users, test assumptions, start small.

اس باب کی مہارتیں اسٹارٹ اپ اور ڈیجیٹل جدت کے شعبوں میں روزگار کے دروازے کھولتی ہیں، جیسے اسٹارٹ اپ بانی، پراڈکٹ مینیجر، بزنس اینا...

Key terms

Term	Meaning	اردو
Entrepreneur	A person who identifies a problem, thinks of a solution, and turns it into a product or service.	کاروباری شخص
Entrepreneurship	The process of planning, starting, testing and improving a business idea.	کاروبار کاری
Calculated risk	A risk taken after carefully thinking about the possible loss and keeping it small.	سوچا سمجھا خطرہ
Digital entrepreneurship	Using digital tools to start, test, improve and grow a business idea.	ڈیجیٹل کاروبار کاری
Business idea	A proposed product or service that solves a real problem for real users.	کاروباری خیال
Target users	The people who will use the product first.	ہدف صارفین
Prototype	An early model of a product, service, website or app made to test the idea and collect feedback.	ابتدائی نمونہ
Fidelity	How close a prototype is to the final product.	اصل سے مطابقت کا درجہ
Low-fidelity prototype	A quick, rough paper model showing the main idea and flow.	کم تفصیل والا نمونہ
Mid-fidelity prototype	A clearer model in PowerPoint, Canva or Figma showing layout and navigation without final graphics.	درمیانی تفصیل والا نمونہ
High-fidelity prototype	A detailed model close to the final product with realistic colours, buttons, data and working features.	زیادہ تفصیل والا نمونہ
Wireframe	A simple outline drawing of a screen showing where items go, without colours or pictures.	اسکرین کا خاکہ
Navigation	How a user moves from one screen to another.	اسکرینوں کے درمیان آمد و رفت
Iteration	Repeating the cycle of creating, testing and improving.	بار بار دہرانا
Feedback	Users' responses and comments about a product or idea.	رائے
Minimum Viable Product (MVP)	The simplest working version of a product with only the essential features, used to test the idea with real users.	کم از کم قابل عمل پراڈکٹ

Term	Meaning	اردو
Viable	Usable enough to actually solve the main problem.	قابلِ عمل
Market demand	Whether enough people want and will use or pay for a product.	بازار کی طلب
Assumption	Something we believe to be true but have not tested yet.	مفروضہ
Riskiest assumption	The belief that would make the whole business fail if it turned out to be wrong.	سب سے خطرناک مفروضہ
MVP plan	A plan stating problem, target users, main feature, design, cost, marketing, feedback and improvement.	ایم وی پی کا منصوبہ
Number-based feedback	Feedback in counts, e.g. how many tried the MVP or would use it again.	عددی رائے
Comment-based feedback	Feedback in words explaining likes, dislikes or confusion.	تبصراتی رائے
Observation	Watching users to see where they face difficulty.	مشاہدہ
Beachhead market	A small, focused group of first users with a clear need who are likely to try the product early.	ابتدائی مخصوص مارکیٹ
Rubric	A marking guide describing levels of quality for each criterion.	جانچ کا معیار نامہ
Privacy	A person's right to control their personal information.	رازداری
Equity	Making sure people with fewer resources or language difficulty can also use a solution.	مساوی رسائی
Startup	A new, small business built around a new product or service idea.	نیا کاروبار

Chapter in one look

- **Entrepreneur:** a person who identifies a problem, thinks of a solution, and turns it into a product or service; observes people, takes calculated risks, improves step by step.
- **Entrepreneurship:** the process of planning, starting, testing and improving a business idea until it becomes useful (Problem → Solution → Product).
- **Digital age:** entrepreneurs use accessible digital tools (social media, forms, websites, WhatsApp, digital payments) to start and test ideas quickly with low investment; often no coding is needed.
- A digital idea must solve a **real problem for real users**, not just sound modern (Table 6.1 examples: notices, embroidery shop, test dates, canteen queues).
- **Business ideas** come from observing daily life and talking directly to users: Observe → Ask Users → Select One Problem → Suggest Solution → Test Early. Start with a small, clear problem.
- **Prototype:** an early model used to show and test the design of an idea; not the final product; makes the idea visible, finds mistakes early, saves time and money.
- **Fidelity** is how close a prototype is to the final product: Low (paper), Mid (PowerPoint/Canva/Figma, tests navigation), High (realistic, Figma/HTML, only after the idea is tested).
- **Development cycle:** Design → Build → Test → Iterate, repeated (iteration). Feedback is crucial; a prototype fails only if useful feedback is ignored.
- **MVP:** the simplest working version with only essential features (Minimum) that still solves the main problem (Viable); it tests real use and market demand.
- **Prototype vs MVP:** prototype asks "Does it look right?"; MVP asks "Will people actually use it?" A prototype may not work; an MVP must.
- **Riskiest assumption:** the untested belief that, if wrong, will make the business fail; test it early with real behaviour.
- **MVP plan:** problem, target users, ONE main feature, basic design, cost, marketing, feedback, improvement plan. Do not build everything at once.
- **Canteen case study:** paper prototype → Google Form MVP with 5 items for 3 days → staff found too many items hard → kept 5 items, separate line, orders close 20 minutes before break.
- **MVP testing:** plan users, duration and data in advance; collect numbers (40 tried, 30 would return = 75%) and comments/observations; count, look for repeated problems, fix the most important first.
- **Beachhead market:** a small, focused, reachable group of first users with a strong, immediate need; win it first, then expand.
- **Ethics and equity:** respect privacy, truthfulness and fairness; collect no unnecessary data; share nothing without permission; design for weak internet, limited devices and Urdu/Sindhi speakers.

NIGHT BEFORE

One-page recaps

Read each recap, close the booklet, say it aloud. Then stop.

Chapter 1 recap — Human Computer Interaction

- HCI = study of how people and computers work together so systems are usable, useful and satisfying.
- Traditional interaction = keyboard, mouse, menus. Natural interaction = touch, voice, gesture, gaze.
- Components: user, computer, interaction (input + output + feedback) inside an environment.
- UI = how it looks and the controls. UX = how the whole experience feels. Wireframe → prototype (Figma) → test.
- Evaluate with usability tests, heuristic evaluation, A/B tests and automated checks. Golden: accessibility, HCI problems, UI vs UX, testing methods.

Chapter 2 recap — Algorithms and Data Structures

- Correctness: trace table (variable values after every step) or stepwise reasoning.
- Clarity: modularity (small named parts) and readability (names, comments, indentation).
- Efficiency: count steps, conditions, repetitions. Big O: $O(1)$ constant, $O(\log n)$ binary search, $O(n)$ linear, $O(n \log n)$ good sorts, $O(n^2)$ nested loops.
- Linear DS: array, linked list, stack (LIFO, push/pop), queue (FIFO, enqueue/dequeue). Non-linear: tree (one root, parent-child), graph (nodes + edges).
- Pick the structure from the problem: undo → stack; printer jobs → queue; hierarchy → tree; maps → graph; fast index → array.

Chapter 3 recap — Python Data Structures, Functions and Files

- list [] ordered, mutable. tuple () ordered, immutable. set {} unordered, unique. dict {k:v} key-value, mutable.
- Functions: def, parameters, return. Local vs global scope. Built-in vs user-defined.
- Files: "r" read, "w" overwrite, "a" append. Always prefer with open(...) as f so the file closes itself.
- Exceptions: try / except / else / finally. Catch specific errors (ValueError, FileNotFoundError).

Chapter 4 recap — Data and Analysis

- SQLite in Python: connect → cursor → execute → commit → close. Always commit writes.
- Pandas DataFrame = table. load CSV/Excel, isnull, dropna, fillna, sort, filter, groupby.
- Charts: bar (categories), line (trend over time), pie (parts of a whole, few slices), histogram (distribution), scatter (relationship of two variables), box (spread and outliers).
- Mean = average. Median = middle after sorting. Mode = most frequent. Range = max-min. Variance / std-dev measure spread. pandas .std() uses sample n-1.

Chapter 5 recap — AI, Security and Collaboration

- AI ⊃ Machine Learning ⊃ Deep Learning. A neuron: weighted sum + bias, then an activation function. Deep = many hidden layers.
- Secure collaboration: authentication (passwords, MFA), access control (who may do what).
- Protect data: encryption, strong passwords, 3-2-1 backup, firewall. Threats: malware, phishing, DoS/DDoS, social engineering.
- Equity: everyone can take part (devices, language, disability, bandwidth). Tools: shared docs, video, LMS, versioning.

Chapter 6 recap — Entrepreneurship in the Digital Age

- Entrepreneur = person who turns a problem into a venture, taking risk. Digital tools make testing cheap (WhatsApp, Forms, a Facebook page).
- Prototype = experimental model. Low / mid / high fidelity. Cycle: Design → Build → Test → Iterate.
- MVP = smallest product that tests the riskiest assumption with real users. Prototype ≠ MVP (MVP is used in the real market).
- Beachhead market = the first, narrow group you can win. Measure retention, not only sign-ups. Use data ethically.